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UW1

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2.16 Karp’s minimum mean-weight cycle algorithm

$G = (V, E)$ -directed graph with weight function $w : E \rightarrow \mathbb{R}$ $n = |V|$
Assume that every vertex is reachable from $s \in V$. $\delta_k(s, v)$ shortest k -path from s to v (simple dp) Minimum mean-weight cycle is

$$\min_{v \in V} \max_{0 \leq k \leq n-1} \frac{\delta_n(s, v) - \delta_k(s, v)}{n - k}$$

Matma (3)

berlekamp-massey

#0b7946, includes: simple-module
 $\mathcal{O}(n^2 \log k)$, BerlekampMassey<mod> bm(x) zgaduje rekurencję ciągu x ,
bm.get(k) zwraca k -ty wyraz ciągu x (index 0)

```
struct BerlekampMassey {
    int n;
    vi x, c;
    BerlekampMassey(C vi &x_) : x(_x) {
        auto B = c = {1};
        int b = 1, m = 0;
        REP(i, ssize(x)) {
            m++; int d = x[i];
            FOR(j, 1, ssize(c) - 1)
                d = add(d, mul(c[j], x[i - j]));
            if(d == 0) continue;
            auto _B = c;
            c.resize(max(ssize(c), m + ssize(B)));
            int coef = mul(d, inv(b));
            FOR(j, m, m + ssize(B) - 1)
                c[j] = sub(c[j], mul(coef, B[j - m]));
            if(ssize(_B) < m + ssize(B)) { B = _B; b = d; m = 0; }
        }
        c.erase(c.begin());
        for(int &t : c) t = sub(0, t);
        n = ssize(c);
    }
    vi combine(vi a, vi b) {
        vi ret(n * 2 + 1);
        REP(i, n + 1) REP(j, n + 1)
            ret[i + j] = add(ret[i + j], mul(a[i], b[j]));
        for(int i = 2 * n; i > n; i--) REP(j, n)
            ret[i - j - 1] = add(ret[i - j - 1], mul(ret[i], c[j]));
        return ret;
    }
    int get(ll k) {
        if (!n) return 0;
        vi r(n + 1), pw(n + 1);
        r[0] = pw[1] = 1;
        for(k++; k; k /= 2) {
            if(k % 2) r = combine(r, pw);
            pw = combine(pw, pw);
        }
        int ret = 0;
        REP(i, n) ret = add(ret, mul(r[i + 1], x[i]));
        return ret;
    }
};
```

bignum

#140b6c
Podstawa wynosi 1e9. Mnożenie, dzielenie, nwd oraz modulo jest kwadratowe, wersje operatorX(Num, int) liniowe. Podstawę można zmieniać (ma zachodzić base == 10^digits_per_elem).

```
// BEGIN HASH dcf8cf
struct Num {
    static constexpr int digits_per_elem = 9, base = int(1e9);
```

berlekamp-massey bignum binsearch-ster

```
int sign = 0;
vi x;
Num& shorten() {
    while(ssize(x) and x.back() == 0)
        x.pop_back();
    for(int a : x)
        assert(0 <= a and a < base);
    if(x.empty())
        sign = 0;
    return *this;
}
Num(string s) {
    sign = ssize(s) and s[0] == '-' ? s.erase(s.begin())
        , -1 : 1;
    for(int i = ssize(s); i > 0; i -= digits_per_elem)
        if(i < digits_per_elem)
            x.eb(stoi(s.substr(0, i)));
        else
            x.eb(stoi(s.substr(i - digits_per_elem,
                digits_per_elem)));
    shorten();
}
Num() {}
Num(ll s) : Num(to_string(s)) {}
}; // END HASH
// BEGIN HASH f6944d
string to_string(C Num& n) {
    stringstream s;
    s << (n.sign == -1 ? "-" : "") << (ssize(n.x) ? n.x.back() : 0);
    for(int i = ssize(n.x) - 2; i >= 0; --i)
        s << setfill('0') << setw(n.digits_per_elem) << n.x[i];
    return s.str();
}
ostream& operator<<(ostream &o, C Num& n) {
    return o << to_string(n).c_str();
} // END HASH
// BEGIN HASH 2c9227
auto operator<=>(C Num& a, C Num& b) {
    if(a.sign != b.sign or ssize(a.x) != ssize(b.x))
        return ssize(a.x) * a.sign <=> ssize(b.x) * b.sign;
    for(int i = ssize(a.x) - 1; i >= 0; --i)
        if(a.x[i] != b.x[i])
            return a.x[i] * a.sign <=> b.x[i] * b.sign;
    return strong_ordering::equal;
}
bool operator==(C Num& a, C Num& b) {
    return a.x == b.x and a.sign == b.sign;
} // END HASH
// BEGIN HASH 57d66a
Num abs(Num n) { n.sign &= 1; return n; }
Num operator+(Num a, Num b) {
    int mode = a.sign * b.sign >= 0 ? a.sign != b.sign, 1
        : abs(b) > abs(a) ? swap(a, b), -1 : -1, carry = 0;
    for(int i = 0; i < max(ssize((mode == 1 ? a : b).x),
        ssize(b.x)) or carry; ++i) {
        if(mode == 1 and i == ssize(a.x))
            a.x.eb(0);
        a.x[i] += mode * (carry + (i < ssize(b.x) ? b.x[i] : 0));
        carry = a.x[i] >= a.base or a.x[i] < 0;
        a.x[i] -= mode * carry * a.base;
    }
    return a.shorten();
} // END HASH
Num operator-(Num a) { a.sign *= -1; return a; }
Num operator-(Num a, Num b) { return a + -b; }
// BEGIN HASH 32f87a
Num operator*(Num a, int b) {
    assert(abs(b) < a.base);
    int carry = 0;
    for(int i = 0; i < ssize(a.x) or carry; ++i) {
        if(i == ssize(a.x))
            a.x.eb(0);
```

```
ll cur = a.x[i] * ll(abs(b)) + carry;
a.x[i] = int(cur % a.base);
carry = int(cur / a.base);
}
if(b < 0)
    a.sign *= -1;
return a.shorten();
} // END HASH
// BEGIN HASH ca88a0
Num operator*(C Num& a, C Num& b) {
    Num c;
    c.x.resize(ssize(a.x) + ssize(b.x));
    REP(i, ssize(a.x))
        for(int j = 0, carry = 0; j < ssize(b.x) or carry; ++j) {
            ll cur = c.x[i + j] + a.x[i] * ll(j < ssize(b.x)
                ? b.x[j] : 0) + carry;
            c.x[i + j] = int(cur % a.base);
            carry = int(cur / a.base);
        }
    c.sign = a.sign * b.sign;
    return c.shorten();
} // END HASH
// BEGIN HASH 520797
Num operator/(Num a, int b) {
    assert(b != 0 and abs(b) < a.base);
    int carry = 0;
    for(int i = ssize(a.x) - 1; i >= 0; --i) {
        ll cur = a.x[i] + carry * ll(a.base);
        a.x[i] = int(cur / abs(b));
        carry = int(cur % abs(b));
    }
    if(b < 0)
        a.sign *= -1;
    return a.shorten();
} // END HASH
// BEGIN HASH c2ef8e
// zwraca a * pow(a.base, b)
Num shift(Num a, int b) {
    V v(b, 0);
    a.x.insert(a.x.begin(), v.begin(), v.end());
    return a.shorten();
}
Num operator/(Num a, Num b) {
    assert(ssize(b.x));
    int s = a.sign * b.sign;
    Num c;
    a = abs(a);
    b = abs(b);
    for(int i = ssize(a.x) - ssize(b.x); i >= 0; --i) {
        if (a < shift(b, i)) continue;
        int l = 0, r = a.base - 1;
        while (l < r) {
            int m = (l + r + 1) / 2;
            if (shift(b * m, i) <= a)
                l = m;
            else
                r = m - 1;
        }
        c = c + shift(l, i);
        a = a - shift(b * l, i);
    }
    c.sign = s;
    return c.shorten();
} // END HASH
// BEGIN HASH cb30ff
template<typename T>
Num operator%(C Num& a, C T& b) { return a - ((a / b) * b); }
Num nwd(C Num& a, C Num& b) { return b == Num() ? a : nwd(b, a % b); }
// END HASH
```

binsearch-ster

#3dce62

$\mathcal{O}(\log \max_val)$, szuka największego a/b , że $is_ok(a/b)$ oraz $0 \leq a, b \leq \max_value$. Zakłada, że $is_ok(0) == true$.

```
using Frac = pair<ll, ll>;
Frac my_max(Frac l, Frac r) {
    return l.fi * __int128_t(r.se) > r.fi * __int128_t(l.se) ? l : r;
}
Frac binsearch(ll max_value, function<bool (Frac)> is_ok) {
    assert(is_ok(pair(0, 1)) == true);
    Frac left = {0, 1}, right = {1, 0}, best_found = left;
    int current_dir = 0;
    while(max(left.fi, left.se) <= max_value) {
        best_found = my_max(best_found, left);
        auto get_frac = [&](ll mul) {
            ll mull = current_dir ? 1 : mul;
            ll mulr = current_dir ? mul : 1;
            return pair(left.fi * mull + right.fi * mulr,
                left.se * mull + right.se * mulr);
        };
        auto is_good_mul = [&](ll mul) {
            Frac mid = get_frac(mul);
            return is_ok(mid) == current_dir and max(mid.fi, mid.se) <= max_value;
        };
        ll power = 1;
        for(; is_good_mul(power); power *= 2) {}
        ll bl = power / 2 + 1, br = power;
        while(bl != br) {
            ll bm = (bl + br) / 2;
            if(not is_good_mul(bm))
                br = bm;
            else
                bl = bm + 1;
        }
        tie(left, right) = pair(get_frac(bl - 1), get_frac(bl));
        if(current_dir == 0)
            swap(left, right);
        current_dir ^= 1;
    }
    return best_found;
}
```

crt

#e3fa03, includes: extended-gcd
 $\mathcal{O}(\log n)$, crt(a, m, b, n) zwraca takie x , że $x \bmod m = a$ oraz $x \bmod n = b$, m oraz n nie muszą być względnie pierwsze, ale może nie być wtedy rozwiązania (assert wywali, ale można zmienić na return -1).

```
ll crt(ll a, ll m, ll b, ll n) {
    if(n > m) swap(a, b), swap(m, n);
    auto [d, x, y] = extended_gcd(m, n);
    assert((a - b) % d == 0);
    ll ret = (b - a) % n * x % n / d * m + a;
    return ret < 0 ? ret + m * n / d : ret;
}
```

determinant

#448aca, includes: matrix-header
 $\mathcal{O}(n^3)$, wyznacznik macierzy (modulo lub double)

```
T determinant(V<V<T>>& a) {
    int n = ssize(a);
    T res = 1;
    REP(i, n) {
        int b = i;
        FOR(j, i + 1, n - 1)
            if(abs(a[j][i]) > abs(a[b][i]))
                b = j;
        if(i != b)
            swap(a[i], a[b]), res = sub(0, res);
        res = mul(res, a[i][i]);
    }
```

```
    if (equal(res, 0))
        return 0;
    FOR(j, i + 1, n - 1) {
        T v = divide(a[j][i], a[i][i]);
        if (not equal(v, 0))
            FOR(k, i + 1, n - 1)
                a[j][k] = sub(a[j][k], mul(v, a[i][k]));
    }
    return res;
}
```

discrete-log
#466b80, includes: simple-modulo
 $\mathcal{O}(\sqrt{m} \log n)$ czasowo, $\mathcal{O}(\sqrt{n})$ pamięciowo, dla liczby pierwszej mod oraz $a, b \nmid mod$ znajdzie e takie że $a^e \equiv b \pmod{mod}$. Jak zwróci -1 to nie istnieje.

```
int discrete_log(int a, int b) {
    int n = int(sqrt(mod)) + 1;
    int an = 1;
    REP(i, n)
        an = mul(an, a);
    unordered_map<int, int> vals;
    int cur = b;
    FOR(q, 0, n) {
        vals[cur] = q;
        cur = mul(cur, a);
    }
    cur = 1;
    FOR(p, 1, n) {
        cur = mul(cur, an);
        if(vals.count(cur)) {
            int ans = n * p - vals[cur];
            return ans;
        }
    }
    return -1;
}
```

discrete-root
#7a0737, includes: primitive-root, discrete-log
Dla pierwszego mod oraz $a \perp mod$, k znajduje b takie, że $b^k = a$ (pierwiastek k -tego stopnia z a). Jak zwróci -1 to nie istnieje.

```
int discrete_root(int a, int k) {
    int g = primitive_root();
    int y = discrete_log(powi(g, k), a);
    if(y == -1)
        return -1;
    return powi(g, y);
}
```

extended-gcd
#c499ae
 $\mathcal{O}(\log(\min(a, b)))$, dla danego (a, b) znajduje takie $(gcd(a, b), x, y)$, że $ax + by = gcd(a, b)$. auto [gcd, x, y] = extended_gcd(a, b);

```
tuple<ll, ll, ll> extended_gcd(ll a, ll b) {
    if(a == 0)
        return {b, 0, 1};
    auto [gcd, x, y] = extended_gcd(b % a, a);
    return {gcd, y - x * (b / a), x};
}
```

fft-mod
#a03d84, includes: fft
 $\mathcal{O}(n \log n)$, conv_mod(a, b) zwraca iloczyn wielomianów modulo, ma większą dokładność niż zwykłe fft.

```
vi conv_mod(vi a, vi b, int M) {
    if(a.empty() or b.empty()) return {};
    vi res(ssize(a) + ssize(b) - 1);
    C int CUTOFF = 125;
```

```
    if (min(ssize(a), ssize(b)) <= CUTOFF) {
        if (ssize(a) > ssize(b))
            swap(a, b);
        REP(i, ssize(a))
            REP(j, ssize(b))
                res[i + j] = int((res[i + j] + ll(a[i] * b[j]) % M);
        return res;
    }
    int B = 32 - __builtin_clz(ssize(res)), n = 1 << B;
    int cut = int(sqrt(M));
    V<Complex> L(n), R(n), outl(n), outs(n);
    REP(i, ssize(a)) L[i] = Complex((int) a[i] / cut, (int) a[i] % cut);
    REP(i, ssize(b)) R[i] = Complex((int) b[i] / cut, (int) b[i] % cut);
    fft(L), fft(R);
    REP(i, n) {
        int j = -i & (n - 1);
        outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
        outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / 1i;
    }
    fft(outl), fft(outs);
    REP(i, ssize(res)) {
        ll av = ll(real(outl[i]) + 0.5), cv = ll(imag(outs[i]) + 0.5);
        ll bv = ll(imag(outl[i]) + 0.5) + ll(real(outs[i]) + 0.5);
        res[i] = int(((av % M * cut + bv) % M * cut + cv) % M);
    }
    return res;
}
```

fft
#b0cf54
 $\mathcal{O}(n \log n)$, conv(a, b) to iloczyn wielomianów.

```
// BEGIN HASH 8b009c
using Complex = complex<double>;
void fft(V<Complex> &a) {
    int n = ssize(a), L = 31 - __builtin_clz(n);
    static V<complex<long double>> R(2, 1);
    static V<Complex> rt(2, 1);
    for(static int k = 2; k < n; k *= 2) {
        R.resize(n), rt.resize(n);
        auto x = polar(1.0L, acos(-1) / k);
        FOR(i, k, 2 * k - 1)
            rt[i] = R[i] = i & 1 ? R[i / 2] * x : R[i / 2];
    }
    vi rev(n);
    REP(i, n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
    REP(i, n) if(i < rev[i]) swap(a[i], a[rev[i]]);
    for(int k = 1; k < n; k *= 2) {
        for(int i = 0; i < n; i += 2 * k) REP(j, k) {
            Complex z = rt[j + k] * a[i + j + k]; // mozna zoptowac rozpisujac
            a[i + j + k] = a[i + j] - z;
            a[i + j] += z;
        }
    }
} // END HASH
V<double> conv(V<double> &a, V<double> &b) {
    if(a.empty() || b.empty()) return {};
    V<double> res(ssize(a) + ssize(b) - 1);
    int L = 32 - __builtin_clz(ssize(res)), n = (1 << L);
    V<Complex> in(n), out(n);
    copy(all(a), in.begin());
    REP(i, ssize(b)) in[i].imag(b[i]);
    fft(in);
    for(auto &x : in) x *= x;
    REP(i, n) out[i] = in[-i & (n - 1)] - conj(in[i]);
    fft(out);
    REP(i, ssize(res)) res[i] = imag(out[i]) / (4 * n);
    return res;
}
```

```
    }

floor-sum
#b06d8c
 $\mathcal{O}(\log a)$ , liczy  $\sum_{i=0}^{n-1} \left\lfloor \frac{a \cdot i + b}{c} \right\rfloor$ . Działa dla  $0 \leq a, b < c$  oraz  $1 \leq c, n \leq 10^9$ . Dla innych  $n, a, b, c$  trzeba uważać lub użyć __int128.

ll floor_sum(ll n, ll a, ll b, ll c) {
    ll ans = 0;
    if (a >= c) {
        ans += (n - 1) * n * (a / c) / 2;
        a %= c;
    }
    if (b >= c) {
        ans += n * (b / c);
        b %= c;
    }
    ll d = (a * (n - 1) + b) / c;
    if (d == 0) return ans;
    ans += d * (n - 1) - floor_sum(d, c, c - b - 1, a);
    return ans;
}
```

fwht
#c01d10
 $\mathcal{O}(n \log n)$, n musi być potęgą dwójki, fwht_or(a)[i] = suma(j będące podmaską i) a[j], ifwht_or(fwht_or(a)) = a, convolution_or(a, b)[i] = suma(j | k == i) a[j] * b[k], fwht_and(a)[i] = suma(j będące nadmaską i) a[j], ifwht_and(fwht_and(a)) = a, convolution_and(a, b)[i] = suma(j & k == i) a[j] * b[k], fwht_xor(a)[i] = suma(j oraz i mają parzystości wspólnie zapalonych bitów) a[j] - suma(j oraz i mają nieparzystości) a[j], ifwht_xor(fwht_xor(a)) = a, convolution_xor(a, b)[i] = suma(j k⊖ == i) a[j] * b[k].

```
// BEGIN HASH f58aba
vi fwht_or(vi a) {
    int n = ssize(a);
    assert((n & (n - 1)) == 0);
    for(int s = 1; 2 * s <= n; s *= 2)
        for(int l = 0; l < n; l += 2 * s)
            for(int i = l; i < l + s; ++i)
                a[i + s] += a[i];
    return a;
}
vi ifwht_or(vi a) {
    int n = ssize(a);
    assert((n & (n - 1)) == 0);
    for(int s = n / 2; s >= 1; s /= 2)
        for(int l = 0; l < n; l += 2 * s)
            for(int i = l; i < l + s; ++i)
                a[i + s] -= a[i];
    return a;
}
vi convolution_or(vi a, vi b) {
    int n = ssize(a);
    assert((n & (n - 1)) == 0 and ssize(b) == n);
    a = fwht_or(a);
    b = fwht_or(b);
    REP(i, n)
        a[i] *= b[i];
    return ifwht_or(a);
} // END HASH
// BEGIN HASH 4bbc88
vi fwht_and(vi a) {
    int n = ssize(a);
    assert((n & (n - 1)) == 0);
    for(int s = 1; 2 * s <= n; s *= 2)
        for(int l = 0; l < n; l += 2 * s)
            for(int i = l; i < l + s; ++i)
                a[i] += a[i + s];
    return a;
}
vi ifwht_and(vi a) {
    int n = ssize(a);
```

```
    assert((n & (n - 1)) == 0);
    for(int s = n / 2; s >= 1; s /= 2)
        for(int l = 0; l < n; l += 2 * s)
            for(int i = l; i < l + s; ++i)
                a[i] -= a[i + s];
    return a;
}
vi convolution_and(vi a, vi b) {
    int n = ssize(a);
    assert((n & (n - 1)) == 0 and ssize(b) == n);
    a = fwht_and(a);
    b = fwht_and(b);
    REP(i, n)
        a[i] *= b[i];
    return ifwht_and(a);
} // END HASH
// BEGIN HASH 6606b1
vi fwht_xor(vi a) {
    int n = ssize(a);
    assert((n & (n - 1)) == 0);
    for(int s = 1; 2 * s <= n; s *= 2)
        for(int l = 0; l < n; l += 2 * s)
            for(int i = l; i < l + s; ++i) {
                int t = a[i + s];
                a[i + s] = a[i] - t;
                a[i] += t;
            }
    return a;
}
vi ifwht_xor(vi a) {
    int n = ssize(a);
    assert((n & (n - 1)) == 0);
    for(int s = n / 2; s >= 1; s /= 2)
        for(int l = 0; l < n; l += 2 * s)
            for(int i = l; i < l + s; ++i) {
                int t = a[i + s];
                a[i + s] = (a[i] - t) / 2;
                a[i] = (a[i] + t) / 2;
            }
    return a;
}
vi convolution_xor(vi a, vi b) {
    int n = ssize(a);
    assert((n & (n - 1)) == 0 and ssize(b) == n);
    a = fwht_xor(a);
    b = fwht_xor(b);
    REP(i, n)
        a[i] *= b[i];
    return ifwht_xor(a);
} // END HASH
```

gauss
#482bf4, includes: matrix-header
 $\mathcal{O}(nm(n + m))$, Wrzucam n vectorów {wsp_x0, wsp_x1, ..., wsp_xm - 1, suma}, gauss wtedy zwraca liczbę rozwiązań (0, 1 albo 2 (tzn. nieskończoność) oraz jedno poprawne rozwiązanie (o ile istnieje). Przykład gauss({2, -1, 1, 7}, {1, 1, 1, 1}, {0, 1, -1, 6.5}) zwraca 1, {6.75, 0.375, -6.125}).

```
pair<int, V<T>> gauss(V<V<T>> &a) {
    int n = ssize(a); // liczba wierszy
    int m = ssize(a[0]) - 1; // liczba zmiennych
    vi where(m, -1); // w ktorym wierszu jest zdefiniowana i-ta zmienna
    for(int col = 0, row = 0; col < m and row < n; ++col)
    {
        int sel = row;
        for(int y = row; y < n; ++y)
            if(abs(a[y][col]) > abs(a[sel][col]))
                sel = y;
        if(equal(a[sel][col], 0))
            continue;
        for(int x = col; x <= m; ++x)
            swap(a[sel][x], a[row][x]);
        // teraz sel jest nieaktualne
        where[col] = row;
```

```
for(int y = 0; y < n; ++y)
    if(y != row) {
        T wspolczynnik = divide(a[y][col], a[row][col]);
        ;
        for(int x = col; x <= m; ++x)
            a[y][x] = sub(a[y][x], mul(wspolczynnik, a[
                row][x]));
    }
    ++row;
}
V<T> answer(m);
for(int col = 0; col < m; ++col)
    if(where[col] != -1)
        answer[col] = divide(a[where[col]][m], a[where[
            col]][col]);
for(int row = 0; row < n; ++row) {
    T got = 0;
    for(int col = 0; col < m; ++col)
        got = add(got, mul(answer[col], a[row][col]));
    if(not equal(got, a[row][m]))
        return {0, answer};
}
for(int col = 0; col < m; ++col)
    if(where[col] == -1)
        return {2, answer};
return {1, answer};
}
```

integral

```
#fad4ef
ℳ (idk), zwraca całkę f na [l, r].

using D = long double;
D simpson(function<D (D)> f, D l, D r) {
    return (f(l) + 4 * f((l + r) / 2) + f(r)) * (r - l) /
        6;
}
D integrate(function<D (D)> f, D l, D r, D s, D eps) {
    D m = (l + r) / 2;
    D sl = simpson(f, l, m), sr = simpson(f, m, r), s2 =
        sl + sr;
    if(abs(s2 - s) < 15 * eps or r - l < 1e-10)
        return s2 + (s2 - s) / 15;
    return integrate(f, l, m, sl, eps / 2)
        + integrate(f, m, r, sr, eps / 2);
}
D integrate(function<D (D)> f, D l, D r) {
    return integrate(f, l, r, simpson(f, l, r), 1e-8);
}
```

lagrange-consecutive

#04d4e8, includes: simple-modulo
 $\mathcal{O}(n)$, przyjmuje wartości wielomianu w punktach $0, 1, \dots, n - 1$ i wylicza jego wartość w x . lagrange_consecutive([2, 3, 4], 3) == 5

```
int lagrange_consecutive(vi y, int x) {
    int n = ssize(y), fac = 1, pref = 1, suff = 1, ret =
        0;
    FOR(i, 1, n) fac = mul(fac, i);
    fac = inv(fac);
    REP(i, n) {
        fac = mul(fac, n - i);
        y[i] = mul(y[i], mul(pref, fac));
        y[n - 1 - i] = mul(y[n - 1 - i], mul(suff, mul(i %
            2 ? mod - 1 : 1, fac)));
        pref = mul(pref, sub(x, i));
        suff = mul(suff, sub(x, n - 1 - i));
    }
    REP(i, n) ret = add(ret, y[i]);
    return ret;
}
```

matrix-header

#954c9d

integral lagrange-consecutive matrix-header matrix-inverse miller-rabin multiplicative ntt pell

Funkcje pomocnicze do algorytmów macierzowych.

```
#if 1
#ifdef CHANGABLE_MOD
int mod = 998'244'353;
#else
constexpr int mod = 998'244'353;
#endif
// BEGIN HASH 38b0c9
bool equal(int a, int b) {
    return a == b;
}
int mul(int a, int b) {
    return int(a * ll(b) % mod);
}
int add(int a, int b) {
    a += b;
    return a >= mod ? a - mod : a;
}
int powi(int a, int b) {
    for(int ret = 1; b /= 2) {
        if(b == 0)
            return ret;
        if(b & 1)
            ret = mul(ret, a);
        a = mul(a, a);
    }
}
int inv(int x) {
    return powi(x, mod - 2);
}
int divide(int a, int b) {
    return mul(a, inv(b));
}
int sub(int a, int b) {
    return add(a, mod - b);
}
using T = int;
// END HASH
#else
// BEGIN HASH a32baf
constexpr double eps = 1e-9;
bool equal(double a, double b) {
    return abs(a - b) < eps;
}
#define OP(name, op) double name(double a, double b) {
    return a op b; }
OP(mul, *)
OP(add, +)
OP(divide, /)
OP(sub, -)
using T = double;
// END HASH
#endif
```

matrix-inverse

#86d4aa, includes: matrix-header
 $\mathcal{O}(n^3)$, odwrotność macierzy (moduło lub double). Zwraca rząd macierzy. Dla odwracalnych macierzy (rząd = n) w a znajdzie się jej odwrotność.

```
int inverse(V<V<T>>& a) {
    int n = ssize(a);
    vi col(n);
    V h(n, V<T>(n));
    REP(l, n)
        h[l][l] = 1, col[l] = i;
    REP(i, n) {
        int r = i, c = i;
        FOR(j, i, n - 1) FOR(k, i, n - 1)
            if(abs(a[j][k]) > abs(a[r][c]))
                r = j, c = k;
        if (equal(a[r][c], 0))
            return i;
        a[i].swap(a[r]);
        h[i].swap(h[r]);
```

```
REP(j, n)
    swap(a[j][i], a[j][c]), swap(h[j][i], h[j][c]);
swap(col[i], col[c]);
T v = a[i][i];
FOR(j, i + 1, n - 1) {
    T f = divide(a[j][i], v);
    a[j][i] = 0;
    FOR(k, i + 1, n - 1)
        a[j][k] = sub(a[j][k], mul(f, a[i][k]));
    REP(k, n)
        h[j][k] = sub(h[j][k], mul(f, h[i][k]));
}
FOR(j, i + 1, n - 1)
    a[i][j] = divide(a[i][j], v);
REP(j, n)
    h[i][j] = divide(h[i][j], v);
a[i][i] = 1;
}
for(int i = n - 1; i > 0; --i) REP(j, i) {
    T v = a[j][i];
    REP(k, n)
        h[j][k] = sub(h[j][k], mul(v, h[i][k]));
}
REP(i, n)
    REP(j, n)
        a[col[i]][col[j]] = h[i][j];
return n;
}
```

miller-rabin

#ae0853
 $\mathcal{O}(\log^2 n)$ test pierwszości Millera-Rabina, działa dla long longów.

```
ll llmul(ll a, ll b, ll m) {
    return ll(__int128_t(a) * b % m);
}
ll llpowi(ll a, ll n, ll m) {
    for (ll ret = 1; n /= 2) {
        if (n == 0)
            return ret;
        if (n % 2)
            ret = llmul(ret, a, m);
        a = llmul(a, a, m);
    }
}
bool miller_rabin(ll n) {
    if(n < 2) return false;
    int r = 0;
    ll d = n - 1;
    while(d % 2 == 0)
        d /= 2, r++;
    for(int a : {2, 325, 9375, 28178, 450775, 9780504,
        1795265022}) {
        if (a % n == 0) continue;
        ll x = llpowi(a, d, n);
        if(x == 1 || x == n - 1)
            continue;
        bool composite = true;
        REP(i, r - 1) {
            x = llmul(x, x, n);
            if(x == n - 1) {
                composite = false;
                break;
            }
        }
        if(composite) return false;
    }
    return true;
}
```

multiplicative

#3070a7, includes: sieve
 $\mathcal{O}(n)$, mobius(n) oblicza funkcję Möbiusa na $[0..n]$, totient(n) oblicza funkcję Eulera na $[0..n]$, wartości w 0 niezdefiniowane.

```
// BEGIN HASH 882c54
vi mobius(int n) {
    sieve(n);
    vi ans(n + 1, 0);
    if (n) ans[1] = 1;
    FOR(i, 2, n) {
        int p = prime_div[i];
        if (i / p % p) ans[i] = -ans[i / p];
    }
    return ans;
} // END HASH
// BEGIN HASH e94976
vi totient(int n) {
    sieve(n);
    vi ans(n + 1, 1);
    FOR(i, 2, n) {
        int p = prime_div[i];
        ans[i] = ans[i / p] * (p - bool(i / p % p));
    }
    return ans;
} // END HASH
```

ntt

#0a21fe, includes: simple-modulo
 $\mathcal{O}(n \log n)$ mnożenie wielomianów mod 998244353.

```
// BEGIN HASH bcb639
using vi = vi;
constexpr int root = 3;
void ntt(vi& a, int n, bool inverse = false) {
    assert((n & (n - 1)) == 0);
    a.resize(n);
    vi b(n);
    for(int w = n / 2; w /= 2, swap(a, b)) {
        int r = powi(root, (mod - 1) / n * w), m = 1;
        for(int i = 0; i < n; i += w * 2, m = mul(m, r))
            REP(j, w) {
                int u = a[i + j], v = mul(a[i + j + w], m);
                b[i / 2 + j] = add(u, v);
                b[i / 2 + j + n / 2] = sub(u, v);
            }
    }
    if(inverse) {
        reverse(a.begin() + 1, a.end());
        int invn = inv(n);
        for(int& e : a) e = mul(e, invn);
    }
} // END HASH
vi conv(vi a, vi b) {
    if(a.empty() or b.empty()) return {};
    int l = ssize(a) + ssize(b) - 1, sz = 1 << __lg(2 * l
        - 1);
    ntt(a, sz), ntt(b, sz);
    REP(l, sz) a[l] = mul(a[l], b[l]);
    ntt(a, sz, true), a.resize(l);
    return a;
}
```

pell

#9071bf
 $\mathcal{O}(\log n)$, pell(n) oblicza rozwiązanie fundamentalne $x^2 - ny^2 = 1$, zwraca $(0, 0)$ jeżeli nie istnieje (n jest kwadratem lub wynik przekracza ll), all_pell(n, limit) wyznacza wszystkie rozwiązania $x^2 - ny^2 = 1$ z $x \leq \text{limit}$, w razie potrzeby można przepisać na pythona lub użyć bignumów.

```
pair<ll, ll> pell(ll n) {
    ll s = ll(sqrtl(n));
    if (s * s == n) return {0, 0};
    ll m = 0, d = 1, a = s;
    __int128 num1 = 1, num2 = a, den1 = 0, den2 = 1;
    while (num2 * num2 - n * den2 * den2 != 1) {
        m = d * a - m;
        d = (n - m * m) / d;
        a = (s + m) / d;
```

```
    if (num2 > (1ll << 62) / a) return {0, 0};
    tie(num1, num2) = pair(num2, a * num2 + num1);
    tie(den1, den2) = pair(den2, a * den2 + den1);
}
return {num2, den2};
}
V<pair<ll, ll>> all_pell(ll n, ll limit) {
    auto [x0, y0] = pell(n);
    if (!x0) return {};
    V<pair<ll, ll>> ret;
    __int128 x = x0, y = y0;
    while (x <= limit) {
        ret.eb(x, y);
        if (y0 * y > (1ll << 62) / n) break;
        tie(x, y) = pair(x0 * x + n * y0 * y, x0 * y + y0 * x);
    }
    return ret;
}
```

pi
#f777db
 $\mathcal{O}\left(n^{\frac{3}{4}}\right)$, liczba liczb pierwszych na przedziale $[1, n]$. Pi pi(n);
pi.query(d); // musi zachodzić $d \mid n$

```
struct Pi {
    vll w, dp;
    int id(ll v) {
        if (v <= w.back() / v)
            return int(v - 1);
        return ssize(w) - int(w.back() / v);
    }
    Pi(ll n) {
        for (ll i = 1; i * i <= n; ++i) {
            w.pb(i);
            if (n / i != i)
                w.eb(n / i);
        }
        sort(all(w));
        for (ll i : w)
            dp.eb(i - 1);
        for (ll i = 1; (i + 1) * (i + 1) <= n; ++i) {
            if (dp[i] == dp[i - 1])
                continue;
            for (int j = ssize(w) - 1; w[j] >= (i + 1) * (i + 1); --j)
                dp[j] -= dp[id(w[j] / (i + 1))] - dp[i - 1];
        }
        ll query(ll v) {
            assert(w.back() % v == 0);
            return dp[id(v)];
        }
    };
};
```

polynomial
#8a2d5d, includes: ntt
Operacje na wielomianach mod 998244353, deriv, integr $\mathcal{O}(n)$, powi_deg $\mathcal{O}(n \cdot \deg)$, sqrt, inv, log, exp, powi, div $\mathcal{O}(n \log n)$, powi_slow, eval, inter $\mathcal{O}(n \log^2 n)$ Ogólnie to przepisujemy co chcemy. Funkcje oznaczone jako KONIECZNE są wymagane. Funkcje oznaczone WYMAGA ABC wymagają wcześniejszego przepisania ABC. deriv(a) zwraca a' , integr(a) zwraca $\int a$, powi(_deg_slow)(a, k, n) zwraca $a^k \pmod{x^n}$, sqrt(a, n) zwraca $a^{\frac{1}{2}} \pmod{x^n}$, inv(a, n) zwraca $a^{-1} \pmod{x^n}$, log(a, n) zwraca $\ln(a) \pmod{x^n}$, exp(a, n) zwraca $\exp(a) \pmod{x^n}$, div(a, b) zwraca (q, r) takie, że $a = qb + r$, eval(a, x) zwraca y taki, że $a(x_i) = y_i$, inter(x, y) zwraca a taki, że $a(x_i) = y_i$.
// BEGIN HASH f824a3
vi mod_xn(C vi& a, int n) { // KONIECZNE
 return vi(a.begin(), a.begin() + min(n, ssize(a)));
}
void sub(vi& a, C vi& b) { // KONIECZNE
 a.resize(max(ssize(a), ssize(b)));

```
    REP(i, ssize(b)) a[i] = sub(a[i], b[i]);
} // END HASH
// BEGIN HASH 2c8fbb
vi deriv(vi a) {
    REP(i, ssize(a)) a[i] = mul(a[i], i);
    if(ssize(a)) a.erase(a.begin());
    return a;
}
vi integr(vi a) {
    int n = ssize(a);
    a.insert(a.begin(), 0);
    static vi f{1};
    FOR(i, ssize(f), n) f.eb(mul(f[i - 1], i));
    int r = inv(f[n]);
    for(int i = n; i > 0; --i)
        a[i] = mul(a[i], mul(r, f[i - 1])), r = mul(r, i);
    return a;
} // END HASH
// BEGIN HASH d6d6d4
vi powi_deg(C vi& a, int k, int n) {
    assert(ssize(a) and a[0] != 0);
    vi v(n), f(n, 1);
    v[0] = powi(a[0], k);
    REP(i, n - 1) f[i + 1] = mul(f[i], n - i);
    int r = inv(mul(f[n - 1], a[0]));
    FOR(i, 1, n - 1) {
        FOR(j, 1, min(ssize(a) - 1, i)) {
            v[i] = add(v[i], mul(a[j], mul(v[i - j], sub(mul(k, j), i - j))));
        }
        v[i] = mul(v[i], mul(r, f[n - i]));
        r = mul(r, i);
    }
    return v;
} // END HASH
// BEGIN HASH 57a01a
vi powi_slow(C vi& a, int k, int n) {
    vi v{1}, b = mod_xn(a, n);
    int x = 1; while(x < n) x *= 2;
    while(k) {
        ntt(b, 2 * x);
        if(k & 1) {
            ntt(v, 2 * x);
            REP(i, 2 * x) v[i] = mul(v[i], b[i]);
            ntt(v, 2 * x, true);
            v.resize(x);
        }
        REP(i, 2 * x) b[i] = mul(b[i], b[i]);
        ntt(b, 2 * x, true);
        b.resize(x);
        k /= 2;
    }
    return mod_xn(v, n);
} // END HASH
// BEGIN HASH 504d4e
vi sqrt(C vi& a, int n) {
    auto at = [&](int i) { if(i < ssize(a)) return a[i];
        else return 0; };
    assert(ssize(a) and a[0] == 1);
    C int inv2 = inv(2);
    vi v{1}, f{1}, g{1};
    for(int x = 1; x < n; x *= 2) {
        vi z = v;
        ntt(z, x);
        vi b = g;
        REP(i, x) b[i] = mul(b[i], z[i]);
        ntt(b, x, true);
        REP(i, x / 2) b[i] = 0;
        ntt(b, x);
        REP(i, x) b[i] = mul(b[i], g[i]);
        ntt(b, x, true);
        REP(i, x / 2) f.eb(sub(0, b[i + x / 2]));
        REP(i, x) z[i] = mul(z[i], z[i]);
        ntt(z, x, true);
        vi c(2 * x);
```

```
    REP(i, x) c[i + x] = sub(add(at(i), at(i + x)), z[i
    ]);
    ntt(c, 2 * x);
    g = f;
    ntt(g, 2 * x);
    REP(i, 2 * x) c[i] = mul(c[i], g[i]);
    ntt(c, 2 * x, true);
    REP(i, x) v.eb(mul(c[i + x], inv2));
}
return mod_xn(v, n);
} // END HASH
// BEGIN HASH 02cc82
vi inv(C vi& a, int n) {
    assert(ssize(a) and a[0] != 0);
    vi v{inv(a[0])};
    for(int x = 1; x < n; x *= 2) {
        vi f = mod_xn(a, 2 * x), g = v;
        ntt(g, 2 * x);
        REP(k, 2) {
            ntt(f, 2 * x);
            REP(i, 2 * x) f[i] = mul(f[i], g[i]);
            ntt(f, 2 * x, true);
            REP(i, x) f[i] = 0;
        }
        sub(v, f);
    }
    return mod_xn(v, n);
} // END HASH
// BEGIN HASH 6635b5
vi log(C vi& a, int n) { // WYMAGA deriv, integr, inv
    assert(ssize(a) and a[0] == 1);
    return integr(mod_xn(conv(deriv(mod_xn(a, n)), inv(a, n)), n - 1));
} // END HASH
// BEGIN HASH 7b9b7f
vi exp(C vi& a, int n) { // WYMAGA deriv, integr
    assert(a.empty() or a[0] == 0);
    vi v{1}, f{1}, g, h{0}, s;
    for(int x = 1; x < n; x *= 2) {
        g = v;
        REP(k, 2) {
            ntt(g, (2 - k) * x);
            if(!k) s = g;
            REP(i, x) g[i] = mul(g[(2 - k) * i], h[i]);
            ntt(g, x, true);
            REP(i, x / 2) g[i] = 0;
        }
        sub(f, g);
        vi b = deriv(mod_xn(a, x));
        ntt(b, x);
        REP(i, x) b[i] = mul(s[2 * i], b[i]);
        ntt(b, x, true);
        vi c = deriv(v);
        sub(c, b);
        rotate(all(c) - 1, c.end());
        ntt(c, 2 * x);
        h = f;
        ntt(h, 2 * x);
        REP(i, 2 * x) c[i] = mul(c[i], h[i]);
        ntt(c, 2 * x, true);
        c.resize(x);
        vi t(x - 1);
        c.insert(c.begin(), t.begin(), t.end());
        vi d = mod_xn(a, 2 * x);
        sub(d, integr(c));
        d.erase(d.begin(), d.begin() + x);
        ntt(d, 2 * x);
        REP(i, 2 * x) d[i] = mul[d[i], s[i]);
        ntt(d, 2 * x, true);
        REP(i, x) v.eb(d[i]);
    }
    return mod_xn(v, n);
} // END HASH
// BEGIN HASH 802699
vi powi(C vi& a, int k, int n) { // WYMAGA log, exp
    vi v = mod_xn(a, n);
```

```
    int cnt = 0;
    while(cnt < ssize(v) and !v[cnt])
        ++cnt;
    if(ll(cnt) * k >= n)
        return {};
    v.erase(v.begin(), v.begin() + cnt);
    if(v.empty())
        return k ? vi{} : vi{1};
    int pow0 = powi(v[0], k);
    int inv0 = inv(v[0]);
    for(int& e : v) e = mul(e, inv0);
    v = log(v, n - cnt * k);
    for(int& e : v) e = mul(e, k);
    v = exp(v, n - cnt * k);
    for(int& e : v) e = mul(e, powi0);
    vi t(cnt * k, 0);
    v.insert(v.begin(), t.begin(), t.end());
    return v;
} // END HASH
// BEGIN HASH 748a86
pair<vi, vi> div_slow(vi a, C vi& b) {
    vi x;
    while(ssize(a) >= ssize(b)) {
        x.eb(mul(a.back(), inv(b.back())));
        if(x.back() != 0)
            REP(i, ssize(b))
                a.end()[-i - 1] = sub(a.end()[-i - 1], mul(x.back(), b.end()[-i - 1]));
        a.pop_back();
    }
    reverse(all(x));
    return {x, a};
}
pair<vi, vi> div(vi a, C vi& b) { // WYMAGA inv, div_slow
    C int d = ssize(a) - ssize(b) + 1;
    if (d <= 0)
        return {{}, a};
    if (min(d, ssize(b)) < 250)
        return div_slow(a, b);
    vi x = mod_xn(conv(mod_xn({a.rbegin(), a.rend()}, d), inv({b.rbegin(), b.rend()}, d)), d);
    reverse(all(x));
    sub(a, conv(x, b));
    return {x, mod_xn(a, ssize(b))};
} // END HASH
// BEGIN HASH 6a6b92
vi build(V<vi> &tree, int v, auto l, auto r) {
    if (r - l == 1) {
        return tree[v] = vi{sub(0, *l), 1};
    } else {
        auto M = l + (r - l) / 2;
        return tree[v] = conv(build(tree, 2 * v, l, M), build(tree, 2 * v + 1, M, r));
    }
} // END HASH
// BEGIN HASH c3c4fc
int eval_single(C vi& a, int x) {
    int y = 0;
    RFOR(i, ssize(a)-1, 0) {
        y = mul(y, x);
        y = add(y, a[i]);
    }
    return y;
}
vi eval_helper(C vi& a, V<vi>& tree, int v, auto l, auto r) {
    if (r - l == 1) {
        return {eval_single(a, *l)};
    } else {
        auto m = l + (r - l) / 2;
        vi A = eval_helper(div(a, tree[2 * v])).se, tree, 2 * v, l, m);
        vi B = eval_helper(div(a, tree[2 * v + 1])).se, tree, 2 * v + 1, m, r);
        A.insert(A.end(), B.begin(), B.end());
```

```
        return A;
    }
}
vi eval(C vi& a, C vi& x) { // WYMAGA div, eval_single,
    build, eval_helper
    if (x.empty())
        return {};
    V<vi> tree(4 * ssize(x));
    build(tree, 1, begin(x), end(x));
    return eval_helper(a, tree, 1, begin(x), end(x));
} // END HASH
// BEGIN HASH 87c63d
vi inter_helper(C vi& a, V<vi>& tree, int v, auto l,
    auto r, auto ly, auto ry) {
    if (r - l == 1) {
        return {mul(*ly, inv(a[0]))};
    }
    else {
        auto m = l + (r - l) / 2;
        auto my = ly + (ry - ly) / 2;
        vi A = inter_helper(div(a, tree[2 * v]).se, tree, 2
            * v, l, m, ly, my);
        vi B = inter_helper(div(a, tree[2 * v + 1]).se,
            tree, 2 * v + 1, m, r, my, ry);
        vi L = conv(A, tree[2 * v + 1]);
        vi R = conv(B, tree[2 * v]);
        REP(i, ssize(R))
            L[i] = add(L[i], R[i]);
        return L;
    }
}
vi inter(C vi& x, C vi& y) { // WYMAGA deriv, div,
    build, inter_helper
    assert(ssize(x) == ssize(y));
    if (x.empty())
        return {};
    V<vi> tree(4 * ssize(x));
    return inter_helper(deriv(build(tree, 1, begin(x),
        end(x))), tree, 1, begin(x), end(x), begin(y), end
        (y));
} // END HASH
```

power-sum

#32d0ba, includes: lagrange-consecutive
power_monomial_sum $\mathcal{O}(k \log k)$, power_binomial_sum $\mathcal{O}(k)$.
power_monomial_sum(a, k, n) liczy $\sum_{i=0}^{n-1} a^i \cdot i^k$,
power_binomial_sum(a, k, n) liczy $\sum_{i=0}^{n-1} a^i \cdot \binom{i}{k}$. Działa dla $0 \leq n$ oraz $a \neq 1$.

```
// BEGIN HASH 74870f
int power_monomial_sum(int a, int k, int n) {
    if (n == 0) return 0;
    int p = 1, b = 1, c = 0, d = a, inva = inv(a);
    vi v(k + 1, k == 0);
    FOR(i, 1, k) v[i] = add(v[i - 1], mul(p = mul(p, a),
        powi(i, k)));
    BinomCoeff bc(k + 1);
    REP(i, k + 1) {
        c = add(c, mul(bc(k + 1, i), mul(v[k - i], b)));
        b = mul(b, sub(0, a));
    }
    c = mul(c, inv(powi(sub(1, a), k + 1)));
    REP(i, k + 1) v[i] = mul(sub(v[i], c), d = mul(d,
        inva));
    return add(c, mul(lagrange_consecutive(v, n - 1),
        powi(a, n - 1)));
} // END HASH
// BEGIN HASH 7f9702
int power_binomial_sum(int a, int k, int n) {
    int p = powi(a, n), inva1 = inv(sub(a, 1)), binom =
        1, ans = 0;
    BinomCoeff bc(k + 1);
    REP(i, k + 1) {
        ans = sub(mul(p, binom), mul(ans, a));
        if(!i) ans = sub(ans, 1);
        ans = mul(ans, inva1);
    }
```

power-sum primitive-root pythagorean-triples rho-pollard same-div sieve simple-modulo simplex

```
    binom = mul(binom, mul(n - i, mul(bc.rev[i + 1], bc
        .fac[i])));
    }
    return ans;
} // END HASH
```

primitive-root

#9f409a, includes: simple-modulo, rho-pollard
 $\mathcal{O}(\log^2(mod))$, dla pierwszego mod znajduje generator modulo mod
(z być może sporą stałą).

```
int primitive_root() {
    if(mod == 2)
        return 1;
    int q = mod - 1;
    vll v = factor(q);
    vi fact;
    REP(i, ssize(v))
        if(!i or v[i] != v[i - 1])
            fact.eb(v[i]);
    while(true) {
        int g = rd(2, q);
        auto is_good = [&] {
            for(auto &f : fact)
                if(powi(g, q / f) == 1)
                    return false;
            return true;
        };
        if(is_good())
            return g;
    }
}
```

pythagorean-triples

#a0b5a2
Wyznacza wszystkie trójki (a, b, c) takie, że $a^2 + b^2 = c^2$,
 $gcd(a, b, c) = 1$ oraz $c \leq$ limit. Zwraca tylko jedną $z(a, b, c)$ oraz $z(a, b, c)$.

```
V<tuple<int, int, int>> pythagorean_triples(int limit)
{
    V<tuple<int, int, int>> ret;
    function<void(int, int, int)> gen = [&](int a, int b,
        int c) {
        if (c > limit)
            return;
        ret.eb(a, b, c);
        REP(i, 3) {
            gen(a + 2 * b + 2 * c, 2 * a + b + 2 * c, 2 * a +
                2 * b + 3 * c);
            a = -a;
            if (i) b = -b;
        }
    };
    gen(3, 4, 5);
    return ret;
}
```

rho-pollard

#db8f43, includes: miller-rab
 $\mathcal{O}(n^{\frac{1}{4}})$, factor(n) zwraca V dzielników pierwszych n , niekoniecznie
posortowany, get_pairs(n) zwraca posortowany V par (dzielnik
pierwszych, krotność) dla liczby n, all_factors(n) zwraca V wszystkich
dzielników n , niekoniecznie posortowany, factor(12) = {2, 2, 3},
factor(545423) = {53, 41, 251};, get_pairs(12) = {(2, 2), (3, 1)},
all_factors(12) = {1, 3, 2, 6, 4, 12}.

```
// BEGIN HASH 6d1d12
ll rho_pollard(ll n) {
    if(n % 2 == 0) return 2;
    for(ll i = 1; i++) {
        auto f = [&](ll x) { return (llmul(x, x, n) + i) %
            n; };
        ll x = 2, y = f(x), p;
        while((p = __gcd(n - x + y, n)) == 1)
```

```
        x = f(x), y = f(f(y));
        if(p != n) return p;
    }
}
vll factor(ll n) {
    if(n == 1) return {};
    if(miller_rabin(n)) return {n};
    ll x = rho_pollard(n);
    auto l = factor(x), r = factor(n / x);
    l.insert(l.end(), r.begin(), r.end());
    return l;
} // END HASH
V<pair<ll, int>> get_pairs(ll n) {
    auto v = factor(n);
    sort(all(v));
    V<pair<ll, int>> ret;
    REP(i, ssize(v)) {
        int x = i + 1;
        while (x < ssize(v) and v[x] == v[i])
            ++x;
        ret.eb(v[i], x - i);
        i = x - 1;
    }
    return ret;
}
vll all_factors(ll n) {
    auto v = get_pairs(n);
    vll ret;
    function<void(ll, int)> gen = [&](ll val, int p) {
        if (p == ssize(v)) {
            ret.eb(val);
            return;
        }
        auto [x, cnt] = v[p];
        gen(val, p + 1);
        REP(i, cnt) {
            val *= x;
            gen(val, p + 1);
        }
    };
    gen(1, 0);
    return ret;
}
```

same-div

#94bc3b
 $\mathcal{O}(\sqrt{n})$, wyznacza przedziały o takiej samej wartości $\lfloor n/x \rfloor$ lub $\lceil n/x \rceil$.
same_floor(8) = {(1, 1), (2, 2), (3, 4), (5, 8)}, same_ceil(8) =
{(8, 8), (4, 7), (3, 3), (2, 2), (1, 1)}, na koncieście raczej
checemy przepisać tylko pętlę i od razu wykonywać obliczenia na parze (l,
r) zamiast grupować wszystkie przedziały w vectorze. Dla n będącego
intem można zmienić wszystkie ll na int, w celu zbicia stałej.

```
// BEGIN HASH 0022a0
V<pair<ll, ll>> same_floor(ll n) {
    V<pair<ll, ll>> v;
    for (ll l = 1, r; l <= n; l = r + 1) {
        r = n / (n / l);
        v.eb(l, r);
    }
    return v;
} // END HASH
// BEGIN HASH 766533
V<pair<ll, ll>> same_ceil(ll n) {
    V<pair<ll, ll>> v;
    for (ll r = n, l; r >= 1; r = l - 1) {
        l = (n + r - 1) / r;
        l = (n + l - 1) / l;
        v.eb(l, r);
    }
    return v;
} // END HASH
```

sieve

#a101b7

$\mathcal{O}(n)$, sieve(n) przetwarza liczby do n włącznie, comp[i] oznacza czy i
jest złożone, primes zawiera wszystkie liczby pierwsze $\leq n$, prime_div[i]
zawiera najmniejszy dzielnik pierwszy i , na CF dla $n = 1e8$ działa w
1.2s.

```
V<bool> comp;
vi primes, prime_div;
void sieve(int n) {
    primes.clear();
    comp.resize(n + 1);
    prime_div.resize(n + 1);
    FOR(i, 2, n) {
        if (!comp[i]) primes.eb(i), prime_div[i] = i;
        for (int p : primes) {
            int x = i * p;
            if (x > n) break;
            comp[x] = true;
            prime_div[x] = p;
            if (i % p == 0) break;
        }
    }
}
```

simple-modulo

#03c593
podstawowe operacje na modulo, pamiętać o constexpr.

```
// BEGIN HASH 6b9273
#ifdef CHANGABLE_MOD
int mod = 998'244'353;
#else
constexpr int mod = 998'244'353;
#endif
int add(int a, int b) {
    a += b;
    return a >= mod ? a - mod : a;
}
int sub(int a, int b) {
    return add(a, mod - b);
}
int mul(int a, int b) {
    return int(a * ll(b) % mod);
}
int powi(int a, int b) {
    for(int ret = 1;; b /= 2) {
        if(b == 0)
            return ret;
        if(b & 1)
            ret = mul(ret, a);
        a = mul(a, a);
    }
}
int inv(int x) {
    return powi(x, mod - 2);
} // END HASH
struct BinomCoeff {
    vi fac, rev;
    BinomCoeff(int n) {
        fac = rev = V(n + 1, 1);
        FOR(i, 1, n) fac[i] = mul(fac[i - 1], i);
        rev[n] = inv(fac[n]);
        for(int i = n; i > 0; --i)
            rev[i - 1] = mul(rev[i], i);
    }
    int operator()(int n, int k) {
        return mul(fac[n], mul(rev[n - k], rev[k]));
    }
};
```

simplex

#37993a
 $\mathcal{O}(szybko)$, Simplex(n, m) tworzy lp solver z n zmiennymi oraz m
ograniczeniami, rozwiązuje max $c x$ przy $A x \leq b$.

```
#define FIND(n, expr) [&] { REP(i, n) if(expr) return i
; return -1; }()
```

```
struct Simplex {
    using T = double;
    C T eps = 1e-9, inf = 1/.0;
    int n, m;
    vi N, B;
    V<V<T>> A;
    V<T> b, c;
    T res = 0;
    Simplex(int vars, int eqs)
        : n(vars), m(eqs), N(n), B(m), A(m, V<T>(n)), b(m),
          c(n) {
        REP(i, n) N[i] = i;
        REP(i, m) B[i] = n + i;
    }
    void pivot(int eq, int var) {
        T coef = 1 / A[eq][var], k;
        REP(i, n)
            if(abs(A[eq][i]) > eps) A[eq][i] *= coef;
        A[eq][var] *= coef, b[eq] *= coef;
        REP(r, m) if(r != eq && abs(A[r][var]) > eps) {
            k = -A[r][var], A[r][var] = 0;
            REP(i, n) A[r][i] += k * A[eq][i];
            b[r] += k * b[eq];
        }
        k = c[var], c[var] = 0;
        REP(i, n) c[i] -= k * A[eq][i];
        res += k * b[eq];
        swap(B[eq], N[var]);
    }
    bool solve() {
        int eq, var;
        while(true) {
            if((eq = FIND(m, b[i] < -eps)) == -1) break;
            if((var = FIND(n, A[eq][i] < -eps)) == -1) {
                res = -inf; // no solution
                return false;
            }
            pivot(eq, var);
        }
        while(true) {
            if((var = FIND(n, c[i] > eps)) == -1) break;
            eq = -1;
            REP(i, m) if(A[i][var] > eps
                && (eq == -1 || b[i] / A[i][var] < b[eq] / A[eq][var]))
                eq = i;
            if(eq == -1) {
                res = inf; // unbound
                return false;
            }
            pivot(eq, var);
        }
        return true;
    }
    V<T> get_vars() {
        V<T> vars(n);
        REP(i, m)
            if(B[i] < n) vars[B[i]] = b[i];
        return vars;
    }
};
```

tonelli-shanks

#4e1b15
 $\mathcal{O}(\log^2(p))$), dla pierwszego p oraz $0 \leq a \leq p-1$ znajduje takie x , że $x^2 \equiv a \pmod{p}$ lub -1 jeżeli takie x nie istnieje, można przepisać by działało dla ll

```
int mul(int a, int b, int p) {
    return int(a * ll(b) % p);
}
int powi(int a, int b, int p) {
    for (int ret = 1; b /= 2) {
        if (!b) return ret;
        if (b & 1) ret = mul(ret, a, p);
        a = mul(a, a, p);
    }
```

tonelli-shanks xor-base associative-queue fenwick-tree2d fenwick-tree find-union hash-map lazy-segment-tree

```
    }
}
int tonelli_shanks(int a, int p) {
    if (a == 0) return 0;
    if (p == 2) return 1;
    if (powi(a, p / 2, p) != 1) return -1;
    int q = p - 1, s = 0, z = 2;
    while (q % 2 == 0) q /= 2, ++s;
    while (powi(z, p / 2, p) == 1) ++z;
    int c = powi(z, q, p), t = powi(a, q, p);
    int r = powi(a, q / 2 + 1, p);
    while (t != 1) {
        int i = 0, x = t;
        while (x != 1) x = mul(x, x, p), ++i;
        c = powi(c, 1 << (s - i - 1), p); // 1ll dla ll
        r = mul(r, c, p), c = mul(c, c, p);
        t = mul(t, c, p), s = i;
    }
    return r;
}
```

xor-base

#788707
 $\mathcal{O}(nB + B^2)$ dla $B = \text{bits}$, dla S wyznacza minimalny zbiór B taki, że każdy element S można zapisać jako xor jakiegoś podzbioru B .

```
int highest_bit(int ai) {
    return ai == 0 ? 0 : __lg(ai) + 1;
}
constexpr int bits = 30;
vi xor_base(vi elems) {
    V<vi> at_bit(bits + 1);
    for(int ai : elems)
        at_bit[highest_bit(ai)].eb(ai);
    for(int b = bits; b >= 1; --b)
        while(ssize(at_bit[b]) > 1) {
            int ai = at_bit[b].back();
            at_bit[b].pop_back();
            ai ^= at_bit[b].back();
            at_bit[highest_bit(ai)].eb(ai);
        }
    at_bit.erase(at_bit.begin());
    REP(b0, bits - 1)
        for(int a0 : at_bit[b0])
            FOR(b1, b0 + 1, bits - 1)
                for(int &a1 : at_bit[b1])
                    if((a1 >> b0) & 1)
                        a1 ^= a0;
    vi ret;
    for(auto &v : at_bit) {
        assert(ssize(v) <= 1);
        for(int ai : v)
            ret.eb(ai);
    }
    return ret;
}
```

Struktury danych (4)

associative-queue

#dd244e
Kolejka wspierająca dowolną operację łączną, $\mathcal{O}(1)$ zamortyzowany. Konstruktor przyjmuje dwuargumentową funkcję oraz jej element neutralny. Dla minów jest AssocQueue<int> q[[]](int a, int b){ return min(a, b); }, numeric_limits<int>::max());

```
template<typename T>
struct AssocQueue {
    using fn = function<T(T, T)>;
    fn f;
    V<pair<T, T>> s1, s2; // {x, f(pref)}
    AssocQueue(fn _f, T e = T()) : f(_f), s1({{e, e}}),
        s2({{e, e}}) {}
}
```

```
void mv() {
    if (ssize(s2) == 1)
        while (ssize(s1) > 1) {
            s2.eb(s1.back().fi, f(s1.back().fi, s2.back().se));
            s1.pop_back();
        }
    void emplace(T x) {
        s1.eb(x, f(s1.back().se, x));
    }
    void pop() {
        mv();
        s2.pop_back();
    }
    T calc() {
        return f(s2.back().se, s1.back().se);
    }
    T front() {
        mv();
        return s2.back().fi;
    }
    int size() {
        return ssize(s1) + ssize(s2) - 2;
    }
    void clear() {
        s1.resize(1);
        s2.resize(1);
    }
};
```

fenwick-tree-2d

#fefc31, includes: fenwick-tree
 $\mathcal{O}(\log^2 n)$, pamięć $\mathcal{O}(n \log n)$, 2D offline, wywołujemy preprocess(x, y) na pozycjach, które chcemy updateować, później init(). update(x, y, val) dodaje val do [x, y], query(x, y) zwraca sumę na prostokącie $(0, 0) - (x, y)$.

```
struct Fenwick2d {
    V<vi> ys;
    V<Fenwick> ft;
    Fenwick2d(int limx) : ys(limx) {}
    void preprocess(int x, int y) {
        for(; x < ssize(ys); x |= x + 1)
            ys[x].pb(y);
    }
    void init() {
        for(auto &v : ys) {
            sort(all(v));
            ft.eb(ssize(v));
        }
    }
    int ind(int x, int y) {
        auto it = lower_bound(all(ys[x]), y);
        return int(distance(ys[x].begin(), it));
    }
    void update(int x, int y, ll val) {
        for(; x < ssize(ys); x |= x + 1)
            ft[x].update(ind(x, y), val);
    }
    ll query(int x, int y) {
        ll sum = 0;
        for(x++; x > 0; x &= x - 1)
            sum += ft[x - 1].query(ind(x - 1, y + 1) - 1);
        return sum;
    }
};
```

fenwick-tree

#7cfd2b
 $\mathcal{O}(\log n)$, indeksowane od 0, update(pos, val) dodaje val do elementu pos, query(pos) zwraca sumę $[0, pos]$.

```
struct Fenwick {
    vll s;
    Fenwick(int n) : s(n) {}
}
```

```
void update(int pos, ll val) {
    for(; pos < ssize(s); pos |= pos + 1)
        s[pos] += val;
}
ll query(int pos) {
    ll ret = 0;
    for(pos++; pos > 0; pos &= pos - 1)
        ret += s[pos - 1];
    return ret;
}
ll query(int l, int r) {
    return query(r) - query(l - 1);
}
};
```

find-union

#22834c
 $\mathcal{O}(\alpha(n))$, mniejszy do większego.

```
struct FindUnion {
    vi rep;
    int size(int x) { return -rep[find(x)]; }
    int find(int x) {
        return rep[x] < 0 ? x : rep[x] = find(rep[x]);
    }
    bool same_set(int a, int b) { return find(a) == find(b); }
    bool join(int a, int b) {
        a = find(a), b = find(b);
        if(a == b)
            return false;
        if(-rep[a] < -rep[b])
            swap(a, b);
        rep[a] += rep[b];
        rep[b] = a;
        return true;
    }
    FindUnion(int n) : rep(n, -1) {}
};
```

hash-map

#a87164, includes: <ext/pb_ds/assoc_container.hpp>
 $\mathcal{O}(1)$, trzeba przed includem dać undef_GLIBCXX_DEBUG.

```
using namespace __gnu_pbds;
struct chash {
    C uint64_t c = ll(2e18 * acos(-1)) + 69;
    C int RANDOM = mt19937(0)();
    size_t operator()(uint64_t x) C {
        return __builtin_bswap64((x^RANDOM) * c);
    }
};
template<class L, class R>
using hash_map = gp_hash_table<L, R, chash>;
```

lazy-segment-tree

#0ec085
Drzewo przedział-przedział, w miarę abstrakcyjne. Wystarczy zmienić Node i funkcje na nim.

```
// BEGIN HASH 97486f
struct Node {
    ll sum = 0, lazy = 0;
    int sz = 1;
};
void push_to_sons(Node &n, Node &l, Node &r) {
    auto push_to_son = [&](Node &c) {
        c.sum += n.lazy * c.sz;
        c.lazy += n.lazy;
    };
    push_to_son(l);
    push_to_son(r);
    n.lazy = 0;
}
Node merge(Node l, Node r) {
```



```
return Node{
    .sum = l.sum + r.sum,
    .lazy = 0,
    .sz = l.sz + r.sz
};
}
void add_to_base(Node &n, int val) {
    n.sum += n.sz * ll(val);
    n.lazy += val;
} // END HASH
// BEGIN HASH f78ac3
struct Tree {
    V<Node> tree;
    int sz = 1;
    Tree(int n) {
        while(sz < n)
            sz *= 2;
        tree.resize(sz * 2);
        for(int v = sz - 1; v >= 1; v--)
            tree[v] = merge(tree[2 * v], tree[2 * v + 1]);
    }
    void push(int v) {
        push_to_sons(tree[v], tree[2 * v], tree[2 * v + 1]);
    }
    Node get(int l, int r, int v = 1) {
        if(l == 0 and r == tree[v].sz - 1)
            return tree[v];
        push(v);
        int m = tree[v].sz / 2;
        if(r < m)
            return get(l, r, 2 * v);
        else if(m <= l)
            return get(l - m, r - m, 2 * v + 1);
        else
            return merge(get(l, m - 1, 2 * v), get(0, r - m,
                2 * v + 1));
    }
    void update(int l, int r, int val, int v = 1) {
        if(l == 0 && r == tree[v].sz - 1) {
            add_to_base(tree[v], val);
            return;
        }
        push(v);
        int m = tree[v].sz / 2;
        if(r < m)
            update(l, r, val, 2 * v);
        else if(m <= l)
            update(l - m, r - m, val, 2 * v + 1);
        else {
            update(l, m - 1, val, 2 * v);
            update(0, r - m, val, 2 * v + 1);
        }
        tree[v] = merge(tree[2 * v], tree[2 * v + 1]);
    }
}; // END HASH
```

lichao-tree

#dec5d3
Dla funkcji, których pary przecinają się co najwyżej raz, oblicza minimum w punkcie x . Podany kod jest dla funkcji liniowych.

```
struct Function {
    int a;
    ll b;
    ll operator()(int x) {
        return x * ll(a) + b;
    }
    Function(int p = 0, ll q = infll) : a(p), b(q) {}
};
ostream& operator<<(ostream &os, Function f) {
    return os << pair(f.a, f.b);
}
struct LiChaoTree {
    int size = 1;
    V<Function> tree;
```

lichao-tree line-container link-cut

```
LiChaoTree(int n) {
    while(size < n)
        size *= 2;
    tree.resize(size << 1);
}
ll get_min(int x) {
    int v = x + size;
    ll ans = infll;
    while(v) {
        chmin(ans, tree[v](x));
        v >>= 1;
    }
    return ans;
}
void add_func(Function new_func, int v, int l, int r) {
    {
        int m = (l + r) / 2;
        bool domin_l = tree[v](l) > new_func(l),
            domin_m = tree[v](m) > new_func(m);
        if(domin_m)
            swap(tree[v], new_func);
        if(l == r)
            return;
        else if(domin_l == domin_m)
            add_func(new_func, v << 1 | 1, m + 1, r);
        else
            add_func(new_func, v << 1, l, m);
    }
    void add_func(Function new_func) {
        add_func(new_func, 1, 0, size - 1);
    }
};
```

line-container

#5316a7
 $\mathcal{O}(\log n)$ set dla funkcji liniowych, add(a, b) dodaje funkcję $y = ax + b$ query(x) zwraca największe y w punkcie x .

```
struct Line {
    mutable ll a, b, p;
    ll eval(ll x) C { return a * x + b; }
    bool operator<(C line &o) C { return a < o.a; }
    bool operator<(ll x) C { return p < x; }
};
struct LineContainer : multiset<Line, less<>> {
    // jak double to inf = 1 / .0, div(a, b) = a / b
    C ll inf = LLONG_MAX;
    ll div(ll a, ll b) { return a / b - ((a ^ b) < 0 && a
        % b); }
    bool intersect(iterator x, iterator y) {
        if(y == end()) { x->p = inf; return false; }
        if(x->a == y->a) x->p = x->b > y->b ? inf : -inf;
        else x->p = div(y->b - x->b, x->a - y->a);
        return x->p >= y->p;
    }
    void add(ll a, ll b) {
        auto z = insert({a, b, 0}); y = z++, x = y;
        while(intersect(y, z)) z = erase(z);
        if(x != begin() && intersect(--x, y))
            intersect(x, erase(y));
        while((y = x) != begin() && (--x)->p >= y->p)
            intersect(x, erase(y));
    }
    ll query(ll x) {
        assert(!empty());
        return lower_bound(x)->eval(x);
    }
};
```

link-cut

#d3d041

$\mathcal{O}(q \log n)$ Link-Cut Tree z wyznaczaniem odległości między wierzchołkami, lca w zakorzenionym drzewie, dodawaniem na ścieżce, dodawaniem na poddrzewie, zwracaniem sumy na ścieżce, zwracaniem sumy na poddrzewie. Przepisać co się chce (logika lazy jest tylko w AdditionalInfo, można np. zostawić puste funkcje). Wywołać konstruktor, potem set_value na wierzchołkach (aby się ustawiło, że nie-nil to nie-nil) i potem jazda.

```
struct AdditionalInfo {
    using T = ll;
    static constexpr T neutral = 0; // Remember that
    there is a nil vertex!
    T node_value = neutral, splay_value = neutral;//,
    splay_value_reversed = neutral;
    T whole_subtree_value = neutral, virtual_value =
    neutral;
    T splay_lazy = neutral; // lazy propagation on paths
    T splay_size = 0; // 0 because of nil
    T whole_subtree_lazy = neutral, whole_subtree_cancel
    = neutral; // lazy propagation on subtrees
    T whole_subtree_size = 0, virtual_size = 0; // 0
    because of nil
    void set_value(T x) {
        node_value = splay_value = whole_subtree_value = x;
        splay_size = 1;
        whole_subtree_size = 1;
    }
    void update_from_sons(AdditionalInfo &l,
        AdditionalInfo &r) {
        splay_value = l.splay_value + node_value + r.
            splay_value;
        splay_size = l.splay_size + 1 + r.splay_size;
        whole_subtree_value = l.whole_subtree_value +
            node_value + virtual_value + r.
                whole_subtree_value;
        whole_subtree_size = l.whole_subtree_size + 1 +
            virtual_size + r.whole_subtree_size;
    }
    void change_virtual(AdditionalInfo &virtual_son, int
        delta) {
        assert(delta == -1 or delta == 1);
        virtual_value += delta * virtual_son.
            whole_subtree_value;
        whole_subtree_value += delta * virtual_son.
            whole_subtree_value;
        virtual_size += delta * virtual_son.
            whole_subtree_size;
        whole_subtree_size += delta * virtual_son.
            whole_subtree_size;
    }
    void push_lazy(AdditionalInfo &l, AdditionalInfo &r,
        bool) {
        l.add_lazy_in_path(splay_lazy);
        r.add_lazy_in_path(splay_lazy);
        splay_lazy = 0;
    }
    void cancel_subtree_lazy_from_parent(AdditionalInfo &
        parent) {
        whole_subtree_cancel = parent.whole_subtree_lazy;
    }
    void pull_lazy_from_parent(AdditionalInfo &parent) {
        if(splay_size == 0) // nil
            return;
        add_lazy_in_subtree(parent.whole_subtree_lazy -
            whole_subtree_cancel);
        cancel_subtree_lazy_from_parent(parent);
    }
    T get_path_sum() {
        return splay_value;
    }
    T get_subtree_sum() {
        return whole_subtree_value;
    }
    void add_lazy_in_path(T x) {
        splay_lazy += x;
        node_value += x;
```

```
        splay_value += x * splay_size;
        whole_subtree_value += x * splay_size;
    }
    void add_lazy_in_subtree(T x) {
        whole_subtree_lazy += x;
        node_value += x;
        splay_value += x * splay_size;
        whole_subtree_value += x * whole_subtree_size;
        virtual_value += x * virtual_size;
    }
};
struct Splay {
    struct Node {
        array<int, 2> child;
        int parent;
        int subtree_splay = 1;
        bool lazy_flip = false;
        AdditionalInfo info;
    };
    V<Node> t;
    C int nil;
    Splay(int n)
    : t(n + 1), nil(n) {
        t[nil].subsize_splay = 0;
        for(Node &v : t)
            v.child[0] = v.child[1] = v.parent = nil;
    }
    void apply_lazy_and_push(int v) {
        auto &[l, r] = t[v].child;
        if(t[v].lazy_flip) {
            for(int c : {l, r})
                t[c].lazy_flip ^= 1;
            swap(l, r);
        }
        t[v].info.push_lazy(t[l].info, t[r].info, t[v].
            lazy_flip);
        for(int c : {l, r})
            if(c != nil)
                t[c].info.pull_lazy_from_parent(t[v].info);
        t[v].lazy_flip = false;
    }
    void update_from_sons(int v) {
        // assumes that v's info is pushed
        auto [l, r] = t[v].child;
        t[v].subsize_splay = t[l].subsize_splay + 1 + t[r].
            subsize_splay;
        for(int c : {l, r})
            apply_lazy_and_push(c);
        t[v].info.update_from_sons(t[l].info, t[r].info);
    }
    // After that, v is pushed and updated
    void splay(int v) {
        apply_lazy_and_push(v);
        auto set_child = [&](int x, int c, int d) {
            if(x != nil and d != -1)
                t[x].child[d] = c;
            if(c != nil) {
                t[c].parent = x;
                t[c].info.cancel_subtree_lazy_from_parent(t[x].
                    info);
            }
        };
        auto get_dir = [&](int x) -> int {
            int p = t[x].parent;
            if(p == nil or (x != t[p].child[0] and x != t[p].
                child[1]))
                return -1;
            return t[p].child[1] == x;
        };
        auto rotate = [&](int x, int d) {
            int p = t[x].parent, c = t[x].child[d];
            assert(c != nil);
            set_child(p, c, get_dir(x));
            set_child(x, t[c].child[!d], d);
            set_child(c, x, !d);
            update_from_sons(x);
```

```

    update_from_sons(c);
};
while(get_dir(v) != -1) {
    int p = t[v].parent; pp = t[p].parent;
    array path_up = {v, p, pp, t[pp].parent};
    for(int i = ssize(path_up) - 1; i >= 0; --i) {
        if(i < ssize(path_up) - 1)
            t[path_up[i]].info.pull_lazy_from_parent(t[
                path_up[i + 1]].info);
        apply_lazy_and_push(path_up[i]);
    }
    int dp = get_dir(v), dpp = get_dir(p);
    if(dpp == -1)
        rotate(p, dp);
    else if(dp == dpp) {
        rotate(pp, dpp);
        rotate(p, dp);
    }
    else {
        rotate(p, dp);
        rotate(pp, dpp);
    }
}
};
struct LinkCut : Splay {
    LinkCut(int n) : Splay(n) {}
    // Cuts the path from x downward, creates path to
    // root, splays x.
    int access(int x) {
        int v = x, cv = nil;
        for(; v != nil; cv = v, v = t[v].parent) {
            splay(v);
            int &right = t[v].child[1];
            t[v].info.change_virtual(t[right].info, +1);
            right = cv;
            t[right].info.pull_lazy_from_parent(t[v].info);
            t[v].info.change_virtual(t[right].info, -1);
            update_from_sons(v);
        }
        splay(x);
        return cv;
    }
    // Changes the root to v.
    // Warning: Linking, cutting, getting the distance,
    // etc, changes the root.
    void reroot(int v) {
        access(v);
        t[v].lazy_flip ^= 1;
        apply_lazy_and_push(v);
    }
    // Returns the root of tree containing v.
    int get_leader(int v) {
        access(v);
        while(apply_lazy_and_push(v), t[v].child[0] != nil)
            v = t[v].child[0];
        splay(v);
        return v;
    }
    bool is_in_same_tree(int v, int u) {
        return get_leader(v) == get_leader(u);
    }
    // Assumes that v and u aren't in same tree and v !=
    // u.
    // Adds edge (v, u) to the forest.
    void link(int v, int u) {
        reroot(v);
        access(u);
        t[u].info.change_virtual(t[v].info, +1);
        assert(t[v].parent == nil);
        t[v].parent = u;
        t[v].info.cancel_subtree_lazy_from_parent(t[u].info
        );
    }
    // Assumes that v and u are in same tree and v != u.
    // Cuts edge going from v to the subtree where is u

```

```

    // (in particular, if there is an edge (v, u), it
    // deletes it).
    // Returns the cut parent.
    int cut(int v, int u) {
        reroot(u);
        access(v);
        int c = t[v].child[0];
        assert(t[c].parent == v);
        t[v].child[0] = nil;
        t[c].parent = nil;
        t[c].info.cancel_subtree_lazy_from_parent(t[nil].
        info);
        update_from_sons(v);
        while(apply_lazy_and_push(c), t[c].child[1] != nil)
            c = t[c].child[1];
        splay(c);
        return c;
    }
    // Assumes that v and u are in same tree.
    // Returns their LCA after a reroot operation.
    int lca(int root, int v, int u) {
        reroot(root);
        if(v == u)
            return v;
        access(v);
        return access(u);
    }
    // Assumes that v and u are in same tree.
    // Returns their distance (in number of edges).
    int dist(int v, int u) {
        reroot(v);
        access(u);
        return t[t[u].child[0]].subsize_splay;
    }
    // Assumes that v and u are in same tree.
    // Returns the sum of values on the path from v to u.
    auto get_path_sum(int v, int u) {
        reroot(v);
        access(u);
        return t[u].info.get_path_sum();
    }
    // Assumes that v and u are in same tree.
    // Returns the sum of values on the subtree of v in
    // which u isn't present.
    auto get_subtree_sum(int v, int u) {
        u = cut(v, u);
        auto ret = t[v].info.get_subtree_sum();
        link(v, u);
        return ret;
    }
    // Applies function f on vertex v (useful for a
    // single add/set operation)
    void apply_on_vertex(int v, function<void (
        AdditionalInfo&> f) {
        access(v);
        f(t[v].info);
    }
    // Assumes that v and u are in same tree.
    // Adds val to each vertex in path from v to u.
    void add_on_path(int v, int u, int val) {
        reroot(v);
        access(u);
        t[u].info.add_lazy_in_path(val);
    }
    // Assumes that v and u are in same tree.
    // Adds val to each vertex in subtree of v that doesn
    // 't have u.
    void add_on_subtree(int v, int u, int val) {
        u = cut(v, u);
        t[v].info.add_lazy_in_subtree(val);
        link(v, u);
    }
}
};

```

majorized-set

#af8039

$\mathcal{O}(\log n)$, w s jest zmajoryzowany set, insert(p) wrzuca parę p do setu, majoryzuje go (zamortyzowany czas) i zwraca, czy podany element został dodany.

```

template<typename A, typename B>
struct MajorizedSet {
    set<pair<A, B>> s;
    bool insert(pair<A, B> p) {
        auto x = s.lower_bound(p);
        if (x != s.end() && x->second >= p.se)
            return false;
        while (x != s.begin() && (--x)->second <= p.se)
            x = s.erase(x);
        s.emplace(p);
        return true;
    }
};

```

ordered-set

#0a779f, includes: <ext/pb_ds/assoc_container.hpp>, <ext/pb_ds/tree_policy.hpp>
insert(x) dodaje element x (nie ma emplace), find_by_order(i) zwraca iterator do i-tego elementu, order_of_key(x) zwraca ile jest mniejszych elementów (x nie musi być w secie). Jeśli chcemy multiseta, to używamy par (val, id).

```

using namespace __gnu_pbds;
template<class T> using ordered_set = tree<
    T,
    null_type,
    less<T>,
    rb_tree_tag,
    tree_order_statistics_node_update
>;

```

persistent-treap

#19b13c

$\mathcal{O}(\log n)$ Implicit Persistent Treap, wszystko indexowane od 0, insert(i, val) insertuje na pozycję i, kopiowanie struktury działa w $\mathcal{O}(1)$, robimy sobie V<Treap> żeby obsługiwać trwałość UPD. uwaga potencjalnie się kwadraci, spytać Bartka kiedy

```

mt19937 rng_i(0);
struct Treap {
    struct Node {
        int val, prio, sub = 1;
        Node *l = nullptr, *r = nullptr;
        Node(int _val) : val(_val), prio(rng_i()) {}
        ~Node() { delete l; delete r; }
    };
    using pNode = Node*;
    pNode root = nullptr;
    int get_sub(pNode n) { return n ? n->sub : 0; }
    void update(pNode n) {
        if(!n) return;
        n->sub = get_sub(n->l) + get_sub(n->r) + 1;
    }
    void split(pNode t, int i, pNode &l, pNode &r) {
        if(!t) l = r = nullptr;
        else {
            t = new Node(*t);
            if(i <= get_sub(t->l))
                split(t->l, i, l, t->l), r = t;
            else
                split(t->r, i - get_sub(t->l) - 1, t->r, r), l
                = t;
        }
        update(t);
    }
    void merge(pNode &t, pNode l, pNode r) {
        if(!l || !r) t = (l ? l : r);
        else if(l->prio > r->prio) {
            l = new Node(*l);
            merge(l->r, l->r, r), t = l;
        }
        else {

```

```

            r = new Node(*r);
            merge(r->l, l, r->l), t = r;
        }
        update(t);
    }
    void insert(pNode &t, int i, pNode it) {
        if(!t) t = it;
        else if(it->prio > t->prio)
            split(t, i, it->l, it->r), t = it;
        else {
            t = new Node(*t);
            if(i <= get_sub(t->l))
                insert(t->l, i, it);
            else
                insert(t->r, i - get_sub(t->l) - 1, it);
        }
        update(t);
    }
    void insert(int i, int val) {
        insert(root, i, new Node(val));
    }
    void erase(pNode &t, int i) {
        if(get_sub(t->l) == i)
            merge(t, t->l, t->r);
        else {
            t = new Node(*t);
            if(i <= get_sub(t->l))
                erase(t->l, i);
            else
                erase(t->r, i - get_sub(t->l) - 1);
        }
        update(t);
    }
    void erase(int i) {
        assert(i < get_sub(root));
        erase(root, i);
    }
};

```

range-add

#5283bf, includes: fenwick-tree

$\mathcal{O}(\log n)$ drzewo przedział-punkt (+, +), wszystko indexowane od 0, update(l, r, val) dodaje val na przedziale [l, r], query(pos) zwraca wartość elementu pos.

```

struct RangeAdd {
    Fenwick f;
    RangeAdd(int n) : f(n) {}
    void update(int l, int r, ll val) {
        f.update(l, val);
        f.update(r + 1, -val);
    }
    ll query(int pos) {
        return f.query(pos);
    }
};

```

rmq

#724ad6

$\mathcal{O}(n \log n)$ czasowo i pamięciowo, Range Minimum Query z użyciem sparse table, zapytanie jest w $\mathcal{O}(1)$.

```

struct RMQ {
    V<vi> st;
    RMQ(C vi &a) {
        int n = ssize(a), lg = 0;
        while((1 << lg) < n) lg++;
        st.resize(lg + 1, a);
        FOR(i, 1, lg) REP(j, n) {
            st[i][j] = st[i - 1][j];
            int q = j + (1 << (i - 1));
            if(q < n) chmin(st[i][j], st[i - 1][q]);
        }
    }
    int query(int l, int r) {
        int q = __lg(r - l + 1), x = r - (1 << q) + 1;

```

```
        return min(st[q][l], st[q][x]);
    }
};
```

segment-tree

#738e4c

Drzewa punkt-przedział. Pierwsze ustawia w punkcie i podaje max na przedziale. Drugie maxuje elementy na przedziale i podaje wartość w punkcie.

```
struct Tree_Get_Max {
    using T = int;
    T f(T a, T b) { return max(a, b); }
    C T zero = 0;
    V<T> tree;
    int sz = 1;
    Tree_Get_Max(int n) {
        while(sz < n)
            sz *= 2;
        tree.resize(sz * 2, zero);
    }
    void update(int pos, T val) {
        tree[pos += sz] = val;
        while(pos /= 2)
            tree[pos] = f(tree[pos * 2], tree[pos * 2 + 1]);
    }
    T get(int l, int r) {
        l += sz, r += sz;
        if(l == r)
            return tree[l];
        T ret_l = tree[l], ret_r = tree[r];
        while(l + 1 < r) {
            if(l % 2 == 0)
                ret_l = f(ret_l, tree[l + 1]);
            if(r % 2 == 1)
                ret_r = f(tree[r - 1], ret_r);
            l /= 2, r /= 2;
        }
        return f(ret_l, ret_r);
    }
};

struct Tree_Update_Max_On_Interval {
    using T = int;
    V<T> tree;
    int sz = 1;
    Tree_Update_Max_On_Interval(int n) {
        while(sz < n)
            sz *= 2;
        tree.resize(sz * 2);
    }
    T get(int pos) {
        T ret = tree[pos += sz];
        while(pos /= 2)
            chmax(ret, tree[pos]);
        return ret;
    }
    void update(int l, int r, T val) {
        l += sz, r += sz;
        chmax(tree[l], val);
        if(l == r)
            return;
        chmax(tree[r], val);
        while(l + 1 < r) {
            if(l % 2 == 0)
                chmax(tree[l + 1], val);
            if(r % 2 == 1)
                chmax(tree[r - 1], val);
            l /= 2, r /= 2;
        }
    }
};
```

treap

#f9c1bb

segment-tree treap 2sat biconnected cactus-cycles centro-decomp

$\mathcal{O}(\log n)$ Implicit Treap, wszystko indexowane od 0, do Node dopisujemy jakie chcemy mieć trzymać dodatkowo dane. Jeśli chcemy robić lazy, to wykonania push należy wstawić tam gdzie oznaczono komentarzem.

```
namespace Treap {
    // BEGIN HASH
    mt19937 rng_key(0);
    struct Node {
        int prio, cnt = 1;
        Node *l = nullptr, *r = nullptr;
        Node() : prio(rng_key()) {}
        ~Node() { delete l; delete r; }
    };
    using pNode = Node*;
    int get_cnt(pNode t) { return t ? t->cnt : 0; }
    void update(pNode t) {
        if (!t) return;
        // push(t);
        t->cnt = get_cnt(t->l) + get_cnt(t->r) + 1;
    }
    void split(pNode t, int i, pNode &l, pNode &r) {
        if (!t) {
            l = r = nullptr;
            return;
        }
        // push(t);
        if (i <= get_cnt(t->l))
            split(t->l, i, l, t->l), r = t;
        else
            split(t->r, i - get_cnt(t->l) - 1, t->r, r), l = t;
        update(t);
    }
    void merge(pNode &t, pNode l, pNode r) {
        if (!l or !r) t = l ?: r;
        else if (l->prio > r->prio) {
            // push(l);
            merge(l->r, l->r, r), t = l;
        }
        else {
            // push(r);
            merge(r->l, l, r->l), t = r;
        }
        update(t);
    }
    // END HASH
    void apply_on_interval(pNode &root, int l, int r,
        function<void (pNode)> f) {
        pNode left, mid, right;
        split(root, r + 1, mid, right);
        split(mid, l, left, mid);
        assert(l <= r and mid);
        f(mid);
        merge(mid, left, mid);
        merge(root, mid, right);
    }
}
```

Grafy (5)

2sat

#8e707e

$\mathcal{O}(n + m)$, Zwraca poprawne przyporządkowanie zmiennym logicznym dla problemu 2-SAT, albo mówi, że takie nie istnieje. Konstruktor przyjmuje liczbę zmiennych, ~ oznacza negację zmiennej. Po wywołaniu solve(), values[0..n-1] zawiera wartości rozwiązania.

```
struct TwoSat {
    int n;
    V<vi> gr;
    vi values;
    TwoSat(int _n = 0) : n(_n), gr(2 * n) {}
    void either(int f, int j) {
        f = max(2 * f, -1 - 2 * f);
        j = max(2 * j, -1 - 2 * j);
    }
};
```

```
gr[f].eb(j ^ 1);
gr[j].eb(f ^ 1);
}
void set_value(int x) { either(x, x); }
void implication(int f, int j) { either(~f, j); }
int add_var() {
    gr.eb();
    gr.eb();
    return n++;
}
void at_most_one(vi& li) {
    if(ssize(li) <= 1) return;
    int cur = ~li[0];
    FOR(i, 2, ssize(li) - 1) {
        int next = add_var();
        either(cur, ~li[i]);
        either(cur, next);
        either(~li[i], next);
        cur = ~next;
    }
    either(cur, ~li[1]);
}
vi val, comp, z;
int t = 0;
int dfs(int i) {
    int low = val[i] = ++t, x;
    z.eb(i);
    for(auto &e : gr[i]) if(!comp[e])
        chmin(low, val[e] ?: dfs(e));
    if(low == val[i]) do {
        x = z.back(); z.pop_back();
        comp[x] = low;
        if (values[x >> 1] == -1)
            values[x >> 1] = x & 1;
    } while (x != i);
    return val[i] = low;
}
bool solve() {
    values.assign(n, -1);
    val.assign(2 * n, 0);
    comp = val;
    REP(i, 2 * n) if(!comp[i]) dfs(i);
    REP(i, n) if(comp[2 * i] == comp[2 * i + 1]) return 0;
    return 1;
}
};
```

biconnected

#8cd55a

$\mathcal{O}(n + m)$, dwuspójne składowe, mosty oraz punkty artykulacji. po skonstruowaniu, bicon = zbiór list id krawędzi, bridges = lista id krawędzi będącymi mostami, arti_points = lista wierzchołków będącymi punktami artykulacji. Tablice są nieposortowane. Wspiera multikrawędzie i wiele spójnych, ale nie pętli.

```
struct Low {
    V<vi> graph;
    vi low, pre;
    V<pii> edges;
    V<vi> bicon;
    vi bicon_stack, arti_points, bridges;
    int gtime = 0;
    void dfs(int v, int p) {
        low[v] = pre[v] = gtime++;
        bool considered_parent = false;
        int son_count = 0;
        bool is_arti = false;
        for(int e : graph[v]) {
            int u = edges[e].fi ^ edges[e].se ^ v;
            if(u == p and not considered_parent)
                considered_parent = true;
            else if(pre[u] == -1) {
                bicon_stack.eb(e);
                dfs(u, v);
                chmin(low[v], low[u]);
            }
        }
    }
};
```

```
if(low[u] >= pre[v]) {
    bicon.eb();
    do {
        bicon.back().eb(bicon_stack.back());
        bicon_stack.pop_back();
    } while(bicon.back().back() != e);
}
++son_count;
if(p != -1 and low[u] >= pre[v])
    is_arti = true;
if(low[u] > pre[v])
    bridges.eb(e);
}
}
else if(pre[v] > pre[u]) {
    chmin(low[v], pre[u]);
    bicon_stack.eb(e);
}
}
if(p == -1 and son_count > 1)
    is_arti = true;
if(is_arti)
    arti_points.eb(v);
}
Low(int n, V<pii> _edges) : graph(n), low(n), pre(n, -1), edges(_edges) {
    REP(i, ssize(edges)) {
        auto [v, u] = edges[i];
    }
    #ifndef LOCAL
        assert(v != u);
    #endif
    graph[v].eb(i);
    graph[u].eb(i);
}
REP(v, n)
    if(pre[v] == -1)
        dfs(v, -1);
}
};
```

cactus-cycles

#c9ef3d

$\mathcal{O}(n)$, wyznaczenie cykli w grafie. Zakłada że jest nieskierowany graf bez pętelek i multikrawędzi, każda krawędź leży na co najwyżej jednym cyklu prostym (silniejsze założenie, niż o wierzchołkach). cactus_cycles(graph) zwraca taką listę cykli, że istnieje krawędź między i -tym, a $(i + 1) \bmod \text{size}(\text{cycle})$ -tym wierzchołkiem.

```
V<vi> cactus_cycles(V<vi> graph) {
    vi state(ssize(graph), 0), stack;
    V<vi> ret;
    function<void (int, int)> dfs = [&](int v, int p) {
        if(state[v] == 2) {
            ret.eb(stack.rbegin(), find(rall(stack), v) + 1);
            return;
        }
        stack.eb(v);
        state[v] = 2;
        for(int u : graph[v])
            if(u != p and state[u] != 1)
                dfs(u, v);
        state[v] = 1;
        stack.pop_back();
    };
    REP(i, ssize(graph))
        if(!state[i])
            dfs(i, -1);
    return ret;
}
```

centro-decomp

#dbd3a2

$\mathcal{O}(n \log n)$, template do Centroid Decomposition Nie używamy podsz, odwi, ani odwi_cnt Konstruktor przyjmuje liczbę wierzchołków i drzewo. Jeśli chcemy mieć rozbudowane krawędzie, to zmienić tam gdzie zaznaczone. Mamy tablicę odwiedzonych z refresem w $\mathcal{O}(1)$ (używać bez skrepowania). visit(v) odznacza v jako odwiedzony. is_vis(v) zwraca, czy v jest odwiedzony. refresh(v) zamienia niezablokowane wierzchołki na nieodwiedzzone. W decomp mamy standardowe wykonanie CD na poziomie spójnej. Tablica par mówi kto jest naszym ojcem w drzewie CD. root to korzeń drzewa CD.

```
struct CentroDecomp {
    C V<vi> &graph; // tu
    vi par, podsz, odwi;
    int odwi_cnt = 1;
    C int INF = int(1e9);
    int root;
    void refresh() { ++odwi_cnt; }
    void visit(int v) { chmax(odwi[v], odwi_cnt); }
    bool is_vis(int v) { return odwi[v] >= odwi_cnt; }
    void dfs_podsz(int v) {
        visit(v);
        podsz[v] = 1;
        for (int u : graph[v]) // tu
            if (!is_vis(u)) {
                dfs_podsz(u);
                podsz[v] += podsz[u];
            }
    }
    int centro(int v) {
        refresh();
        dfs_podsz(v);
        int sz = podsz[v] / 2;
        refresh();
        while (true) {
            visit(v);
            for (int u : graph[v]) // tu
                if (!is_vis(u) && podsz[u] > sz) {
                    v = u;
                    break;
                }
            if (is_vis(v))
                return v;
        }
    }
    void decomp(int v) {
        refresh();
        // Tu kod. Centroid to v, ktory jest juz dozywotnie
        // odwiedzony.
        // Koniec kodu.
        refresh();
        for(int u : graph[v]) // tu
            if (!is_vis(u)) {
                u = centro(u);
                par[u] = v;
                odwi[u] = INF;
                // Opcjonalnie tutaj przekazujemy info synowi w
                // drzewie CD.
                decomp(u);
            }
    }
    CentroDecomp(int n, V<vi> &grph) // tu
        : graph(grph), par(n, -1), podsz(n), odwi(n) {
        root = centro(0);
        odwi[root] = INF;
        decomp(root);
    }
};
```

coloring

#b2d062

$\mathcal{O}(nm)$, wyznacza kolorowanie grafu planaranego. coloring(graph) zwraca 5-kolorowanie grafu coloring(graph, 4) zwraca 4-kolorowanie grafu, jeżeli w każdym momencie procesu usuwania wierzchołka o najmniejszym stopniu jego stopień jest nie większy niż 4

```
vi coloring(C V<vi>& graph, C int limit = 5) {
```

coloring de-brujin directed-mst dominator-tree

```
C int n = ssize(graph);
if (!n) return {};
function<vi(V<bool>)> solve = [&](C V<bool>& active)
{
    if (not *max_element(all(active)))
        return V(n, -1);
    pii best = {n, -1};
    REP(i, n) {
        if (not active[i])
            continue;
        int cnt = 0;
        for (int e : graph[i])
            cnt += active[e];
        chmin(best, pair(cnt, i));
    }
    C int id = best.se;
    auto cp = active;
    cp[id] = false;
    auto col = solve(cp);
    V<bool> used(limit);
    for (int e : graph[id])
        if (active[e])
            used[col[e]] = true;
    REP(i, limit)
        if (not used[i]) {
            col[id] = i;
            return col;
        }
    for (int e0 : graph[id]) {
        for (int e1 : graph[id]) {
            if (e0 >= e1)
                continue;
            V<bool> vis(n);
            function<void(int, int, int)> dfs = [&](int v,
                int c0, int c1) {
                vis[v] = true;
                for (int e : graph[v])
                    if (not vis[e] and (col[e] == c0 or col[e]
                        == c1))
                        dfs(e, c0, c1);
            };
            C int c0 = col[e0], c1 = col[e1];
            dfs(e0, c0, c1);
            if (vis[e1])
                continue;
            REP(i, n)
                if (vis[i])
                    col[i] = col[i] == c0 ? c1 : c0;
            col[id] = c0;
            return col;
        }
    }
    assert(false);
};
return solve(V(n, true));
}
```

de-brujin

#e577d2, includes: eulerian-path $\mathcal{O}(k^n)$, ciag/cykl de Brujina słów długości n nad alfabetem $\{0, 1, \dots, k-1\}$. Jeżeli is_path to zwraca ciąg, wpp. zwraca cykl.

```
vi de_brujin(int k, int n, bool is_path) {
    if (n == 1) {
        vi v(k);
        iota(all(v), 0);
        return v;
    }
    if (k == 1)
        return V(n, 0);
    int N = 1;
    REP(i, n - 1)
        N *= k;
    V<pii> edges;
    REP(i, N)
```

```
    REP(j, k)
        edges.eb(i, i * k % N + j);
    vi path = get<2>(eulerian_path(N, edges, true));
    path.pop_back();
    for(auto& e : path)
        e = e % k;
    if (is_path)
        REP(i, n - 1)
            path.eb(path[i]);
    return path;
}
```

directed-mst

#bdca76 $\mathcal{O}(m \log n)$, dla korzenia i listy krawędzi skierowanych ważonych zwraca najtańszy podzbiór $n-1$ krawędzi taki, że z korzenia istnieje ścieżka do każdego innego wierzchołka, lub -1 gdy nie ma. Zwraca (koszt, ojciec każdego wierzchołka w zwróconym drzewie).

```
struct RollbackUF {
    vi e; V<pii> st;
    RollbackUF(int n) : e(n, -1) {}
    int size(int x) { return -e[find(x)]; }
    int find(int x) { return e[x] < 0 ? x : find(e[x]); }
    int time() { return ssize(st); }
    void rollback(int t) {
        for(int i = time(); i --> t;)
            e[st[i].fi] = st[i].se;
        st.resize(t);
    }
    bool join(int a, int b) {
        a = find(a), b = find(b);
        if (a == b) return false;
        if (e[a] > e[b]) swap(a, b);
        st.pb({a, e[a]});
        st.pb({b, e[b]});
        e[a] += e[b]; e[b] = a;
        return true;
    }
};
};
struct Edge { int a, b; ll w; };
struct Node {
    Edge key;
    Node *l = 0, *r = 0;
    ll delta = 0;
    void prop() {
        key.w += delta;
        if(l) l->delta += delta;
        if(r) r->delta += delta;
        delta = 0;
    }
};
Node* merge(Node *a, Node *b) {
    if(!a || !b) return a ? b : a->prop(), b->prop();
    if(a->key.w > b->key.w) swap(a, b);
    swap(a->l, (a->r = merge(b, a->r)));
    return a;
}
pair<ll, vi> directed_mst(int n, int r, V<Edge> &g) {
    RollbackUF uf(n);
    V<Node*> heap(n);
    V<Node> pool(ssize(g));
    REP(i, ssize(g)) {
        Edge e = g[i];
        heap[e.b] = merge(heap[e.b], &(pool[i] = Node{e}));
    }
    ll res = 0;
    vi seen(n, -1), path(n), par(n);
    seen[r] = r;
    V<Edge> Q(n), in(n, {-1, -1, 0}), comp;
    deque<tuple<int, int, V<Edge>>> cyscs;
    REP(s, n) {
        int u = s, qi = 0, w;
        while(seen[u] != sdom[qi]) {
            Node *hu = heap[u];
```

```
            if(!hu) return {-1, {}};
            hu->prop();
            Edge e = hu->key;
            hu->delta -= e.w; hu->prop(); hu = merge(hu->l,
                hu->r);
            Q[qi] = e, path[qi++] = u, seen[u] = s;
            res += e.w, u = uf.find(e.a);
            if(seen[u] == s) {
                Node *c = 0;
                int end = qi, time = uf.time();
                do c = merge(c, heap[w = path[--qi]]);
                while(uf.join(u, w));
                u = uf.find(u), heap[u] = c, seen[u] = -1;
                cyscs.push_front({u, time, {&Q[qi], &Q[end]}});
            }
            REP(i, qi) in[uf.find(Q[i].b)] = Q[i];
        }
    }
    for(auto [u, t, c] : cyscs) { // restore sol (optional)
        uf.rollback(t);
        Edge inu = in[u];
        for(auto e : c) in[uf.find(e.b)] = e;
        in[uf.find(inu.b)] = inu;
    }
    REP(i, n) par[i] = in[i].a;
    return {res, par};
}
```

dominator-tree

#f79d15 $\mathcal{O}(m \alpha(n))$, dla spójnego DAGu o jednym korzeniu root wyznacza listę synów w dominator tree (które jest drzewem, gdzie ojciec wierzchołka v to najbliższy wierzchołek, którego usunięcie powoduje, że już nie ma ścieżki od korzenia do v). dominator_tree({{1,2},{3},{4},{4},{5},{5}},0) == {{1,4,2},{3},{},{}},{5},{}}

```
V<vi> dominator_tree(V<vi> dag, int root) {
    int n = ssize(dag);
    V<vi> t(n), rg(n), bucket(n);
    vi id(n, -1), sdom = id, par = id, idom = id, dsu =
        id, label = id, rev = id;
    function<int(int, int)> find = [&](int v, int x) {
        if(v == dsu[v]) return x ? -1 : v;
        int u = find(dsu[v], x + 1);
        if(u < 0) return v;
        if(sdom[label[dsu[v]]] < sdom[label[v]]) label[v] =
            label[dsu[v]];
        dsu[v] = u;
        return x ? u : label[v];
    };
    int gtime = 0;
    function<void(int)> dfs = [&](int u) {
        rev[gtime] = u;
        label[gtime] = sdom[gtime] = dsu[gtime] = id[u] =
            gtime;
        gtime++;
        for(int w : dag[u]) {
            if(id[w] == -1) dfs(w), par[id[w]] = id[u];
            rg[id[w]].eb(id[u]);
        }
    };
    dfs(root);
    for(int i = n - 1; i >= 0; i--) {
        for(int u : rg[i]) chmin(sdom[i], sdom[find(u, 0)])
            ;
        if(i > 0) bucket[sdom[i]].pb(i);
        for(int w : bucket[i]) {
            int v = find(w, 0);
            idom[w] = (sdom[v] == sdom[w] ? sdom[w] : v);
        }
        if(i > 0) dsu[i] = par[i];
    }
    FOR(i, 1, n - 1) {
        if(idom[i] != sdom[i]) idom[i] = idom[idom[i]];
        t[rev[idom[i]]].eb(rev[i]);
    }
```

```
    }
    return t;
}
```

dynamic-connectivity

#fd42a9

$\mathcal{O}(q \log^2 n)$ offline, zaczyna z pustym grafem, dla danego zapytania stwierdza czy wierzchołki sa w jednej spójnej. Multikrawędzie oraz pętelki działają.

```
enum Event_type { Add, Remove, Query };
V<bool> dynamic_connectivity(int n, V<tuple<int, int,
Event_type>> events) {
    V<pii> queries;
    for(auto &[v, u, t] : events) {
        if(v > u)
            swap(v, u);
        if(t == Query)
            queries.eb(v, u);
    }
    int leaves = 1;
    while(leaves < ssize(queries))
        leaves *= 2;
    V<V<pii>> edges_to_add(2 * leaves);
    map<pii, deque<int>> edge_longevity;
    int query_i = 0;
    auto add = [&](int l, int r, pii e) {
        if(l > r)
            return;
        debug(l, r, e);
        l += leaves;
        r += leaves;
        while(l <= r) {
            if(l % 2 == 1)
                edges_to_add[l++].eb(e);
            if(r % 2 == 0)
                edges_to_add[r--].eb(e);
            l /= 2;
            r /= 2;
        }
    };
    for(C auto &[v, u, t] : events) {
        auto &que = edge_longevity[pair(v, u)];
        if(t == Add)
            que.eb(query_i);
        else if(t == Remove) {
            if(que.empty())
                continue;
            if(ssize(que) == 1)
                add(que.back(), query_i - 1, pair(v, u));
            que.pop_back();
        }
        else
            ++query_i;
    }
    for(C auto &[e, que] : edge_longevity)
        if(not que.empty())
            add(que.front(), query_i - 1, e);
    V<bool> ret(ssize(queries));
    vi lead(n), leadsz(n, 1);
    iota(all(lead), 0);
    function<int (int)> find = [&](int i) {
        return i == lead[i] ? i : find(lead[i]);
    };
    function<void (int)> dfs = [&](int v) {
        V<tuple<int, int, int, int>> rollback;
        for(auto [e0, e1] : edges_to_add[v]) {
            e0 = find(e0);
            e1 = find(e1);
            if(e0 == e1)
                continue;
            if(leadsz[e0] > leadsz[e1])
                swap(e0, e1);
            rollback.eb(e0, lead[e0], e1, leadsz[e1]);
            leadsz[e1] += leadsz[e0];
            lead[e0] = e1;
        }
    };
    if(v >= leaves) {
        int i = v - leaves;
        assert(i < leaves);
        if(i < ssize(queries))
            ret[i] = find(queries[i].fi) == find(queries[i].se);
    }
    else {
        dfs(2 * v);
        dfs(2 * v + 1);
    }
    reverse(all(rollback));
    for(auto [i, val, j, sz] : rollback) {
        lead[i] = val;
        leadsz[j] = sz;
    }
};
return ret;
}
```

eulerian-path

#d0a611

$\mathcal{O}(n + m)$, ścieżka eulera. Zwraca tupla (exists, ids, vertices). W exists jest informacja czy jest ścieżka/cykl eulera, ids zawiera id kolejnych krawędzi, vertices zawiera listę wierzchołków na tej ścieżce. Dla cyklu, vertices[0] == vertices[m].

```
tuple<bool, vi, vi> eulerian_path(int n, C V<pii> &
edges, bool directed) {
    vi in(n);
    V<vi> adj(n);
    int start = 0;
    REP(i, ssize(edges)) {
        auto [a, b] = edges[i];
        start = a;
        ++in[b];
        adj[a].eb(i);
        if (not directed)
            adj[b].eb(i);
    }
    int cnt_in = 0, cnt_out = 0;
    REP(i, n) {
        if (directed) {
            if (abs(ssize(adj[i]) - in[i]) > 1)
                return {};
            return {};
        }
        if (in[i] < ssize(adj[i]))
            start = i, ++cnt_in;
        else
            cnt_out += in[i] > ssize(adj[i]);
    }
    else if (ssize(adj[i]) % 2)
        start = i, ++cnt_in;
    }
    vi ids, vertices;
    V<bool> used(ssize(edges));
    function<void (int)> dfs = [&](int v) {
        while (ssize(adj[v])) {
            int id = adj[v].back(), u = v ^ edges[id].fi ^
edges[id].se;
            adj[v].pop_back();
            if (used[id]) continue;
            used[id] = true;
            dfs(u);
            ids.eb(id);
        }
    };
    dfs(start);
    if (cnt_in + cnt_out > 2 or not all_of(all(used),
identity{}))
        return {};
    reverse(all(ids));
    if (ssize(ids))
        vertices = {start};
    for (int id : ids)
```

```
vertices.eb(vertices.back() ^ edges[id].fi ^ edges[
id].se);
    return {true, ids, vertices};
}
```

hld

#642525

$\mathcal{O}(q \log n)$ Heavy-Light Decomposition. get_vertex(v) zwraca pozycję odpowiadającą wierzchołkowi. get_path(v, u) zwraca przedziały do obsługi drzewem przedziałowym. get_path(v, u) jeśli robisz operacje na wierzchołkach. get_path(v, u, false) jeśli na krawędziach (nie zawiera lca). get_subtree(v) zwraca przedział preorderów odpowiadający poddrzewu v.

```
struct HLD {
    // BEGIN HASH 0d65c4
    V<vi> &adj;
    vi sz, pre, pos, nxt, par;
    int t = 0;
    void init(int v, int p = -1) {
        par[v] = p;
        sz[v] = 1;
        if(ssize(adj[v]) > 1 && adj[v][0] == p)
            swap(adj[v][0], adj[v][1]);
        for(int &u : adj[v]) if(u != par[v]) {
            init(u, v);
            sz[v] += sz[u];
            if(sz[u] > sz[adj[v][0]])
                swap(u, adj[v][0]);
        }
    }
    void set_paths(int v) {
        pre[v] = t++;
        for(int &u : adj[v]) if(u != par[v]) {
            nxt[u] = (u == adj[v][0] ? nxt[v] : u);
            set_paths(u);
        }
        pos[v] = t;
    }
    HLD(int n, V<vi> &adj)
        : adj(_adj), sz(n), pre(n), pos(n), nxt(n), par(n)
    {
        init(0, set_paths(0));
    } // END HASH
    int lca(int v, int u) {
        while(nxt[v] != nxt[u]) {
            while(pre[v] < pre[u])
                swap(v, u);
            v = par[nxt[v]];
        }
        return (pre[v] < pre[u] ? v : u);
    }
    V<pii> path_up(int v, int u) {
        V<pii> ret;
        while(nxt[v] != nxt[u]) {
            ret.eb(pre[nxt[v]], pre[v]);
            v = par[nxt[v]];
        }
        if(pre[u] != pre[v]) ret.eb(pre[u] + 1, pre[v]);
        return ret;
    }
    int get_vertex(int v) { return pre[v]; }
    V<pii> get_path(int v, int u, bool add_lca = true) {
        int w = lca(v, u);
        auto ret = path_up(v, w);
        auto path_u = path_up(u, w);
        if(add_lca) ret.eb(pre[w], pre[w]);
        ret.insert(ret.end(), path_u.begin(), path_u.end());
        return ret;
    }
    pii get_subtree(int v) { return {pre[v], pos[v] - 1};
    }
```

hld-online-bottom-up

#dc8d43, includes: hld

$\mathcal{O}(q \log^2 n)$, rozwiąza zadania, gdzie wynik to dp bottom-up na drzewie i zmienia się wartość wierzchołka/krawędzi. To zakłada, że da się tak uogólnić tego bottom-up'a, że da się trzymać fragmenty drzewa z "dwoma dziurami" i doczepiać jak LEGO dwa takie fragmenty do siebie.

```
// Information about a single vertex (e.g. color).
// A component contains answers for vertices, not edges
using Value_v = int;
// Probably you want: some information about the up
// vertex, the down vertex,
// answer for whole component, answer containing up,
// answer containing down,
// answer containing both up and down.
struct DpTwoEnds;
// Merge two disjoint-vertex paths. Assume that there
// is an edge
// between "up" vertex of d and "down" vertex od u.
DpTwoEnds merge(DpTwoEnds u, DpTwoEnds d);
// DpOneEnd Contains information about a component
// after forgetting the "down" vertex.
// Probably you want: answer for whole component,
// informations about top vertices.
// It needs a default constructor.
struct DpOneEnd;
// Merge two parallel components. They are vertex-
// disjoint. They do not contain the
// parent (it will be included in the next function).
DpOneEnd merge(DpOneEnd a, DpOneEnd b);
// Assuming that DpOneEnd contain all components of the
// light sons of the parent,
// merge those components once with the parent. It has
// to support passing the
// default/neutral value of DpOneEnd -- it means that
// the vertex doesn't have light sons.
DpTwoEnds merge(DpOneEnd sons, Value_v value_parent);
// From a path that remembers "up" and "down" vertices,
// forget the "down" one.
DpOneEnd two_to_one(DpTwoEnds two);
template<class T> struct Tree {
    int leaves = 1;
    V<T> tree;
    Tree(int n = 0) {
        while(leaves < n)
            leaves *= 2;
        tree.resize(2 * leaves);
    }
    void set(int i, T t) {
        tree[i += leaves] = t;
        while(i /= 2)
            tree[i] = merge(tree[2 * i], tree[2 * i + 1]);
    }
    T get(i) { return tree[i]; }
};
struct DpDynamicBottomUp {
    int n;
    HLD hld;
    V<Tree<DpOneEnd>> tree_sons;
    V<Tree<DpTwoEnds>> tree_path;
    V<Value_v> current_values;
    vi which_on_path, which_light_son;
    DpDynamicBottomUp(V<vi> graph, V<Value_v>
initial_values)
        : n(ssize(graph)), hld(n, graph), tree_sons(n),
tree_path(n), current_values(initial_values),
which_on_path(n, -1), which_light_son(n, -1) {
        function<void (int, int*)> dfs = [&](int v, int *
on_heavy_cnt) {
            int light_sons_cnt = 0, tmp = 0;
            which_on_path[v] = (*on_heavy_cnt = on_heavy_cnt
?: &tmp)++;
            for(int u : hld.adj[v])
                if(u != hld.par[v])
```

```
dfs(u, hld.nxt[u] == u ? which_light_son[u] =
    light_sons_cnt++, nullptr : on_heavy_cnt)
;
tree_sons[v] = Tree<DpOneEnd>(light_sons_cnt);
tree_path[v] = Tree<DpTwoEnds>(tmp);
};
dfs(0, 0);
REP(v, n)
    set(v, initial_values[v]);
}
void set(int v, int value_vertex) {
    current_values[v] = value_vertex;
    while(true) {
        tree_path[hld.nxt[v]].set(which_on_path[v], merge
            (tree_sons[v].get(), current_values[v]));
        v = hld.nxt[v];
        if(hld.par[v] == -1)
            break;
        tree_sons[hld.par[v]].set(which_light_son[v],
            two_to_one(tree_path[hld.nxt[v]].get()));
        v = hld.par[v];
    }
}
DpTwoEnds get() { return tree_path[0].get(); }
};
```

jump-ptr
#86ffd1

$\mathcal{O}((n + q) \log n)$, jump_up(v, k) zwraca wierzchołek o k krawędzi wyżej niż v lub -1 . OperationJumpPtr może otrzymać wynik na ścieżce. Wynik na ścieżce do góry wymaga łączności, wynik dowolnej ścieżki jest poprawny, gdy jest odwrotność wyniku lub przemiennea.

```
// BEGIN HASH a0bbb0
struct SimpleJumpPtr {
    int bits;
    V<vi> graph, jmp;
    vi par, dep;
    void par_dfs(int v) {
        for(int u : graph[v])
            if(u != par[v]) {
                par[u] = v;
                dep[u] = dep[v] + 1;
                par_dfs(u);
            }
    }
    SimpleJumpPtr(V<vi> g = {}, int root = 0) : graph(g)
    {
        int n = ssize(graph);
        bits = _lg(max(1, n)) + 1;
        dep.resize(n);
        par.resize(n, -1);
        if(n > 0)
            par_dfs(root);
        jmp.resize(bits, vi(n, -1));
        jmp[0] = par;
        FOR(b, 1, bits - 1)
            REP(v, n)
                if(jmp[b - 1][v] != -1)
                    jmp[b][v] = jmp[b - 1][jmp[b - 1][v]];
        debug(graph, jmp);
    }
    int jump_up(int v, int h) {
        for(int b = 0; (1 << b) <= h; ++b)
            if((h >> b) & 1)
                v = jmp[b][v];
        return v;
    }
    int lca(int v, int u) {
        if(dep[v] < dep[u])
            swap(v, u);
        v = jump_up(v, dep[v] - dep[u]);
        if(v == u)
            return v;
        for(int b = bits - 1; b >= 0; b--) {
            if(jmp[b][v] != jmp[b][u]) {

```

```
v = jmp[b][v];
u = jmp[b][u];
}
}
return par[v];
}; // END HASH
using PathAns = ll;
PathAns merge(PathAns down, PathAns up) {
    return down + up;
}
struct OperationJumpPtr {
    SimpleJumpPtr ptr;
    V<V<PathAns>> ans_jmp;
    OperationJumpPtr(V<V<pii>> g, int root = 0) {
        debug(g, root);
        int n = ssize(g);
        V<vi> unweighted_g(n);
        REP(v, n)
            for(auto [u, w] : g[v]) {
                (void) w;
                unweighted_g[v].eb(u);
            }
        ptr = SimpleJumpPtr(unweighted_g, root);
        ans_jmp.resize(ptr.bits, V<PathAns>(n));
        REP(v, n)
            for(auto [u, w] : g[v])
                if(u == ptr.par[v])
                    ans_jmp[0][v] = PathAns(w);
        FOR(b, 1, ptr.bits - 1)
            REP(v, n)
                if(ptr.jmp[b - 1][v] != -1 and ptr.jmp[b - 1][
                    ptr.jmp[b - 1][v]] != -1)
                    ans_jmp[b][v] = merge(ans_jmp[b - 1][v],
                        ans_jmp[b - 1][ptr.jmp[b - 1][v]]);
            }
        PathAns path_ans_up(int v, int h) {
            PathAns ret = PathAns();
            for(int b = ptr.bits - 1; b >= 0; b--)
                if((h >> b) & 1) {
                    ret = merge(ret, ans_jmp[b][v]);
                    v = ptr.jmp[b][v];
                }
            return ret;
        }
        PathAns path_ans(int v, int u) { // discards order of
            edges on path
            int l = ptr.lca(v, u);
            return merge(
                path_ans_up(v, ptr.dep[v] - ptr.dep[l]),
                path_ans_up(u, ptr.dep[u] - ptr.dep[l])
            );
        }
    };
};
```

max-clique
#064cbc

$\mathcal{O}(idk)$, działa 1s dla $n=155$ na najgorszych przypadkach (losowe grafy $p=.90$). Działa szybciej dla grafów rzadkich. Zwraca listę wierzchołków w jakiejś max klicie. Pętelki niedozwolone.

```
constexpr int max_n = 500;
vi get_max_clique(V<bitset<max_n>> e) {
    double limit = 0.025, pk = 0;
    V<pii> v;
    V<vi> c(ssize(e) + 1);
    vi qmax, q, S(ssize(c)), old(S);
    REP(i, ssize(e)) v.eb(0, i);
    auto init = [&](V<pii>& r) {
        for (auto& vv : r) for (auto j : r) vv.fi += e[vv.
            se][j.se];
        sort(rall(r));
        int mxD = r[0].fi;
        REP(i, ssize(r)) r[i].fi = min(i, mxD) + 1;
    };
};
```

```
function<void (V<pii>&, int)> expand = [&](V<pii>& R,
    int lev) {
    S[lev] += S[lev - 1] - old[lev];
    old[lev] = S[lev - 1];
    while (ssize(R)) {
        if (ssize(q) + R.back().fi <= ssize(qmax)) return
            ;
        q.eb(R.back().se);
        V<pii> T;
        for(auto [_, vv] : R) if (e[R.back().se][vv]) T.
            eb(0, vv);
        if (ssize(T)) {
            if (S[lev]++ / ++pk < limit) init(T);
            int j = 0, mxk = 1, mnk = max(ssize(qmax) -
                ssize(q) + 1, 1);
            c[1] = c[2] = {};
            for (auto [_, v] : T) {
                int k = 1;
                while (any_of(all(c[k]), [&](int i) { return
                    e[v][i]; }))) k++;
                if (k > mxk) c[(mxk = k) + 1] = {};
                if (k < mnk) T[j++] .se = v;
                c[k].eb(v);
            }
            if (j > 0) T[j - 1].fi = 0;
            FOR(k, mnk, mxk) for (int i : c[k]) T[j++] = {k
                , i};
            expand(T, lev + 1);
        } else if (ssize(q) > ssize(qmax)) qmax = q;
        q.pop_back(), R.pop_back();
    }
};
init(v), expand(v, 1); return qmax;
}
```

negative-cycle
#0ff517

$\mathcal{O}(nm)$ stwierdzanie istnienia i wyznaczenie ujemnego cyklu. cycle spełnia $cycle[i] \rightarrow cycle[(i+1) \% ssize(cycle)]$. Żeby wyznaczyć krawędzie na cyklu, wystarczy wybierać najtańszą krawędź między wierzchołkami.

```
template<class I>
pair<bool, vi> negative_cycle(V<V<pair<int, I>>> graph)
{
    int n = ssize(graph);
    V<I> dist(n);
    vi from(n, -1);
    int v_on_cycle = -1;
    REP(iter, n) {
        v_on_cycle = -1;
        REP(v, n)
            for(auto [u, w] : graph[v])
                if(dist[u] > dist[v] + w) {
                    dist[u] = dist[v] + w;
                    from[u] = v;
                    v_on_cycle = u;
                }
    }
    if(v_on_cycle == -1)
        return {false, {}};
    REP(iter, n)
        v_on_cycle = from[v_on_cycle];
    vi cycle = {v_on_cycle};
    for(int v = from[v_on_cycle]; v != v_on_cycle; v =
        from[v])
        cycle.eb(v);
    reverse(all(cycle));
    return {true, cycle};
}
```

planar-graph-faces
#2ba436

$\mathcal{O}(m \log m)$, zakłada, że każdy punkt ma podane współrzędne, punkty są parami różne oraz krawędzie są nieprzecinającymi się odcinkami. Zwraca wszystkie ściany (wewnętrzne posortowane clockwise, zewnętrzne cc). WAŻNE czasem trzeba złączyć wszystkie ściany zewnętrzne (których może być kilka, gdy jest wiele spójnych) w jedną ścianę. Zewnętrzne ściany mogą wyglądać jak kaktusy, a wewnętrzne zawsze są niezdegenerowanym wielokątem.

```
struct Edge {
    int e, from, to;
    // face is on the right of "from -> to"
};
ostream& operator<<(ostream &o, Edge e) {
    return o << V{e.e, e.from, e.to};
}
struct Face {
    bool is_outside;
    V<Edge> sorted_edges;
    // edges are sorted clockwise for inside and cc for
    outside faces
};
ostream& operator<<(ostream &o, Face f) {
    return o << pair{f.is_outside, f.sorted_edges};
}
V<Face> split_planar_to_faces(V<pii> coord, V<pii>
    edges) {
    int n = ssize(coord);
    int E = ssize(edges);
    V<vi> graph(n);
    REP(e, E) {
        auto [v, u] = edges[e];
        graph[v].eb(e);
        graph[u].eb(e);
    }
    vi lead(2 * E);
    iota(all(lead), 0);
    function<int (int)> find = [&](int v) {
        return lead[v] == v ? v : lead[v] = find(lead[v]);
    };
    auto side_of_edge = [&](int e, int v, bool outward) {
        return 2 * e + ((v != min(edges[e].fi, edges[e].se)
            ) ^ outward);
    };
    REP(v, n) {
        V<pair<pii, int>> sorted;
        for(int e : graph[v]) {
            auto p = coord[edges[e].fi ^ edges[e].se ^ v];
            auto center = coord[v];
            sorted.eb(pair{p.fi - center.fi, p.se - center.se
                }, e);
        }
        sort(all(sorted), [&](pair<pii, int> l0, pair<pii,
            int> r0) {
            auto l = l0.fi;
            auto r = r0.fi;
            bool half_l = l > pair(0, 0);
            bool half_r = r > pair(0, 0);
            if(half_l != half_r)
                return half_l;
            return l.fi * ll(r.se) - l.se * ll(r.fi) > 0;
        });
        REP(i, ssize(sorted)) {
            int e0 = sorted[i].se;
            int e1 = sorted[(i + 1) % ssize(sorted)].se;
            int side_e0 = side_of_edge(e0, v, true);
            int side_e1 = side_of_edge(e1, v, false);
            lead[find(side_e0)] = find(side_e1);
        }
    }
    V<vi> comps(2 * E);
    REP(i, 2 * E)
        comps[find(i)].eb(i);
    V<Face> polygons;
    V<V<pii>> outgoing_for_face(n);
    REP(leader, 2 * E)
        if(ssize(comps[leader])) {

```

```

for(int id : comps[leader]) {
    int v = edges[id / 2].fi;
    int u = edges[id / 2].se;
    if(v > u)
        swap(v, u);
    if(id % 2 == 1)
        swap(v, u);
    outgoing_for_face[v].eb(u, id / 2);
}
V<Edge> sorted_edges;
function<void (int)> dfs = [&](int v) {
    while(ssize(outgoing_for_face[v])) {
        auto [u, e] = outgoing_for_face[v].back();
        outgoing_for_face[v].pop_back();
        dfs(u);
        sorted_edges.eb(e, v, u);
    }
};
dfs(edges[comps[leader].front() / 2].fi);
reverse(all(sorted_edges));
ll area = 0;
for(auto edge : sorted_edges) {
    auto l = coord[edge.from];
    auto r = coord[edge.to];
    area += l.fi * ll(r.se) - l.se * ll(r.fi);
}
polygons.eb(area >= 0, sorted_edges);
}
// Remember that there can be multiple outside faces.
return polygons;
}

```

planarity-check

#d07281

$\mathcal{O}(szybko)$ ale istnieją przykłady $\mathcal{O}(n^2)$, przyjmuje graf nieskierowany bez pętelek i multikrawędzi.

```

bool is_planar(V<vi> graph) {
    int n = ssize(graph), m = 0;
    REP(v, n)
        m += ssize(graph[v]);
    m /= 2;
    if(n <= 3) return true;
    if(m > 3 * n - 6) return false;
    V<vi> up(n), dn(n);
    vi low(n, -1), pre(n);
    REP(start, n)
        if(low[start] == -1) {
            V<pii> e_up;
            int tm = 0;
            function<void (int, int)> dfs_low = [&](int v,
                int p) {
                low[v] = pre[v] = tm++;
                for(int u : graph[v])
                    if(u != p and low[u] == -1) {
                        dn[v].eb(u);
                        dfs_low(u, v);
                        chmin(low[v], low[u]);
                    }
                else if(u != p and pre[u] < pre[v]) {
                    up[v].eb(ssize(e_up));
                    e_up.eb(v, u);
                    chmin(low[v], pre[u]);
                }
            };
            dfs_low(start, -1);
            V<pair<int, bool>> dsu(ssize(e_up));
            REP(v, ssize(dsu)) dsu[v].fi = v;
            function<pair<int, bool> (int)> find = [&](int v)
                {
                    if(dsu[v].fi == v)
                        return pair(v, false);
                    auto [u, ub] = find(dsu[v].fi);
                    return dsu[v] = pair(u, ub ^ dsu[v].se);
                };
            auto onion = [&](int x, int y, bool flip) {

```

```

        auto [v, vb] = find(x);
        auto [u, ub] = find(y);
        if(v == u)
            return not (vb ^ ub ^ flip);
        dsu[v] = {u, vb ^ ub ^ flip};
        return true;
    };
    auto interlace = [&](C vi &ids, int lo) {
        vi ans;
        for(int e : ids)
            if(pre[e_up[e].se] > lo)
                ans.eb(e);
        return ans;
    };
    auto add_fu = [&](C vi &a, C vi &b) {
        FOR(k, 1, ssize(a) - 1)
            if(not onion(a[k - 1], a[k], 0))
                return false;
        FOR(k, 1, ssize(b) - 1)
            if(not onion(b[k - 1], b[k], 0))
                return false;
        return a.empty() or b.empty() or onion(a[0], b[0], 1);
    };
    function<bool (int, int)> dfs_planar = [&](int v,
        int p) {
        for(int u : dn[v])
            if(not dfs_planar(u, v))
                return false;
        REP(i, ssize(dn[v])) {
            FOR(j, i + 1, ssize(dn[v]) - 1)
                if(not add_fu(interlace(up[dn[v][i]], low[
                    dn[v][j]]),
                    interlace(up[dn[v][j]], low[dn[v][i]])))
                    return false;
        }
        for(int j : up[v]) {
            if(e_up[j].fi != v)
                continue;
            if(not add_fu(interlace(up[dn[v][i]], pre[
                e_up[j].se]),
                interlace({j}, low[dn[v][i]])))
                return false;
        }
        return true;
    };
    if(not dfs_planar(start, -1))
        return false;
}
return true;
}

```

SCC

#c5beb2

konstruktor $\mathcal{O}(n)$, $\text{get_compressed } \mathcal{O}(n \log n)$. $\text{group}[v]$ to numer silnie spójnej wierzchołka v , order to toposort, w którym krawędzie idą w lewo (z lewej są liście), $\text{get_compressed}()$ zwraca graf silnie spójnych, $\text{get_compressed}(\text{false})$ nie usuwa multikrawędzi.

```

struct SCC {
    int n;
    V<vi> &graph;
    int group_cnt = 0;
    vi group;
    V<vi> rev_graph;
    vi order;
    void order_dfs(int v) {
        group[v] = 1;
        for(int u : rev_graph[v])

```

```

            if(group[u] == 0)
                order_dfs(u);
            order.eb(v);
        }
    }
    void group_dfs(int v, int color) {
        group[v] = color;
        for(int u : graph[v])
            if(group[u] == -1)
                group_dfs(u, color);
    }
    SCC(V<vi> &_graph) : graph(_graph) {
        n = ssize(graph);
        rev_graph.resize(n);
        REP(v, n)
            for(int u : graph[v])
                rev_graph[u].eb(v);
        group.resize(n);
        REP(v, n)
            if(group[v] == 0)
                order_dfs(v);
        reverse(all(order));
        debug(order);
        group.assign(n, -1);
        for(int v : order)
            if(group[v] == -1)
                group_dfs(v, group_cnt++);
    }
    V<vi> get_compressed(bool delete_same = true) {
        V<vi> ans(group_cnt);
        REP(v, n)
            for(int u : graph[v])
                if(group[v] != group[u])
                    ans[group[v]].eb(group[u]);
        if(not delete_same)
            return ans;
        REP(v, group_cnt) {
            sort(all(ans[v]));
            ans[v].erase(unique(all(ans[v])), ans[v].end());
        }
        return ans;
    }
};

```

toposort

#d0f178

$\mathcal{O}(n)$, $\text{get_toposort_order}(g)$ zwraca listę wierzchołków takich, że krawędzie są od wierzchołków wcześniejszych w liście do późniejszych. $\text{get_new_vertex_id_from_order}(\text{order})$ zwraca odwrotność tej permutacji, tzn. dla każdego wierzchołka trzyma jego nowy numer, aby po przenumerowaniu grafu istniały krawędzie tylko do wierzchołków o większych numerach. $\text{permute}(\text{elems}, \text{new_id})$ zwraca przepermutowaną tablicę elems według nowych numerów wierzchołków (przdatne jak się trzyma informacje o wierzchołkach, a chce się zrobić przenumerowanie topologiczne). $\text{renumerate_vertices}(\dots)$ zwraca nowy graf, w którym wierzchołki są przenumerowane. Nowy graf: $\text{renumerate_vertices}(\text{graph}, \text{get_new_vertex_id_from_order}(\text{get_toposort_order}(\text{graph})))$.

```

// BEGIN HASH 11a0409
vi get_toposort_order(V<vi> graph) {
    int n = ssize(graph);
    vi indeg(n);
    REP(v, n)
        for(int u : graph[v])
            ++indeg[u];
    vi que;
    REP(v, n)
        if(indeg[v] == 0)
            que.eb(v);
    vi ret;
    while(not que.empty()) {
        int v = que.back();
        que.pop_back();
        ret.eb(v);
        for(int u : graph[v])
            if(--indeg[u] == 0)

```

```

                que.eb(u);
            }
        }
        return ret;
    } // END HASH
    vi get_new_vertex_id_from_order(vi order) {
        vi ret(ssize(order), -1);
        REP(v, ssize(order))
            ret[order[v]] = v;
        return ret;
    }
    template<class T>
    V<T> permute(V<T> elems, vi new_id) {
        V<T> ret(ssize(elems));
        REP(v, ssize(elems))
            ret[new_id[v]] = elems[v];
        return ret;
    }
    V<vi> renumerate_vertices(V<vi> graph, vi new_id) {
        int n = ssize(graph);
        V<vi> ret(n);
        REP(v, n)
            for(int u : graph[v])
                ret[new_id[v]].eb(new_id[u]);
        REP(v, n)
            for(int u : ret[v])
                assert(v < u);
        return ret;
    }
}

```

triangles

#5ccda1

$\mathcal{O}(m\sqrt{m})$, liczenie możliwych kształtów podzbiorów trzy- i czterokrawędziowych. Suma zmiennych $\ast 3$ daje liczbę spójnych 3-elementowych podzbiorów krawędzi, analogicznie suma zmiennych $\ast 4$.

```

struct Triangles {
    int triangles3 = 0;
    ll stars3 = 0, paths3 = 0;
    ll ps4 = 0, rectangles4 = 0, paths4 = 0;
    __int128_t ys4 = 0, stars4 = 0;
    Triangles(V<vi> &graph) {
        int n = ssize(graph);
        V<pii> sorted_deg(n);
        REP(i, n)
            sorted_deg[i] = {ssize(graph[i]), i};
        sort(all(sorted_deg));
        vi id(n);
        REP(i, n)
            id[sorted_deg[i].se] = i;
        vi cnt(n);
        REP(v, n) {
            for(int u : graph[v]) if(id[v] > id[u])
                cnt[u] = 1;
            for(int u : graph[v]) if(id[v] > id[u]) for(int w
                : graph[u]) if(id[w] > id[u] and cnt[w]) {
                    ++triangles3;
                    for(int x : {v, u, w})
                        ps4 += ssize(graph[x]) - 2;
                }
            paths3 += -3 * triangles3;
            REP(v, n) for(int u : graph[v]) if(v < u)
                paths3 += (ssize(graph[v]) - 1) * ll(ssize(graph[
                    u]) - 1);
            ys4 += -2 * ps4;
            auto choose2 = [&](int x) { return x * ll(x - 1) /
                2; };

```

```
REP(v, n) for(int u : graph[v])
    ys4 += (ssize(graph[v]) - 1) * choose2(ssize(
        graph[u]) - 1);
paths4 = -(4 * rectangles4 + 2 * ps4 + 3 *
    triangles3);
REP(v, n) {
    int x = 0;
    for(int u : graph[v]) {
        x += ssize(graph[u]) - 1;
        paths4 -= choose2(ssize(graph[u]) - 1);
    }
    paths4 += choose2(x);
}
REP(v, n) {
    int s = ssize(graph[v]);
    stars3 += s * ll(s - 1) * ll(s - 2);
    stars4 += s * ll(s - 1) * ll(s - 2) * __int128_t(
        s - 3);
}
stars3 /= 6;
stars4 /= 24;
};
```

Flowy i matchingi (6)

blossom

#0c4c58
Jeden rabin powie $\mathcal{O}(nm)$, drugi rabin powie, że to nawet nie jest $\mathcal{O}(n^3)$. W grafie nie może być pętelek. Funkcja zwraca match'a, tzn $\text{match}[v] == -1$ albo z kim jest sparowany v . Rozmiar matchingu to $\frac{1}{2} \sum_v \text{int}(\text{match}[v] != -1)$.

```
vi blossom(V<vi> graph) {
    int n = ssize(graph), timer = -1;
    REP(v, n)
        for(int u : graph[v])
            assert(v != u);
    vi match(n, -1), label(n), parent(n), orig(n), aux(n, -1), q;
    auto lca = [&](int x, int y) {
        for(++timer; ; swap(x, y)) {
            if(x == -1)
                continue;
            if(aux[x] == timer)
                return x;
            aux[x] = timer;
            x = (match[x] == -1 ? -1 : orig[parent[match[x]]]);
        }
    };
    auto blossom = [&](int v, int w, int a) {
        while(orig[v] != a) {
            parent[v] = w;
            w = match[v];
            if(label[w] == 1) {
                label[w] = 0;
                q.eb(w);
            }
            orig[v] = orig[w] = a;
            v = parent[w];
        }
    };
    auto augment = [&](int v) {
        while(v != -1) {
            int pv = parent[v], nv = match[pv];
            match[v] = pv;
            match[pv] = v;
            v = nv;
        }
    };
    auto bfs = [&](int root) {
        fill(all(label), -1);
        iota(all(orig), 0);
        label[root] = 0;
    };
```

```
q = {root};
REP(i, ssize(q)) {
    int v = q[i];
    for(int x : graph[v])
        if(label[x] == -1) {
            label[x] = 1;
            parent[x] = v;
            if(match[x] == -1) {
                augment(x);
                return 1;
            }
            label[match[x]] = 0;
            q.eb(match[x]);
        }
        else if(label[x] == 0 and orig[v] != orig[x]) {
            int a = lca(orig[v], orig[x]);
            blossom(x, v, a);
            blossom(v, x, a);
        }
    }
    return 0;
};
REP(i, n)
    if(match[i] == -1)
        bfs(i);
return match;
```

dinic

#da1c73
 $\mathcal{O}(V^2E)$ Dinic bez skalowania. funkcja `get_flow`ing() zwraca dla każdej oryginalnej krawędzi ile przez nią leci.

```
struct Dinic {
    using T = int;
    struct Edge {
        int v, u;
        T flow, cap;
    };
    int n;
    V<vi> graph;
    V<Edge> edges;
    Dinic(int N) : n(N), graph(n) {}
    void add_edge(int v, int u, T cap) {
        debug(v, u, cap);
        int e = ssize(edges);
        graph[v].eb(e);
        graph[u].eb(e + 1);
        edges.eb(v, u, 0, cap);
        edges.eb(u, v, 0, 0);
    }
    vi dist;
    bool bfs(int source, int sink) {
        dist.assign(n, 0);
        dist[source] = 1;
        deque<int> que = {source};
        while(ssize(que) and dist[sink] == 0) {
            int v = que.front();
            que.pop_front();
            for(int e : graph[v])
                if(edges[e].flow != edges[e].cap and dist[edges[e].u] == 0) {
                    dist[edges[e].u] = dist[v] + 1;
                    que.eb(edges[e].u);
                }
        }
        return dist[sink] != 0;
    }
    vi ended_at;
    T dfs(int v, int sink, T flow = numeric_limits<T>::max()) {
        if(flow == 0 or v == sink)
            return flow;
        for(; ended_at[v] != ssize(graph[v]); ++ended_at[v]) {
            Edge &e = edges[graph[v][ended_at[v]]];
```

```
            if(dist[v] + 1 == dist[e.u])
                if(T pushed = dfs(e.u, sink, min(flow, e.cap - e.flow))) {
                    e.flow += pushed;
                    edges[graph[v][ended_at[v]] ^ 1].flow -= pushed;
                    return pushed;
                }
            return 0;
        }
    }
    T operator()(int source, int sink) {
        T answer = 0;
        while(bfs(source, sink)) {
            ended_at.assign(n, 0);
            while(T pushed = dfs(source, sink))
                answer += pushed;
        }
        return answer;
    }
    map<pii, T> get_flowing() {
        map<pii, T> ret;
        REP(v, n)
            for(int i : graph[v]) {
                if(i % 2) // considering only original edges
                    continue;
                Edge &e = edges[i];
                ret[pair(v, e.u)] += e.flow;
            }
        return ret;
    }
};
```

gomory-hu

#8cbc22, includes: dinic
 $\mathcal{O}(n^2 + n \cdot \text{dinic}(n, m))$, zwraca min cięcie między każdą parą wierzchołków w nieskierowanym ważonym grafie o nieujemnych wagach. `gomory_hu(n, edges)[s][t]` == min cut (s, t)

```
pair<Dinic::T, V<bool>> get_min_cut(Dinic &dinic, int s, int t) {
    for(Dinic::Edge &e : dinic.edges)
        e.flow = 0;
    Dinic::T flow = dinic(s, t);
    V<bool> cut(dinic.n);
    REP(v, dinic.n)
        cut[v] = bool(dinic.dist[v]);
    return {flow, cut};
}
V<V<Dinic::T>> get_gomory_hu(int n, V<tuple<int, int, Dinic::T>> edges) {
    Dinic dinic(n);
    for(auto [v, u, cap] : edges) {
        dinic.add_edge(v, u, cap);
        dinic.add_edge(u, v, cap);
    }
    using T = Dinic::T;
    V<V<pair<int, T>>> tree(n);
    vi par(n, 0);
    FOR(v, 1, n - 1) {
        auto [flow, cut] = get_min_cut(dinic, v, par[v]);
        FOR(u, v + 1, n - 1)
            if(cut[u] == cut[v] and par[u] == par[v])
                par[u] = v;
        tree[v].eb(par[v], flow);
        tree[par[v]].eb(v, flow);
    }
    T inf = numeric_limits<T>::max();
    V ret(n, V(n, inf));
    REP(source, n) {
        function<void (int, int, T)> dfs = [&](int v, int p, T mn) {
            ret[source][v] = mn;
            for(auto [u, flow] : tree[v])
                if(u != p)
                    dfs(u, v, min(mn, flow));
        };
```

```
    };
    dfs(source, -1, inf);
}
return ret;
}
```

hopcroft-karp

#489276
 $\mathcal{O}(m\sqrt{n})$ Hopcroft-Karp do liczenia matchingu. Przydaje się głównie w aproksymacji, ponieważ po k iteracjach gwarantuje matching o rozmiarze przynajmniej $k/(k+1)$ ·best matching. Wierzchołki grafu muszą być podzielone na warstwy $[0, n0)$ oraz $[n0, n0 + n1)$. Zwraca rozmiar matchingu oraz przypisanie (lub -1, gdy nie jest zmatchowane).

```
pair<int, vi> hopcroft_karp(V<vi> graph, int n0, int n1) {
    assert(n0 + n1 == ssize(graph));
    REP(v, n0 + n1)
        for(int u : graph[v])
            assert((v < n0) != (u < n0));
    vi matched_with(n0 + n1, -1), dist(n0 + 1);
    constexpr int inf = int(1e9);
    vi manual_queue(n0 + 1);
    auto bfs = [&] {
        int head = 0, tail = -1;
        fill(all(dist), inf);
        REP(v, n0)
            if(matched_with[v] == -1) {
                dist[1 + v] = 0;
                manual_queue[++tail] = v;
            }
        while(head <= tail) {
            int v = manual_queue[head++];
            if(dist[1 + v] < dist[0])
                for(int u : graph[v])
                    if(dist[1 + matched_with[u]] == inf) {
                        dist[1 + matched_with[u]] = dist[1 + v] + 1;
                        manual_queue[++tail] = matched_with[u];
                    }
        }
        return dist[0] != inf;
    };
    function<bool (int)> dfs = [&](int v) {
        if(v == -1)
            return true;
        for(auto u : graph[v])
            if(dist[1 + matched_with[u]] == dist[1 + v] + 1)
                {
                    if(dfs(matched_with[u])) {
                        matched_with[v] = u;
                        matched_with[u] = v;
                        return true;
                    }
                }
        dist[1 + v] = inf;
        return false;
    };
    int answer = 0;
    for(int iter = 0; bfs(); ++iter)
        REP(v, n0)
            if(matched_with[v] == -1 and dfs(v))
                ++answer;
    return {answer, matched_with};
}
```

hungarian

#9a79f8
 $\mathcal{O}(n_0^2 \cdot n_1)$, dla macierzy wag (mogą być ujemne) między dwoma warstwami o rozmiarach $n0$ oraz $n1$ ($n0 \leq n1$) wyznacza minimalną sumę wag skojarzenia pełnego. Zwraca sumę wag oraz matching.

```
pair<ll, vi> hungarian(V<vi> a) {
    if(a.empty())
        return {0, {}};
```



```

int n0 = ssize(a) + 1, n1 = ssize(a[0]) + 1;
assert(n0 <= n1);
vi p(n1), ans(n0 - 1);
vll u(n0), v(n1);
FOR(i, 1, n0 - 1) {
    p[0] = i;
    int j0 = 0;
    vll dist(n1, numeric_limits<ll>::max());
    vi pre(n1, -1);
    V<bool> done(n1 + 1);
    do {
        done[j0] = true;
        int i0 = p[j0], j1 = -1;
        ll delta = numeric_limits<ll>::max();
        FOR(j, 1, n1 - 1)
            if(!done[j]) {
                auto cur = a[i0 - 1][j - 1] - u[i0] - v[j];
                if(cur < dist[j])
                    dist[j] = cur, pre[j] = j0;
                if(dist[j] < delta)
                    delta = dist[j], j1 = j;
            }
        REP(j, n1) {
            if(done[j])
                u[p[j]] += delta, v[j] -= delta;
            else
                dist[j] -= delta;
        }
        j0 = j1;
    } while(p[j0]);
    while(j0) {
        int j1 = pre[j0];
        p[j0] = p[j1], j0 = j1;
    }
}
FOR(j, 1, n1 - 1)
    if(p[j])
        ans[p[j] - 1] = j - 1;
return {-v[0], ans};
}

```

konig-theorem

#c05211, includes: matching

$\mathcal{O}(n + \text{matching}(n, m))$ wyznaczanie w grafie dwudzielnym kolejno minimalnego pokrycia krawędziowego (PK), maksymalnego zbioru niezależnych wierzchołków (NW), minimalnego pokrycia wierzchołkowego (PW) korzystając z maksymalnego zbioru niezależnych krawędzi (NK) (tak zwany matching). Z tw. Koniga zachodzi $|NK|=n-|PK|=n-|NW|=|PW|$.

```

// BEGIN HASH 320322
V<pii> get_min_edge_cover(V<vi> graph) {
    vi match = Matching(graph)().se;
    V<pii> ret;
    REP(v, ssize(match))
        if(match[v] != -1 and v < match[v])
            ret.pb(v, match[v]);
    else if(match[v] == -1 and not graph[v].empty())
        ret.pb(v, graph[v].front());
    return ret;
} // END HASH
// BEGIN HASH f215ab
array<vi, 2> get_coloring(V<vi> graph) {
    int n = ssize(graph);
    vi match = Matching(graph)().se;
    vi color(n, -1);
    function<void (int)> dfs = [&](int v) {
        color[v] = 0;
        for(int u : graph[v])
            if(color[u] == -1) {
                color[u] = true;
                dfs(match[u]);
            }
    };
    REP(v, n)
        if(match[v] == -1)

```

```

        dfs(v);
    REP(v, n)
        if(color[v] == -1)
            dfs(v);
    array<vi, 2> groups;
    REP(v, n)
        groups[color[v]].pb(v);
    return groups;
}
vi get_max_independent_set(V<vi> graph) {
    return get_coloring(graph)[0];
}
vi get_min_vertex_cover(V<vi> graph) {
    return get_coloring(graph)[1];
} // END HASH

```

matching

#d28b80

Średnio około $\mathcal{O}(n \log n)$, najgorzej $\mathcal{O}(n^2)$. Wierzchołki grafu nie muszą być ładnie podzielone na dwa przedziały, musi być po prostu dwudzielny. Na przykład auto [match_size, match] = Matching(graph)();

```

struct Matching {
    V<vi> &adj;
    vi mat, vis;
    int t = 0, ans = 0;
    bool mat_dfs(int v) {
        vis[v] = t;
        for(int u : adj[v])
            if(mat[u] == -1) {
                mat[u] = v;
                mat[v] = u;
                return true;
            }
        for(int u : adj[v])
            if(vis[mat[u]] != t && mat_dfs(mat[u])) {
                mat[u] = v;
                mat[v] = u;
                return true;
            }
        return false;
    }
    Matching(V<vi> &adj) : adj(_adj) {
        mat = vis = vi(ssize(adj), -1);
    }
    pair<int, vi> operator()() {
        int d = -1;
        while(d != 0) {
            d = 0, ++t;
            REP(v, ssize(adj))
                if(mat[v] == -1)
                    d += mat_dfs(v);
            ans += d;
        }
        return {ans, mat};
    }
};

```

mcmf-dijkstra

#4d7e5d

$\mathcal{O}(VE + |flow|E \log V)$, Min-cost max-flow. Można przepisać funkcję get_flow() z Dinic'a. Kiedy wie się coś więcej o początkowym grafie np. że jest DAG-iem lub że ma tylko nieujemne wagi krawędzi, można napisać własne calc_init_dist by usunąć VE ze złożoności. Jeżeli $E = \mathcal{O}(V^2)$, to może być lepiej napisać samemu kwadratową dijkstrę.

```

struct MCMF {
    struct Edge {
        int v, u, flow, cap;
        ll cost;
    };
    friend ostream& operator<<(ostream &os, Edge &e) {
        return os << vll{e.v, e.u, e.flow, e.cap, e.cost};
    };
};

```

```

};
int n;
C ll inf_LL = 1e18;
C int inf_int = 1e9;
V<vi> graph;
V<Edge> edges;
vll init_dist;
MCMF(int N) : n(N), graph(n), init_dist(n) {}
void add_edge(int v, int u, int cap, ll cost) {
    int e = ssize(edges);
    graph[v].pb(e);
    graph[u].pb(e + 1);
    edges.pb(v, u, 0, cap, cost);
    edges.pb(u, v, 0, 0, -cost);
}
void calc_init_dist(int source) {
    fill(all(init_dist), inf_LL);
    V<bool> inside(n);
    inside[source] = true;
    deque<int> que = {source};
    init_dist[source] = 0;
    while(ssize(que)) {
        int v = que.front();
        que.pop_front();
        inside[v] = false;
        for(int i : graph[v]) {
            Edge &e = edges[i];
            if(e.flow < e.cap and init_dist[v] + e.cost < init_dist[e.u]) {
                init_dist[e.u] = init_dist[v] + e.cost;
                if(not inside[e.u]) {
                    inside[e.u] = true;
                    que.pb(e.u);
                }
            }
        }
    }
}
pair<int, ll> augment(int source, int sink) {
    V<bool> vis(n);
    vi from(n, -1);
    vll dist(n, inf_LL);
    priority_queue<pair<ll, int>, V<pair<ll, int>>, greater<>> que;
    que.emplace(0, source);
    dist[source] = 0;
    while(ssize(que)) {
        auto [d, v] = que.top();
        que.pop();
        if(vis[v]) continue;
        vis[v] = true;
        for(int i : graph[v]) {
            Edge &e = edges[i];
            ll new_dist = d + e.cost + init_dist[v];
            if(not vis[e.u] and e.flow != e.cap and new_dist < dist[e.u]) {
                dist[e.u] = new_dist;
                from[e.u] = i;
                que.emplace(new_dist - init_dist[e.u], e.u);
            }
        }
    }
    if(not vis[sink])
        return {0, 0};
    int flow = inf_int, e = from[sink];
    while(e != -1) {
        chmin(flow, edges[e].cap - edges[e].flow);
        e = from[edges[e].v];
    }
    e = from[sink];
    while(e != -1) {
        edges[e].flow += flow;
        edges[e ^ 1].flow -= flow;
        e = from[edges[e].v];
    }
    init_dist.swap(dist);
}

```

```

    return {flow, flow * init_dist[sink]};
}
pair<int, ll> operator()(int source, int sink) {
    calc_init_dist(source);
    int flow = 0;
    ll cost = 0;
    pair<int, ll> got;
    do {
        got = augment(source, sink);
        flow += got.fi;
        cost += got.se;
    } while(got.fi);
    return {flow, cost};
}
};

```

mcmf-spfa

#43da52

$\mathcal{O}(idk)$, Min-cost max-flow z SPFA. Można przepisać funkcję get_flow() z Dinic'a.

```

struct MCMF {
    struct Edge {
        int v, u, flow, cap;
        ll cost;
    };
    friend ostream& operator<<(ostream &os, Edge &e) {
        return os << vll{e.v, e.u, e.flow, e.cap, e.cost};
    };
};
int n;
C ll inf_LL = 1e18;
C int inf_int = 1e9;
V<vi> graph;
V<Edge> edges;
MCMF(int N) : n(N), graph(n) {}
void add_edge(int v, int u, int cap, ll cost) {
    int e = ssize(edges);
    graph[v].pb(e);
    graph[u].pb(e + 1);
    edges.pb(v, u, 0, cap, cost);
    edges.pb(u, v, 0, 0, -cost);
}
pair<int, ll> augment(int source, int sink) {
    vll dist(n, inf_LL);
    vi from(n, -1);
    dist[source] = 0;
    deque<int> que = {source};
    V<bool> inside(n);
    inside[source] = true;
    while(ssize(que)) {
        int v = que.front();
        inside[v] = false;
        que.pop_front();
        for(int i : graph[v]) {
            Edge &e = edges[i];
            if(e.flow != e.cap and dist[e.u] > dist[v] + e.cost) {
                dist[e.u] = dist[v] + e.cost;
                from[e.u] = i;
                if(not inside[e.u]) {
                    inside[e.u] = true;
                    que.pb(e.u);
                }
            }
        }
    }
    if(from[sink] == -1)
        return {0, 0};
    int flow = inf_int, e = from[sink];
    while(e != -1) {
        chmin(flow, edges[e].cap - edges[e].flow);
        e = from[edges[e].v];
    }
    e = from[sink];
    while(e != -1) {

```

```

    edges[e].flow += flow;
    edges[e ^ 1].flow -= flow;
    e = from[edges[e].v];
}
return {flow, flow * dist[sink]};
}
pair<int, ll> operator()(int source, int sink) {
    int flow = 0;
    ll cost = 0;
    pair<int, ll> got;
    do {
        got = augment(source, sink);
        flow += got.fi;
        cost += got.se;
    } while(got.fi);
    return {flow, cost};
}
};

```

weighted-blossom

#85551c

$O(N^3)$ (but fast in practice) Taken from:
<https://judge.yosupo.jp/submission/218005/pdfcompile>,
 weighted_matching::init(n), weighted_matching::add_edge(a, b, c) $V < \pi i >$
 temp, weighted_matching::solve(temp).fi

```

#define pii pii
namespace weighted_matching{
    C int INF = (int)1e9 + 7;
    C int MAXN = 1050; //double of possible N
    struct E{
        int x, y, w;
    };
    int n, m;
    E G[MAXN][MAXN];
    int lab[MAXN], match[MAXN], slack[MAXN], st[MAXN], pa[
        MAXN], flo_from[MAXN][MAXN], S[MAXN], vis[MAXN];
    vi flo[MAXN];
    queue<int> Q;
    void init(int _n) {
        n = _n;
        for(int x = 1; x <= n; ++x)
            for(int y = 1; y <= n; ++y)
                G[x][y] = E{x, y, 0};
    }
    void add_edge(int x, int y, int w) {
        G[x][y].w = G[y][x].w = w;
    }
    int e_delta(E e) {
        return lab[e.x] + lab[e.y] - G[e.x][e.y].w * 2;
    }
    void update_slack(int u, int x) {
        if(!slack[x] || e_delta(G[u][x]) < e_delta(G[slack[x]
            ][x]))
            slack[x] = u;
    }
    void set_slack(int x) {
        slack[x] = 0;
        for(int u = 1; u <= n; ++u)
            if(G[u][x].w > 0 && st[u] != x && S[st[u]] == 0)
                update_slack(u, x);
    }
    void q_push(int x) {
        if(x <= n) Q.push(x);
        else for(int i = 0; i < (int)flo[x].size(); ++i)
            q_push(flo[x][i]);
    }
    void set_st(int x, int b) {
        st[x] = b;
        if(x > n) for(int i = 0; i < (int)flo[x].size(); ++i)
            set_st(flo[x][i], b);
    }
    int get_pr(int b, int xr) {
        int pr = find(all(flo[b]), xr) - flo[b].begin();
        if(pr & 1) {
            reverse(flo[b].begin() + 1, flo[b].end());

```

```

        return (int)flo[b].size() - pr;
    }
    else return pr;
}
void set_match(int x, int y) {
    match[x] = G[x][y].y;
    if(x <= n) return;
    E e = G[x][y];
    int xr = flo_from[x][e.x], pr = get_pr(x, xr);
    for(int i = 0; i < pr; ++i) set_match(flo[x][i], flo[
        x][i^1]);
    set_match(xr, y);
    rotate(flo[x].begin(), flo[x].begin() + pr, flo[x].
        end());
}
void augment(int x, int y) {
    while(1) {
        int ny = st[match[x]];
        set_match(x, y);
        if(!ny) return;
        set_match(ny, st[pa[ny]]);
        x = st[pa[ny]], y = ny;
    }
}
int get_lca(int x, int y) {
    static int t = 0;
    for(++t; x || y; swap(x, y)) {
        if(x == 0) continue;
        if(vis[x] == t) return x;
        vis[x] = t;
        x = st[match[x]];
        if(x) x = st[pa[x]];
    }
    return 0;
}
void add_blossom(int x, int l, int y) {
    int b = n + 1;
    while(b <= m && st[b]) ++b;
    if(b > m) ++m;
    lab[b] = 0, S[b] = 0;
    match[b] = match[l];
    flo[b].clear();
    flo[b].pb(l);
    for(int u = x, v; u != l; u = st[pa[v]])
        flo[b].pb(u), flo[b].pb(v = st[match[u]]), q_push(v
            );
    reverse(flo[b].begin() + 1, flo[b].end());
    for(int u = y, v; u != l; u = st[pa[v]])
        flo[b].pb(u), flo[b].pb(v = st[match[u]]), q_push(v
            );
    set_st(b, b);
    for(int i = 1; i <= m; ++i) G[b][i].w = G[i][b].w =
        0;
    for(int i = 1; i <= n; ++i) flo_from[b][i] = 0;
    for(int i = 0; i < (int)flo[b].size(); ++i) {
        int us = flo[b][i];
        for(int u = 1; u <= m; ++u)
            if(G[b][u].w == 0 || e_delta(G[us][u]) < e_delta(
                G[b][u]))
                G[b][u] = G[us][u], G[u][b] = G[u][us];
        for(int u = 1; u <= n; ++u)
            if(flo_from[us][u])
                flo_from[b][u] = us;
    }
    set_slack(b);
}
void expand_blossom(int b) {
    for(int i = 0; i < (int)flo[b].size(); ++i)
        set_st(flo[b][i], flo[b][i]);
    int xr = flo_from[b][G[b][pa[b]].x], pr = get_pr(b,
        xr);
    for(int i = 0; i < pr; i += 2) {
        int xs = flo[b][i], xns = flo[b][i + 1];
        pa[xs] = G[xns][xs].x;
        S[xs] = 1, S[xns] = 0;
        slack[xs] = 0, set_slack(xns);

```

```

        q_push(xns);
    }
    S[xr] = 1, pa[xr] = pa[b];
    for(int i = pr + 1; i < (int)flo[b].size(); ++i) {
        int xs = flo[b][i];
        S[xs] = -1, set_slack(xs);
    }
    st[b] = 0;
}
bool on_found_edge(E e) {
    int x = st[e.x], y = st[e.y];
    if(S[y] == -1) {
        pa[y] = e.x, S[y] = 1;
        int ny = st[match[y]];
        slack[y] = slack[ny] = 0;
        S[ny] = 0, q_push(ny);
    }
    else if(S[y] == 0) {
        int l = get_lca(x, y);
        if(!l) return augment(x, y), augment(y, x), true;
        else add_blossom(x, l, y);
    }
    return false;
}
bool matching() {
    fill(S + 1, S + m + 1, -1);
    fill(slack + 1, slack + m + 1, 0);
    Q = queue<int>();
    for(int x = 1; x <= m; ++x)
        if(st[x] == x && !match[x]) pa[x] = 0, S[x] = 0,
            q_push(x);
    if(Q.empty()) return false;
    while(1) {
        while(Q.size()) {
            int x = Q.front(); Q.pop();
            if(S[st[x]] == 1) continue;
            for(int y = 1; y <= n; ++y) {
                if(G[x][y].w > 0 && st[x] != st[y]) {
                    if(e_delta(G[x][y]) == 0) {
                        if(on_found_edge(G[x][y])) return true;
                    }
                    else update_slack(x, st[y]);
                }
            }
        }
        int d = INF;
        for(int b = n + 1; b <= m; ++b)
            if(st[b] == b && S[b] == 1) chmin(d, lab[b] / 2);
        for(int x = 1; x <= m; ++x)
            if(st[x] == x && slack[x]) {
                if(S[x] == -1) chmin(d, e_delta(G[slack[x]][x])
                    );
                else if(S[x] == 0) chmin(d, e_delta(G[slack[x]
                    ][x]) / 2);
            }
        for(int x = 1; x <= n; ++x) {
            if(S[st[x]] == 0) {
                if(lab[x] <= d) return 0;
                lab[x] -= d;
            }
            else if(S[st[x]] == 1) lab[x] += d;
        }
        for(int b = n + 1; b <= m; ++b)
            if(st[b] == b) {
                if(S[st[b]] == 0) lab[b] += d * 2;
                else if(S[st[b]] == 1) lab[b] -= d * 2;
            }
        Q = queue<int>();
        for(int x = 1; x <= m; ++x)
            if(st[x] == x && slack[x] && st[slack[x]] != x &&
                e_delta(G[slack[x]][x]) == 0)
                if(on_found_edge(G[slack[x]][x])) return true;
        for(int b = n + 1; b <= m; ++b)
            if(st[b] == b && S[b] == 1 && lab[b] == 0)
                expand_blossom(b);
    }
}

```

```

    return false;
}
pair<ll, int> solve(V<pii> &ans) {
    fill(match + 1, match + n + 1, 0);
    m = n;
    int cnt = 0; ll sum = 0;
    for(int u = 0; u <= n; ++u) st[u] = u, flo[u].clear()
        ;
    int mx = 0;
    for(int x = 1; x <= n; ++x)
        for(int y = 1; y <= n; ++y){
            flo_from[x][y] = (x == y ? x : 0);
            chmax(mx, G[x][y].w);
        }
    for(int x = 1; x <= n; ++x) lab[x] = mx;
    while(matching()) ++cnt;
    for(int x = 1; x <= n; ++x)
        if(match[x] && match[x] < x) {
            sum += G[x][match[x]].w;
            ans.pb({x, G[x][match[x]].y});
        }
    return {sum, cnt};
}
}

```

Geometria (7)

advanced-complex

#bcc8b5, includes: point

Większość nie działa dla intów.

```

constexpr D pi = acosl(-1);
// nachylenie k-> y = kx + m
D slope(P a, P b) { return tan(arg(b - a)); }
// rzut p na ab
P project(P p, P a, P b) {
    return a + (b - a) * dot(p - a, b - a) / norm(a - b);
}
// odbicie p względem ab
P reflect(P p, P a, P b) {
    return a + conj((p - a) / (b - a)) * (b - a);
}
// obrot a względem p o theta radianow
P rotate(P a, P p, D theta) {
    return (a - p) * polar(1.0L, theta) + p;
}
// kat ABC, w radianach z przedziału [0..pi]
D angle(P a, P b, P c) {
    return abs(remainder(arg(a - b) - arg(c - b), 2.0 *
        pi));
}
// szybkie przeciecie prostych, nie działa dla
rownoległych
P intersection(P a, P b, P p, P q) {
    D c1 = cross(p - a, b - a), c2 = cross(q - a, b - a);
    return (c1 * q - c2 * p) / (c1 - c2);
}
// check czy sa rownolegle
bool is_parallel(P a, P b, P p, P q) {
    P c = (a - b) / (p - q); return equal(c, conj(c));
}
// check czy sa prostopadłe
bool is_perpendicular(P a, P b, P p, P q) {
    P c = (a - b) / (p - q); return equal(c, -conj(c));
}
// zwraca takie q, ze (p, q) jest rownolegle do (a, b)
P parallel(P a, P b, P p) {
    return p + a - b;
}
// zwraca takie q, ze (p, q) jest prostopadłe do (a, b)
P perpendicular(P a, P b, P p) {
    return reflect(p, a, b);
}
// przeciecie srodkowych trojkata
P centro(P a, P b, P c) {

```

```
    return (a + b + c) / 3.0L;
}
```

angle-sort

#032856, includes: point
 $\mathcal{O}(n \log n)$, zwraca wektory P posortowane kątowno zgodnie z ruchem wskazówek zegara od najbliższego kątowno do wektora $(0, 1)$ włącznie. Aby posortować po argumente (kącie) swapujemy x, y , używamy angle-sort i ponownie swapujemy x, y . Zakłada że nie ma punktu $(0, 0)$ na wejściu.

```
V<P> angle_sort(V<P> t) {
    for(P p : t) assert(not equal(p, P(0, 0)));
    auto it = partition(all(t), [](P a){ return P(0, 0) < a; });
    auto cmp = [&](P a, P b) {
        return sign(cross(a, b)) == -1;
    };
    sort(t.begin(), it, cmp);
    sort(it, t.end(), cmp);
    return t;
}
```

angle180-intervals

#9e4d50, includes: angle-sort
 $\mathcal{O}(n)$, ZAKŁADA że punkty są posortowane kątowno. Zwraca n par $[i, r]$, gdzie r jest maksymalnym cyklicznie indeksem, że wszystkie punkty w tym cyklicznym przedziale są ściśle „po prawej” stronie wektora $(0, 0) - in[i]$, albo są na tej półprostej.

```
V<pii> angle180_intervals(V<P> in) {
    // in must be sorted by angle
    int n = ssize(in);
    vi nxt(n);
    iota(all(nxt), 1);
    int r = nxt[n - 1] = 0;
    V<pii> ret(n);
    REP(l, n) {
        if(nxt[r] == l) r = nxt[r];
        auto good = [&](int i) {
            auto c = cross(in[l], in[i]);
            if(not equal(c, 0)) return c < 0;
            if((P(0, 0) < in[l]) != (P(0, 0) < in[i]))
                return false;
            return l < i;
        };
        while(nxt[r] != l and good(nxt[r]))
            r = nxt[r];
        ret[l] = {l, r};
    }
    return ret;
}
```

area

#7b2943, includes: point
Pole wielokąta, niekoniecznie wypukłego. W vectorze muszą być wierzchołki zgodnie z kierunkiem ruchu zegara. Jeśli D jest intem to może się psuć / 2. area(a, b, c) zwraca pole trójkąta o takich długościach boku.

```
D area(V<P> pts) {
    int n = ssize(pts);
    D ans = 0;
    REP(i, n) ans += cross(pts[i], pts[(i + 1) % n]);
    return fabs1(ans / 2);
}
D area(D a, D b, D c) {
    D p = (a + b + c) / 2;
    return sqrt1(p * (p - a) * (p - b) * (p - c));
}
```

circle-intersection

#eeabd8, includes: point

Przecięcia okręgu oraz prostej $ax + by + c = 0$ oraz przecięcia okręgu oraz okręgu. Gdy ssize(circle_circle(...)) == 3 to jest nieskończenie wiele rozwiązań.

```
// BEGIN HASH 4774d4
V<P> circle_line(D r, D a, D b, D c) {
    D len_ab = a * a + b * b,
      x0 = -a * c / len_ab,
      y0 = -b * c / len_ab,
      d = r * r - c * c / len_ab,
      mult = sqrt(d / len_ab);
    if(sign(d) < 0)
        return {};
    else if(sign(d) == 0)
        return {{x0, y0}};
    return {
        {x0 + b * mult, y0 - a * mult},
        {x0 - b * mult, y0 + a * mult}
    };
}
V<P> circle_line(D x, D y, D r, D a, D b, D c) {
    V<P> ret = circle_line(r, a, b, c + (a * x + b * y));
    for(P & p : ret) p = P(p.x() + x, p.y() + y);
    return ret;
} // END HASH
// BEGIN HASH c5d0a6
V<P> circle_circle(D x1, D y1, D r1, D x2, D y2, D r2)
{
    x2 -= x1;
    y2 -= y1;
    // now x1 = y1 = 0;
    if(sign(x2) == 0 and sign(y2) == 0) {
        if(equal(r1, r2))
            return {{0, 0}, {0, 0}, {0, 0}}; // inf points
        else
            return {};
    }
    auto vec = circle_line(r1, -2 * x2, -2 * y2,
        x2 * x2 + y2 * y2 + r1 * r1 - r2 * r2);
    for(P & p : vec)
        p += P(x1, y1);
    return vec;
} // END HASH
```

circle-tangents

#f03bcb, includes: point
 $\mathcal{O}(1)$, dla dwóch okręgów zwraca dwie styczne (wewnętrzne lub zewnętrzne, zależnie od wartości inner). Zwraca $1 + \text{sign}(\text{dist}(p_0, p_1) - (\text{inside} ? r_0 + r_1 : \text{abs}(r_0 - r_1)))$ rozwiązań, albo 0 gdy $p_1 = p_2$. Działa gdy jakiś promień jest 0 – przydatne do policzenia stycznej punktu do okręgu.

```
V<pair<P, P>> circle_tangents(P p1, D r1, P p2, D r2,
    bool inner) {
    if(inner) r2 *= -1;
    P d = p2 - p1;
    D dr = r1 - r2, d2 = dot(d, d), h2 = d2 - dr * dr;
    if(equal(d2, 0) or sign(h2) < 0)
        return {};
    V<pair<P, P>> ret;
    for(D sign : {-1, 1}) {
        P v = (d * dr + P(-d.y(), d.x()) * sqrt(max(D(0), h2)) * sign) / d2;
        ret.eb(p1 + v * r1, p2 + v * r2);
    }
    ret.resize(1 + (sign(h2) > 0));
    return ret;
}
```

closest-pair

#16e742, includes: point
 $\mathcal{O}(n \log n)$, zakłada ssize(in) > 1.

```
pair<P, P> closest_pair(V<P> in) {
    set<P> s;
```

```
    sort(all(in), [](P a, P b) { return a.y() < b.y(); });
    ;
    pair<D, pair<P, P>> ret(1e18, {P(), P()});
    int j = 0;
    for (P p : in) {
        P d(1 + sqrt(ret.fi), 0);
        while (in[j].y() <= p.y() - d.x()) s.erase(in[j++] );
        auto lo = s.lower_bound(p - d), hi = s.upper_bound(
            p + d);
        for ( ; lo != hi; ++lo)
            chmin(ret, pair(pow(dist(*lo, p), 2), pair(*lo, p
                )));
        s.insert(p);
    }
    return ret.se;
}
```

convex-gen

#7f3cac, includes: point, angle-sort, headers/gen
Generatorka wielokątów wypukłych. Zwraca wielokąt z co najmniej n - PROC punktami w zakresie $[-\text{range}, \text{range}]$. Jeśli $n (n > 2)$ jest okoto $\frac{2}{3}$, to powinno chodzić $\mathcal{O}(n \log n)$. Dla większych n może nie dać rady. Ostatni punkt jest zawsze w $(0, 0)$ - można dodać przesunięcie o wektor dla pełnej losowości.

```
vi num_split(int value, int n) {
    vi v(n, value);
    REP(i, n - 1)
        v[i] = rd(0, value);
    sort(all(v));
    adjacent_difference(all(v), v.begin());
    return v;
}
vi capped_zero_split(int cap, int n) {
    int m = rd(1, n - 1);
    auto lf = num_split(cap, m);
    auto rg = num_split(cap, n - m);
    for (int i : rg)
        lf.eb(-i);
    return lf;
}
V<P> gen_convex_polygon(int n, int range, bool
    strictly_convex = false) {
    assert(n > 2);
    V<P> t;
    C double PROC = 0.9;
    do {
        t.clear();
        auto dx = capped_zero_split(range, n);
        auto dy = capped_zero_split(range, n);
        shuffle(all(dx), rng);
        REP (i, n)
            if (dx[i] || dy[i])
                t.eb(dx[i], dy[i]);
        t = angle_sort(t);
        if (strictly_convex) {
            V<P> nt(1, t[0]);
            FOR (i, 1, ssize(t) - 1) {
                if (!sign(cross(t[i], nt.back())))
                    nt.back() += t[i];
                else
                    nt.eb(t[i]);
            }
            while (!nt.empty() && !sign(cross(nt.back(), nt
                [0]))) {
                nt[0] += nt.back();
                nt.pop_back();
            }
            t = nt;
        }
    } while (ssize(t) < n * PROC);
    partial_sum(all(t), t.begin());
    return t;
}
```

convex-hull-online

#c74f71
 $\mathcal{O}(\log n)$ na każdą operację dodania, Wyznacza górną otoczkę wypukłą online.

```
using P = pii;
ll operator*(P l, P r) {
    return l.fi * ll(r.se) - l.se * ll(r.fi);
}
P operator-(P l, P r) {
    return {l.fi - r.fi, l.se - r.se};
}
int sign(ll x) {
    return x > 0 ? 1 : x < 0 ? -1 : 0;
}
int dir(P a, P b, P c) {
    return sign((b - a) * (c - b));
}
struct UpperConvexHull {
    set<P> hull;
    void add_point(P p) {
        if(hull.empty()) {
            hull = {p};
            return;
        }
        auto it = hull.lower_bound(p);
        if(*hull.begin() < p and p < *prev(hull.end())) {
            assert(it != hull.end() and it != hull.begin());
            if(dir(*prev(it), p, *it) >= 0)
                return;
        }
        it = hull.emplace(p).fi;
        auto have_to_rm = [&](auto iter) {
            if(iter == hull.end() or next(iter) == hull.end()
                or iter == hull.begin())
                return false;
            return dir(*prev(iter), *iter, *next(iter)) >= 0;
        };
        while(have_to_rm(next(it)))
            it = prev(hull.erase(next(it)));
        while(it != hull.begin() and have_to_rm(prev(it)))
            it = hull.erase(prev(it));
    }
};
```

convex-hull

#31845a, includes: point
 $\mathcal{O}(n \log n)$, top_bot_hull zwraca osobno górę i dół, hull zwraca punkty na otoczek clockwise gdzie pierwszy jest najbardziej lewym.

```
array<V<P>, 2> top_bot_hull(V<P> in) {
    sort(all(in));
    array<V<P>, 2> ret;
    REP(d, 2) {
        for(auto p : in) {
            while(ssize(ret[d]) > 1 and dir(ret[d].end()[-2],
                ret[d].back(), p) >= 0)
                ret[d].pop_back();
            ret[d].eb(p);
        }
        reverse(all(in));
    }
    return ret;
}
V<P> hull(V<P> in) {
    if(ssize(in) <= 1) return in;
    auto ret = top_bot_hull(in);
    REP(d, 2) ret[d].pop_back();
    ret[0].insert(ret[0].end(), ret[1].begin(), ret[1].
        end());
    return ret[0];
}
```

delaunay-triangulation

#26839a

$\mathcal{O}(n \log n)$, zwraca zbiór trójkątów sumujący się do otoczki wypukłej, gdzie każdy trójkąt nie zawiera żadnego innego punktu wewnątrz okręgu opisanego (czyli maksymalizuje minimalny kąt trójkątów). Zakłada brak identycznych punktów. W przypadku współliniowości wszystkich punktów zwraca pusty V. Zwraca V rozmiaru $3X$, gdzie wartości $3i$, $3i+1$, $3i+2$ tworzą counter-clockwise trójkąt. Wśród sąsiadów zawsze jest najbliższy wierzchołek. Euclidean min. spanning tree to podzbiór krawędzi.

```
using PI = pii;
typedef struct Quad* Q;
PI distinct(INT_MAX, INT_MAX);
ll dist2(PI p) {
    return p.fi * ll(p.fi)
        + p.se * ll(p.se);
}
ll operator*(PI a, PI b) {
    return a.fi * ll(b.se)
        - a.se * ll(b.fi);
}
PI operator-(PI a, PI b) {
    return {a.fi - b.fi,
        a.se - b.se};
}
ll cross(PI a, PI b, PI c) { return (a - b) * (b - c);
}
struct Quad {
    Q rot, o = nullptr;
    PI p = distinct;
    bool mark = false;
    Quad(Q _rot) : rot(_rot) {}
    PI& F() { return r()->p; }
    Q& r() { return rot->rot; }
    Q prev() { return rot->o->rot; }
    Q next() { return r()->prev(); }
} *H; // it's safe to use in multitests
V<Q> to_dealloc;
bool is_p_inside_circle(PI p, PI a, PI b, PI c) {
    __int128_t p2 = dist2(p), a2 = dist2(a)-p2,
        b2 = dist2(b)-p2, c2 = dist2(c)-p2;
    return cross(p,a,b) * c2 + cross(p,b,c) * a2 + cross(
        p,c,a) * b2 > 0;
}
Q makeEdge(PI orig, PI dest) {
    Q r = H;
    if (!r) {
        r = new Quad(new Quad(new Quad(new Quad(0))));
        Q del = r;
        REP(i, 4) {
            to_dealloc.eb(del);
            del = del->rot;
        }
    }
    H = r->o; r->r()->r() = r;
    REP(i, 4) {
        r = r->rot, r->p = distinct;
        r->o = i & 1 ? r : r->r();
    }
    r->p = orig; r->F() = dest;
    return r;
}
void splice(Q a, Q b) {
    swap(a->o->rot->o, b->o->rot->o);
    swap(a->o, b->o);
}
Q connect(Q a, Q b) {
    Q q = makeEdge(a->F(), b->p);
    splice(q, a->next());
    splice(q->r(), b);
    return q;
}
pair<Q, Q> rec(C V<PI>& s) {
    if (ssize(s) <= 3) {
        Q a = makeEdge(s[0], s[1]);
        Q b = makeEdge(s[1], s.back());
        if (ssize(s) == 2) return {a, a->r()};
    }
    splice(a->r(), b);
    auto valid = [&](Q e, Q base) {
        return cross(e->F(), base->F(), base->p) > 0;
    };
    int half = ssize(s) / 2;
    auto [ra, A] = rec({s.begin(), s.end() - half});
    auto [B, rb] = rec({ssize(s) - half + s.begin(), s.
        end()});
    while ((cross(B->p, A->F(), A->p) < 0
        and (A = A->next()))
        or (cross(A->p, B->F(), B->p) > 0
        and (B = B->r()->o))) {}
    Q base = connect(B->r(), A);
    if (A->p == ra->p) ra = base->r();
    if (B->p == rb->p) rb = base;
    auto del = [&](Q init, function<Q (Q)> dir) {
        Q e = dir(init);
        if (valid(e, base))
            while (is_p_inside_circle(dir(e)->F(), base->F(),
                base->p, e->F())) {
                Q t = dir(e);
                splice(e, e->prev());
                splice(e->r(), e->r()->prev());
                e->o = H; H = e; e = t;
            }
        return e;
    };
    while(true) {
        Q LC = del(base->r(), [&](Q q) { return q->o; });
        Q RC = del(base, [&](Q q) { return q->prev(); });
        if (!valid(LC, base) and !valid(RC, base)) break;
        if (!valid(LC, base) or (valid(RC, base)
            and is_p_inside_circle(RC->F(), RC->p, LC->F
                (), LC->p)))
            base = connect(RC, base->r());
        else
            base = connect(base->r(), LC->r());
    }
    return {ra, rb};
}
V<PI> triangulate(V<PI> in) {
    sort(all(in));
    assert(unique(all(in)) == in.end());
    if (ssize(in) < 2) return {};
    Q e = rec(in).fi;
    V<Q> q = {e};
    int qi = 0;
    while (cross(e->o->F(), e->F(), e->p) < 0)
        e = e->o;
    auto add = [&] {
        Q c = e;
        do {
            c->mark = 1;
            in.eb(c->p);
            q.eb(c->r());
            c = c->next();
        } while (c != e);
    };
    add(); in.clear();
    while (qi < ssize(q))
        if (!(e = q[qi++]->mark) add());
    for (Q x : to_dealloc) delete x;
    to_dealloc.clear();
    return in;
}
}

furthest-pair
#f8538c, includes: convex-hull
 $\mathcal{O}(n)$  po poszczeniu otoczki, zakładam  $n \geq 2$ .
```

```
splice(a->r(), b);
auto side = cross(s[0], s[1], s[2]);
Q c = side ? connect(b, a) : 0;
return {side < 0 ? c->r() : a,
    side < 0 ? c : b->r()};
}
auto valid = [&](Q e, Q base) {
    return cross(e->F(), base->F(), base->p) > 0;
};
int half = ssize(s) / 2;
auto [ra, A] = rec({s.begin(), s.end() - half});
auto [B, rb] = rec({ssize(s) - half + s.begin(), s.
    end()});
while ((cross(B->p, A->F(), A->p) < 0
    and (A = A->next()))
    or (cross(A->p, B->F(), B->p) > 0
    and (B = B->r()->o))) {}
Q base = connect(B->r(), A);
if (A->p == ra->p) ra = base->r();
if (B->p == rb->p) rb = base;
auto del = [&](Q init, function<Q (Q)> dir) {
    Q e = dir(init);
    if (valid(e, base))
        while (is_p_inside_circle(dir(e)->F(), base->F(),
            base->p, e->F())) {
            Q t = dir(e);
            splice(e, e->prev());
            splice(e->r(), e->r()->prev());
            e->o = H; H = e; e = t;
        }
    return e;
};
while(true) {
    Q LC = del(base->r(), [&](Q q) { return q->o; });
    Q RC = del(base, [&](Q q) { return q->prev(); });
    if (!valid(LC, base) and !valid(RC, base)) break;
    if (!valid(LC, base) or (valid(RC, base)
        and is_p_inside_circle(RC->F(), RC->p, LC->F
            (), LC->p)))
        base = connect(RC, base->r());
    else
        base = connect(base->r(), LC->r());
}
return {ra, rb};
}
V<PI> triangulate(V<PI> in) {
    sort(all(in));
    assert(unique(all(in)) == in.end());
    if (ssize(in) < 2) return {};
    Q e = rec(in).fi;
    V<Q> q = {e};
    int qi = 0;
    while (cross(e->o->F(), e->F(), e->p) < 0)
        e = e->o;
    auto add = [&] {
        Q c = e;
        do {
            c->mark = 1;
            in.eb(c->p);
            q.eb(c->r());
            c = c->next();
        } while (c != e);
    };
    add(); in.clear();
    while (qi < ssize(q))
        if (!(e = q[qi++]->mark) add());
    for (Q x : to_dealloc) delete x;
    to_dealloc.clear();
    return in;
}
}

furthest-pair
#f8538c, includes: convex-hull
 $\mathcal{O}(n)$  po poszczeniu otoczki, zakładam  $n \geq 2$ .
```

```
pair<P, P> furthest_pair(V<P> in) {
```

```
in = hull(in);
int n = ssize(in), j = 1;
pair<D, pair<P, P>> ret;
REP(i, j)
    for(;; j = (j + 1) % n) {
        chmax(ret, pair(dist(in[i], in[j]), pair(in[i],
            in[j])));
        if (sign(cross(in[(j + 1) % n] - in[j], in[i + 1]
            - in[j])) <= 0)
            break;
    }
return ret.se;
}
```

geo3d

#4d58a5

Geo3d od Warsaw Eagles.

```
using LD = long double;
C LD kEps = 1e-9;
C LD kPi = acosl(-1);
LD Sq(LD x) { return x * x; }
struct Point {
    LD x, y;
    Point() {}
    Point(LD a, LD b) : x(a), y(b) {}
    Point(C Point& a) : Point(a.x, a.y) {}
    void operator=(C Point& a) { x = a.x; y = a.y; }
    Point operator+(C Point& a) C { Point p(x + a.x, y +
        a.y); return p; }
    Point operator-(C Point& a) C { Point p(x - a.x, y -
        a.y); return p; }
    Point operator*(LD a) C { Point p(x * a, y * a);
        return p; }
    Point operator/(LD a) C { assert(abs(a) > kEps);
        Point p(x / a, y / a); return p; }
    Point &operator+=(C Point& a) { x += a.x; y += a.y;
        return *this; }
    Point &operator-=(C Point& a) { x -= a.x; y -= a.y;
        return *this; }
    LD CrossProd(C Point& a) C { return x * a.y - y * a.x
        ; }
    LD CrossProd(Point a, Point b) C { a -= *this; b -= *
        this; return a.CrossProd(b); }
};
struct Line {
    Point p[2];
    Line(Point a, Point b) { p[0] = a; p[1] = b; }
    Point &operator[](int a) { return p[a]; }
};
struct P3 {
    LD x, y, z;
    P3 operator+(P3 a) { P3 p{x + a.x, y + a.y, z + a.z};
        return p; }
    P3 operator-(P3 a) { P3 p{x - a.x, y - a.y, z - a.z};
        return p; }
    P3 operator*(LD a) { P3 p{x * a, y * a, z * a};
        return p; }
    P3 operator/(LD a) { assert(a > kEps); P3 p{x / a, y
        / a, z / a}; return p; }
    P3 &operator+=(P3 a) { x += a.x; y += a.y; z += a.z;
        return *this; }
    P3 &operator-=(P3 a) { x -= a.x; y -= a.y; z -= a.z;
        return *this; }
    P3 &operator*=(LD a) { x *= a; y *= a; z *= a; return
        *this; }
    P3 &operator/=(LD a) { assert(a > kEps); x /= a; y /=
        a; z /= a; return *this; }
    LD operator[](int a) {
        if (a == 0) return x;
        if (a == 1) return y;
        return z;
    }
    bool IsZero() { return abs(x) < kEps && abs(y) < kEps
        && abs(z) < kEps; }
}
```

```
LD DotProd(P3 a) { return x * a.x + y * a.y + z * a.z
    ; }
LD Norm() { return sqrt(x * x + y * y + z * z); }
LD SqNorm() { return x * x + y * y + z * z; }
void NormalizeSelf() { *this /= Norm(); }
P3 Normalize() {
    P3 res(*this); res.NormalizeSelf();
    return res;
}
LD Dis(P3 a) { return (*this - a).Norm(); }
pair<LD, LD> SphericalAngles() {
    return {atan2(z, sqrt(x * x + y * y)), atan2(y, x)
        };
}
LD Area(P3 p) { return Norm() * p.Norm() * sin(Angle(
    p)) / 2; }
LD Angle(P3 p) {
    LD a = Norm();
    LD b = p.Norm();
    LD c = Dis(p);
    return acos((a * a + b * b - c * c) / (2 * a * b));
}
LD Angle(P3 p, P3 q) { return p.Angle(q); }
P3 CrossProd(P3 p) {
    P3 q(*this);
    return {q[1] * p[2] - q[2] * p[1], q[2] * p[0] - q
        [0] * p[2],
        q[0] * p[1] - q[1] * p[0]};
}
bool LexCmp(P3 &a, C P3 &b) {
    if (abs(a.x - b.x) > kEps) return a.x < b.x;
    if (abs(a.y - b.y) > kEps) return a.y < b.y;
    return a.z < b.z;
}
};
struct Line3 {
    P3 p[2];
    P3 &operator[](int a) { return p[a]; }
    friend ostream &operator<<(ostream &out, Line3 m);
};
struct Plane {
    P3 p[3];
    P3 &operator[](int a) { return p[a]; }
    P3 GetNormal() {
        P3 cross = (p[1] - p[0]).CrossProd(p[2] - p[0]);
        return cross.Normalize();
    }
    void GetPlaneEq(LD &a, LD &b, LD &c, LD &d) {
        P3 normal = GetNormal();
        a = normal[0];
        b = normal[1];
        c = normal[2];
        d = normal.DotProd(p[0]);
        assert(abs(d - normal.DotProd(p[1])) < kEps);
        assert(abs(d - normal.DotProd(p[2])) < kEps);
    }
    V<P3> GetOrthonormalBase() {
        P3 normal = GetNormal();
        P3 cand = {-normal.y, normal.x, 0};
        if (abs(cand.x) < kEps && abs(cand.y) < kEps) {
            cand = {0, -normal.z, normal.y};
        }
        cand.NormalizeSelf();
        P3 third = Plane{P3{0, 0, 0}, normal, cand}.
            GetNormal();
        assert(abs(normal.DotProd(cand)) < kEps &&
            abs(normal.DotProd(third)) < kEps &&
            abs(cand.DotProd(third)) < kEps);
        return {normal, cand, third};
    }
}
};
struct Circle3 {
    Plane pl; P3 o; LD r;
};
struct Sphere {
    P3 o;
```

```
LD r;
};
// angle PQR
LD Angle(P3 P, P3 Q, P3 R) { return (P - Q).Angle(R - Q); }
P3 ProjPtToLine3(P3 p, Line3 l) { // ok
P3 diff = l[1] - l[0];
diff.NormalizeSelf();
return l[0] + diff * (p - l[0]).DotProd(diff);
}
LD DisPtLine3(P3 p, Line3 l) { // ok
// LD area = Area(p, l[0], l[1]); LD dis1 = 2 * area / l[0].Dis(l[1]);
LD dis2 = p.Dis(ProjPtToLine3(p, l)); // assert(abs(dis1 - dis2) < kEps);
return dis2;
}
LD DisPtPlane(P3 p, Plane pl) {
P3 normal = pl.GetNormal();
return abs(normal.DotProd(p - pl[0]));
}
P3 ProjPtToPlane(P3 p, Plane pl) {
P3 normal = pl.GetNormal();
return p - normal * normal.DotProd(p - pl[0]);
}
bool PtBelongToLine3(P3 p, Line3 l) { return DisPtLine3(p, l) < kEps; }
bool Lines3Equal(Line3 p, Line3 l) {
return PtBelongToLine3(p[0], l) && PtBelongToLine3(p[1], l);
}
bool PtBelongToPlane(P3 p, Plane pl) { return DisPtPlane(p, pl) < kEps; }
Point PlanePtTo2D(Plane pl, P3 p) { // ok
assert(PtBelongToPlane(p, pl));
V<P3> base = pl.GetOrthonormalBase();
P3 control{0, 0, 0};
REP(tr, 3) { control += base[tr] * p.DotProd(base[tr]); }
assert(PtBelongToPlane(pl[0] + base[1], pl));
assert(PtBelongToPlane(pl[0] + base[2], pl));
assert((p - control).IsZero());
return {p.DotProd(base[1]), p.DotProd(base[2])};
}
Line PlaneLineTo2D(Plane pl, Line3 l) {
return {PlanePtTo2D(pl, l[0]), PlanePtTo2D(pl, l[1])};
}
P3 PlanePtTo3D(Plane pl, Point p) { // ok
V<P3> base = pl.GetOrthonormalBase();
return base[0] * base[0].DotProd(pl[0]) + base[1] * p.x + base[2] * p.y;
}
Line3 PlaneLineTo3D(Plane pl, Line l) {
return {PlanePtTo3D(pl, l[0]), PlanePtTo3D(pl, l[1])};
}
Line3 ProjLineToPlane(Line3 l, Plane pl) { // ok
return {ProjPtToPlane(l[0], pl), ProjPtToPlane(l[1], pl)};
}
bool Line3BelongToPlane(Line3 l, Plane pl) {
return PtBelongToPlane(l[0], pl) && PtBelongToPlane(l[1], pl);
}
LD Det(P3 a, P3 b, P3 d) { // ok
P3 pts[3] = {a, b, d};
LD res = 0;
for (int sign : {-1, 1}) {
REP(st_col, 3) {
int c = st_col;
LD prod = 1;
REP(r, 3) {
prod *= pts[r][c];
c = (c + sign + 3) % 3;
}
}
```

```
res += sign * prod;
}
}
return res;
}
LD Area(P3 p, P3 q, P3 r) {
q -= p; r -= p;
return q.Area(r);
}
V<Point> InterLineLine(Line &a, Line &b) { // working fine
Point vec_a = a[1] - a[0];
Point vec_b1 = b[1] - a[0];
Point vec_b0 = b[0] - a[0];
LD tr_area = vec_b1.CrossProd(vec_b0);
LD quad_area = vec_b1.CrossProd(vec_a) + vec_a.CrossProd(vec_b0);
if (abs(quad_area) < kEps) { // parallel or coinciding
if (abs(b[0].CrossProd(b[1], a[0])) < kEps) {
return {a[0], a[1]};
} else return {};
}
return {a[0] + vec_a * (tr_area / quad_area)};
}
V<P3> InterLineLine(Line3 k, Line3 l) {
if (Lines3Equal(k, l)) return {k[0], k[1]};
if (PtBelongToLine3(l[0], k)) return {l[0]};
Plane pl{l[0], k[0], k[1]};
if (!PtBelongToPlane(l[1], pl)) return {};
Line k2 = PlaneLineTo2D(pl, k);
Line l2 = PlaneLineTo2D(pl, l);
V<Point> inter = InterLineLine(k2, l2);
V<P3> res;
for (auto P : inter) res.pb(PlanePtTo3D(pl, P));
return res;
}
LD DisLineLine(Line3 l, Line3 k) { // ok
Plane together{l[0], l[1], l[0] + k[1] - k[0]}; // parallel FIXME
Line3 proj = ProjLineToPlane(k, together);
P3 inter = (InterLineLine(l, proj))[0];
P3 on_k_inter = k[0] + inter - proj[0];
return inter.Dis(on_k_inter);
}
Plane ParallelPlane(Plane pl, P3 A) { // plane parallel to pl going through A
P3 diff = A - ProjPtToPlane(A, pl);
return {pl[0] + diff, pl[1] + diff, pl[2] + diff};
}
// image of B in rotation wrt line passing through origin s.t. A1->A2
// implemented in more general case with similarity instead of rotation
P3 RotateAccordingly(P3 A1, P3 A2, P3 B1) { // ok
Plane pl{A1, A2, {0, 0, 0}};
Point A12 = PlanePtTo2D(pl, A1);
Point A22 = PlanePtTo2D(pl, A2);
complex<LD> rat = complex<LD>(A22.x, A22.y) / complex<LD>(A12.x, A12.y);
Plane plb = ParallelPlane(pl, B1);
Point B2 = PlanePtTo2D(plb, B1);
complex<LD> Brot = rat * complex<LD>(B2.x, B2.y);
return PlanePtTo3D(plb, {Brot.real(), Brot.imag()});
}
V<CircleS> IntersSpherePlane(Sphere s, Plane pl) { // ok
P3 proj = ProjPtToPlane(s.o, pl);
LD dis = s.o.Dis(proj);
if (dis > s.r + kEps) return {};
if (dis > s.r - kEps) return {{pl, proj, 0}}; // is it best choice?
return {{pl, proj, sqrt(s.r * s.r - dis * dis)}};
}
bool PtBelongToSphere(Sphere s, P3 p) { return abs(s.r - s.o.Dis(p)) < kEps; }
```

```
struct PointS { // just for conversion purposes, probably toEucl suffices
LD lat, lon;
P3 toEucl() { return P3(cos(lat) * cos(lon), cos(lat) * sin(lon), sin(lat)); }
PointS(P3 p) {
p.NormalizeSelf();
lat = asin(p.z);
lon = acos(p.y / cos(lat));
}
};
LD DistS(P3 a, P3 b) { return atan2l(b.CrossProd(a).Norm(), a.DotProd(b)); }
struct CircleS {
P3 o; // center of circle on sphere
LD r; // arc len
LD area() C { return 2 * kPi * (1 - cos(r)); }
};
CircleS From3(P3 a, P3 b, P3 c) { // any three different points
int tmp = 1;
if ((a - b).Norm() > (c - b).Norm()) { swap(a, c); tmp = -tmp; }
if ((b - c).Norm() > (a - c).Norm()) { swap(a, b); tmp = -tmp; }
P3 v = (c - b).CrossProd(b - a);
v = v * (tmp / v.Norm());
return CircleS{v, DistS(a, v)};
}
CircleS From2(P3 a, P3 b) { // neither the same nor the opposite
P3 mid = (a + b) / 2;
mid = mid / mid.Norm();
return From3(a, mid, b);
}
LD SphAngle(P3 a, P3 b, P3 c) { // angle at A, no two points opposite
LD a2 = b.DotProd(c);
LD b2 = c.DotProd(a);
LD c2 = a.DotProd(b);
return acos((b2 - a2 * c2) / sqrt((1 - Sq(a2)) * (1 - Sq(c2))));
}
LD TriangleArea(P3 a, P3 b, P3 c) { // no two points opposite
LD a2 = SphAngle(c, a, b);
LD b2 = SphAngle(a, b, c);
LD c2 = SphAngle(b, c, a);
return a2 + b2 + c2 - kPi;
}
```

halfplane-intersection

#5e7cbf, includes: point

$\mathcal{O}(n \log n)$ wyznaczenie punktów na brzegu/otoczące przecięcia podanych półpłaszczyzn. Halfplane(a, b) tworzy półpłaszczyznę wzdłuż prostej $a \rightarrow b$ z obszarem po lewej stronie wektora $a \rightarrow b$. Jeżeli zostało zwróconych mniej, niż trzy punkty, to pole przecięcia jest puste. Na przykład halfplane_intersection({Halfplane(P(2, 1), P(4, 2)), Halfplane(P(6, 3), P(2, 4)), Halfplane(P(-4, 7), P(4, 2))}) == {(4, 2), (6, 3), (0, 4.5)}. Pole przecięcia jest zawsze ograniczone, ponieważ w kodzie są dodawane cztery półpłaszczyzny o współrzędnych w +/--inf, ale nie należy na tym polegać przez eps oraz błędy precyzji (najlepiej jest zmniejszyć inf tyle, ile się da).

```
struct Halfplane {
P p, pq;
D angle;
Halfplane() {}
Halfplane(P a, P b) : p(a), pq(b - a) {
angle = atan2l(pq.imag(), pq.real());
}
};
ostream& operator<<(ostream&o, Halfplane h) {
```

```
return o << '(' << h.p << ", " << h.pq << ", " << h.angle << ')';
}
bool is_outside(Halfplane hi, P p) {
return sign(cross(hi.pq, p - hi.p)) == -1;
}
P inter(Halfplane s, Halfplane t) {
D alpha = cross(t.p - s.p, t.pq) / cross(s.pq, t.pq);
return s.p + s.pq * alpha;
}
V<P> halfplane_intersection(V<Halfplane> h) {
for (int i = 0; i < 4; ++i) {
constexpr D inf = 1e9;
array box = {P(-inf, -inf), P(inf, -inf), P(inf, inf), P(-inf, inf)};
h.eb(box[i], box[(i + 1) % 4]);
}
sort(all(h), [&](Halfplane l, Halfplane r) {
return l.angle < r.angle;
});
deque<Halfplane> dq;
for (auto &hi : h) {
while (ssize(dq) >= 2 and is_outside(hi, inter(dq.end()[-1], dq.end()[-2])))
dq.pop_back();
while (ssize(dq) >= 2 and is_outside(hi, inter(dq[0], dq[1])))
dq.pop_front();
if (ssize(dq) and sign(cross(hi.pq, dq.back().pq)) == 0) {
if (sign(dot(hi.pq, dq.back().pq)) < 0)
return {};
if (is_outside(hi, dq.back().p))
dq.pop_back();
else
continue;
}
}
dq.eb(hi);
while (ssize(dq) >= 3 and is_outside(dq[0], inter(dq.end()[-1], dq.end()[-2])))
dq.pop_back();
while (ssize(dq) >= 3 and is_outside(dq.end()[-1], inter(dq[0], dq[1])))
dq.pop_front();
V<P> ret;
REP(i, ssize(dq))
ret.eb(inter(dq[i], dq[(i + 1) % ssize(dq)]));
ret.erase(unique(all(ret), [&](P l, P r) { return equal(l, r); }), ret.end());
if (ssize(ret) >= 2 and equal(ret.front(), ret.back()))
ret.pop_back();
for (Halfplane hi : h)
if (ssize(ret) <= 2 and is_outside(hi, ret[0]))
return {};
return ret;
}
```

intersect-lines

#6a7387, includes: point

$\mathcal{O}(1)$ ale intersect_segments ma sporą stałą (ale działa na wszystkich edge-case'ach). Jeżeli intersect_segments zwróci dwa punkty to wszystkie inf rozwiązań są pomiędzy.

```
// BEGIN HASH 95db50
P intersect_lines(P a, P b, P c, P d) {
D c1 = cross(c - a, b - a), c2 = cross(d - a, b - a);
// c1 == c2 => równolege
return (c1 * d - c2 * c) / (c1 - c2);
} // END HASH
// BEGIN HASH 65e219
bool on_segment(P a, P b, P p) {
return equal(cross(a - p, b - p), 0) and sign(dot(a - p, b - p)) <= 0;
} // END HASH
```

```
// BEGIN HASH 2b171b
bool is_intersection_segment(P a, P b, P c, P d) {
    auto aux = [&](D q, D w, D e, D r) {
        return sign(max(q, w) - min(e, r)) >= 0;
    };
    return aux(c.x(), d.x(), a.x(), b.x()) and aux(a.x(),
        b.x(), c.x(), d.x())
        and aux(c.y(), d.y(), a.y(), b.y()) and aux(a.y(),
        b.y(), c.y(), d.y())
        and dir(a, d, c) * dir(b, d, c) != 1
        and dir(d, b, a) * dir(c, b, a) != 1;
} // END HASH
// BEGIN HASH c969b4
V<P> intersect_segments(P a, P b, P c, P d) {
    D acd = cross(c - a, d - c), bcd = cross(c - b, d - c
    ),
        cab = cross(a - c, b - a), dab = cross(a - d, b
        - a);
    if(sign(acd) * sign(bcd) < 0 and sign(cab) * sign(dab
        ) < 0)
        return {(a * bcd - b * acd) / (bcd - acd)};
    set<P> s;
    if(on_segment(c, d, a)) s.emplace(a);
    if(on_segment(c, d, b)) s.emplace(b);
    if(on_segment(a, b, c)) s.emplace(c);
    if(on_segment(a, b, d)) s.emplace(d);
    return {s.begin(), s.end()};
} // END HASH
```

is-in-hull

#5eea9f, includes: intersect-lines

$\mathcal{O}(\log n)$, zwraca czy punkt jest wewnątrz otoczki h. Zakłada że punkty są clockwise oraz nie ma trzech współliniowych (dziłała na convex-hull).

```
bool is_in_hull(V<P> h, P p, bool can_on_edge) {
    if(ssize(h) < 3) return can_on_edge and on_segment(h
    [0], h.back(), p);
    int l = 1, r = ssize(h) - 1;
    if(dir(h[0], h[l], p) >= can_on_edge or dir(h[0], h[r
    ], p) <= -can_on_edge)
        return false;
    while(r - l > 1) {
        int m = (l + r) / 2;
        (dir(h[0], h[m], p) < 0 ? l : r) = m;
    }
    return dir(h[l], h[r], p) < can_on_edge;
}
```

line

#033da1, includes: point

Konwersja różnych postaci prostej.

```
struct Line {
    D a, b, c;
    // postac ogolna ax + by + c = 0
    Line(D _a, D _b, D _c) : a(_a), b(_b), c(_c) {}
    tuple<D, D, D> get_tuple() { return {a, b, c}; }
    // postac kierunkowa ax + b = y
    Line(D _a, D _b) : a(_a), b(-1), c(_b) {}
    pair<D, D> get_dir() { return {- a / b, - c / b}; }
    // prosta pq
    Line(P p, P q) {
        assert(not equal(p, q));
        if(not equal(p.x(), q.x())) {
            a = (q.y() - p.y()) / (p.x() - q.x());
            b = 1, c = -(a * p.x() + b * p.y());
        }
        else a = 1, b = 0, c = -p.x();
    }
    pair<P, P> get_pts() {
        if(!equal(b, 0)) return { P(0, - c / b), P(1, - (a
            + c) / b) };
        return { P(- c / a, 0), P(- c / a, 1) };
    }
    D directed_dist(P p) {
```

```
        return (a * p.x() + b * p.y() + c) / sqrt(a * a + b
            * b);
    }
    D dist(P p) {
        return abs(directed_dist(p));
    }
};
```

point

#a14c07

Wrapper na std::complex, definiy trzeba dać nad bitsami, wtedy istnieje p.x() oraz p.y(). abs długość, arg kąt $(-\pi, \pi]$ gdzie $(0, 1)$ daje $\frac{\pi}{2}$, polar(len, angle) tworzy P . Istnieją atan2, asin, sinh.

```
// Before include bits:
// #define real x
// #define imag y
using D = long double;
using P = complex<D>;
constexpr D eps = 1e-9;
bool equal(D a, D b) { return abs(a - b) < eps; }
bool equal(P a, P b) { return equal(a.x(), b.x()) and
    equal(a.y(), b.y()); }
int sign(D a) { return equal(a, 0) ? 0 : a > 0 ? 1 :
    -1; }
namespace std { bool operator<(P a, P b) { return sign(
    a.x() - b.x()) == 0 ? sign(a.y() - b.y()) < 0 : a.x
    () < b.x(); } }
// cross({1, 0}, {0, 1}) = 1
D cross(P a, P b) { return a.x() * b.y() - a.y() * b.x
    (); }
D dot(P a, P b) { return a.x() * b.x() + a.y() * b.y(); }
D dist(P a, P b) { return abs(a - b); }
int dir(P a, P b, P c) { return sign(cross(b - a, c - b
    )); }
```

polygon-gen

#4470a8, includes: point, intersect-lines, headers/gen

Generatorka wielokątów niekoniecznie-wypukłych. Zwraca wielokąt o n punktach w zakresie $[-r, r]$, który nie zawiera jakiegolwiek trójki współliniowych punktów. Ciągnie do ~ 80 . Dla $n < 3$ zwraca zdegenerowane.

```
V<P> gen_polygon(int n, int r) {
    V<P> t;
    while (ssize(t) < n) {
        P p(rd(-r, r), rd(-r, r));
        if ([&]() {
            REP (l, ssize(t))
                REP (j, l)
                    if (dir(t[l], t[j], p) == 0)
                        return false;
            return find(all(t), p) == t.end();
        }())
            t.eb(p);
    }
    bool go = true;
    while (go) {
        go = false;
        REP (i, n)
            REP (j, i - 1)
                if ((i + 1) % n != j && ssize(
                    intersect_segments(t[i], t[(i + 1) % n], t[j
                    ], t[j + 1])) {
                    swap(t[(i + rd(0, 1)) % n], t[(j + rd(0, 1)
                        ) % n]);
                    go = true;
                }
    }
    return t;
}
```

polygon-print

#de8102, includes: point

Należy przekierować stdout do pliku i otworzyć go np. w przeglądarce. n zwiększa obrazek, d zmniejsza rozmiar napisów/wierzchołków.

```
void polygon_print(V<P> v, int r = 10) {
    int m = 350 / r, d = 50;
    auto ori = v;
    for (auto &p : v)
        p = P((p.x() + r * 1.1) * m, (p.y() + r * 1.1)
            * m);
    r = int(r * m * 2.5);
    printf("<svg height='%d' width='%d'>rect width
        ='100%' height='100%' fill='white' />", r, r);
    int n = ssize(v);
    REP (i, n) {
        printf("<line x1='%Lf' y1='%Lf' x2='%Lf' y2='%
            Lf' style='stroke:black' />", v[i].x(), v[i
                ].y(), v[(i + 1) % n].x(), v[(i + 1) % n].y
                ());
        printf("<circle cx='%Lf' cy='%Lf' r='%f' fill='
            red' />", v[i].x(), v[i].y(), r / d / 10.0);
        printf("<text x='%Lf' y='%Lf' font-size='%d'
            fill='violet'%d (%.1Lf, %.1Lf)</text>", v[i
                ].x() + 5, v[i].y() - 5, r / d, i + 1, ori[i
                    ].x(), ori[i].y());
    }
    printf("</svg>\n");
}
```

voronoi-diagram

#1f8a8f, includes: delaunay-triangulatio, convex-hull

$\mathcal{O}(n \log n)$, dla każdego punktu zwraca odpowiadającą mu ścianę będącą otoczką wypukłą. Suma otoczek w całości zawiera kwadrat $(-mx, mx) - (mx, mx)$, ale może zawierać więcej. Współrzędne ścian mogą być kilka rzędów wielkości większe niż te na wejściu. Max abs wartości współrzędnych to 3e8.

```
using Frac = pair<__int128_t, __int128_t>;
D to_d(Frac f) { return D(f.fi) / D(f.se); }
Frac create_frac(__int128_t a, __int128_t b) {
    assert(b != 0);
    if (b < 0) a *= -1, b *= -1;
    __int128_t d = __gcd(a, b);
    return {a / d, b / d};
}
using P128 = pair<Frac, Frac>;
ll sq(int x) { return x * ll(x); }
__int128_t dist128(PI p) { return sq(p.fi) + sq(p.se); }
pair<Frac, Frac> calc_mid(PI a, PI b, PI c) {
    __int128_t ux = dist128(a) * (b.se - c.se)
        + dist128(b) * (c.se - a.se)
        + dist128(c) * (a.se - b.se),
        uy = dist128(a) * (c.fi - b.fi)
        + dist128(b) * (a.fi - c.fi)
        + dist128(c) * (b.fi - a.fi),
        d = 2 * (a.fi * ll(b.se - c.se)
            + b.fi * ll(c.se - a.se)
            + c.fi * ll(a.se - b.se));
    return {create_frac(ux, d), create_frac(uy, d)};
}
V<V<P>> voronoi_faces(V<PI> in, C int max_xy = int(3e8)
    ) {
    int n = ssize(in);
    map<PI, int> id_of_in;
    REP(i, n)
        id_of_in[in[i]] = i;
    for(int sx : {-1, 1})
        for(int sy : {-1, 1}) {
            int mx = 3 * max_xy + 100;
            in.eb(mx * sx, mx * sy);
        }
    V<PI> triangles = triangulate(in);
    debug(triangles);
    assert(not triangles.empty());
    int tn = ssize(triangles) / 3;
    V<P128> mids(tn);
```

```
map<pair<PI, PI>, V<P128>> on_sides;
REP(l, tn) {
    array<PI, 3> ps = {triangles[3 * l], triangles[3 *
        l + 1], triangles[3 * l + 2]};
    mids[l] = calc_mid(ps[0], ps[1], ps[2]);
    REP(j, 3) {
        PI a = ps[j], b = ps[(j + 1) % 3];
        on_sides[pair(min(a, b), max(a, b))].eb(mids[l]);
    }
}
V<V<P128>> faces128(n);
for(auto [edge, sides] : on_sides)
    if(ssize(sides) == 2)
        for(PI e : {edge.fi, edge.se})
            if(id_of_in.find(e) != id_of_in.end())
                for(auto m : sides)
                    faces128[id_of_in[e]].eb(m);
V<V<P>> faces(n);
REP(i, ssize(faces128)) {
    auto &f = faces128[i];
    sort(all(f));
    f.erase(unique(all(f)), f.end());
    for(auto [x, y] : f)
        faces[i].eb(to_d(x), to_d(y));
    faces[i] = hull(faces[i]);
}
return faces;
}
```

Tekstówki (8)

aho-corasick

#be512e

$\mathcal{O}(|s|\alpha)$, Konstruktor tworzy sam korzeń w node[0], add(s) dodaje słowo, convert() zamienia nieodwracalnie trie w automat Aho-Corasick, link(x) zwraca suffix link, go(x, c) zwraca następnik x przez literę c, najpierw dodajemy słowa, potem robimy convert(), a na koniec używamy goilink.

```
constexpr int alpha = 26;
struct AhoCorasick {
    struct Node {
        array<int, alpha> next, go;
        int p, pch, link = -1;
        bool is_word_end = false;
        Node(int _p = -1, int ch = -1) : p(_p), pch(ch) {
            fill(all(next), -1);
            fill(all(go), -1);
        }
    };
    V<Node> node;
    bool converted = false;
    AhoCorasick() : node(1) {}
    void add(C vi &s) {
        assert(!converted);
        int v = 0;
        for (int c : s) {
            if (node[v].next[c] == -1) {
                node[v].next[c] = ssize(node);
                node.eb(v, c);
            }
            v = node[v].next[c];
        }
        node[v].is_word_end = true;
    }
    int link(int v) {
        assert(converted);
        return node[v].link;
    }
    int go(int v, int c) {
        assert(converted);
        return node[v].go[c];
    }
    void convert() {
        assert(!converted);
```

```
converted = true;
deque<int> que = {};
while (not que.empty()) {
    int v = que.front();
    que.pop_front();
    if (v == 0 or node[v].p == 0)
        node[v].link = 0;
    else
        node[v].link = go(link(node[v].p), node[v].pch)
        ;
    REP (c, alpha) {
        if (node[v].next[c] != -1) {
            node[v].go[c] = node[v].next[c];
            que.eb(node[v].next[c]);
        }
        else
            node[v].go[c] = v == 0 ? 0 : go(link(v), c);
    }
}
};
```

eertree

#a21027

$\mathcal{O}(n\alpha)$ konstrukcja, $\mathcal{O}(n)$ DP oraz odzyskanie. Eertree ma korzeń „pusty” w 0 oraz „ujemny” w 1. Z wierzchołka wychodzi krawędź z literą, gdy jego słowo można otoczyć z obu stron tą literą. Funkcja `add_letter` zwraca wierzchołek odpowiadający za największy palindromiczny suffix aktualnego słowa. Suffix link prowadzi do najdłuższego palindromicznego suffixu słowa wierzchołka. Linki tworzą drzewo z 1 jako korzeń (który ma syna 0). Żeby policzyć liczbę wystąpień wierzchołka, po każdym dodaniu litery „wystarczy” dodać +1 każdemu na ścieżce od last do korzenia po linkach. `palindromic_split_dp` zwraca na każdym prefixie (min podział palindromiczny, indeks do odzyskania min podziału, liczbę podziałów). Gdy `only_even_lens` to może nie istnieć odpowiedź, wtedy `.mn == n + 1`, `.cnt == 0`. `construct_min_palindromic_split` zwraca palindromiczne przedziały pokrywające słowo.

```
// BEGIN HASH b1ff16
constexpr int alpha = 26;
struct Eertree {
    V<array<int, alpha>> edge;
    array<int, alpha> empty;
    vi str = {-1}, link = {1, 0}, len = {0, -1};
    int last = 0;
    Eertree() {
        empty.fill(0);
        edge.resize(2, empty);
    }
    int find(int v) {
        while(str.end()[v-1] != str.end()[v-len[v] - 2])
            v = link[v];
        return v;
    }
    int add_letter(int c) {
        str.eb(c);
        last = find(last);
        if(edge[last][c] == 0) {
            edge.eb(empty);
            len.eb(len[last] + 2);
            link.eb(edge[find(link[last])][c]);
            edge[last][c] = ssize(edge) - 1;
        }
        return last = edge[last][c];
    }
}; // END HASH
int add(int a, int b) { return a + b; } // ċDopisa
modulo ́jeeli trzeba.
// BEGIN HASH c44f07
struct Dp { int mn, mn_i, cnt; };
Dp operator+(Dp l, Dp r) {
    return {min(l.mn, r.mn), l.mn < r.mn ? l.mn_i : r.
        mn_i, add(l.cnt, r.cnt)};
}
V<Dp> palindromic_split_dp(vi str, bool only_even_lens
= false) {
```

```
int n = ssize(str);
Eertree t;
vi big_link(2), diff(2);
V<Dp> series_ans(2), ans(n, {n + 1, -1, 0});
REP(i, n) {
    int last = t.add_letter(str[i]);
    if(last >= ssize(big_link)) {
        diff.eb(t.len.back() - t.len[t.link.back()]);
        big_link.eb(diff.back() == diff[t.link.back()] ?
            big_link[t.link.back()] : t.link.back());
        series_ans.eb(c);
    }
    for(int v = last; t.len[v] > 0; v = big_link[v]) {
        int j = i - t.len[big_link[v]] - diff[v];
        series_ans[v] = j == -1 ? Dp{0, j, 1} : Dp{ans[j]
            .mn, j, ans[j].cnt};
        if(diff[v] == diff[t.link[v]])
            series_ans[v] = series_ans[v] + series_ans[t.
                link[v]];
        if(i % 2 == 1 or not only_even_lens)
            ans[i] = ans[i] + Dp{series_ans[v].mn + 1,
                series_ans[v].mn_i, series_ans[v].cnt};
    }
}
return ans;
} // END HASH
// BEGIN HASH e17097
V<pii> construct_min_palindromic_split(V<Dp> ans) {
    if(ans.back().mn == ssize(ans) + 1)
        return {};
    V<pii> split = {{0, ssize(ans) - 1}};
    while(ans[split.back().se].mn_i != -1)
        split.eb(0, ans[split.back().se].mn_i);
    reverse(all(split));
    REP(i, ssize(split) - 1)
        split[i + 1].fi = split[i].se + 1;
    return split;
} // END HASH
```

hashing

#781b34

Hashowanie z małą stałą. Można zmienić bazę (jeśli serio trzeba). `opensl` prime -generate -bits 60 generuje losową liczbę pierwszą o 60 bitach ($\leq 1.15 \cdot 10^{18}$).

```
struct Hashing {
    vll ha, pw;
    static constexpr ll mod = (1ll << 61) - 1;
    ll reduce(ll x) { return x >= mod ? x - mod : x; }
    ll mul(ll a, ll b) {
        C auto c = __int128(a) * b;
        return reduce(ll(c & mod) + ll(c >> 61));
    }
    Hashing(C vi &str, C int base = 37) {
        int len = ssize(str);
        ha.resize(len + 1);
        pw.resize(len + 1, 1);
        REP(i, len) {
            ha[i + 1] = reduce(mul(ha[i], base) + str[i] + 1)
                ;
            pw[i + 1] = mul(pw[i], base);
        }
    }
    ll operator()(int l, int r) {
        return reduce(ha[r + 1] - mul(ha[l], pw[r - l + 1])
            + mod);
    }
};
```

kmp

#6cf4ba

$\mathcal{O}(n)$, zachodzi $[0, pi[i]] = (i - pi[i], i]$. `get_kmp` $\{0, 1, 0, 0, 1, 0, 1, 0, 0, 1\}$ == $\{0, 0, 1, 1, 2, 3, 2, 3, 4, 5\}$, `get_borders` $\{0, 1, 0, 0, 1, 0, 1, 0, 0, 1\}$ == $\{2, 5, 10\}$.

// BEGIN HASH d38133

```
vi get_kmp(vi str) {
    int len = ssize(str);
    vi ret(len);
    for(int i = 1; i < len; i++) {
        int pos = ret[i - 1];
        while(pos and str[i] != str[pos])
            pos = ret[pos - 1];
        ret[i] = pos + (str[i] == str[pos]);
    }
    return ret;
} // END HASH
vi get_borders(vi str) {
    vi kmp = get_kmp(str), ret;
    int len = ssize(str);
    while(len) {
        ret.eb(len);
        len = kmp[len - 1];
    }
    return vi(rall(ret));
}
```

lyndon-min-cyclic-rot

#51b6f0

$\mathcal{O}(n)$, wyznaczenie faktoryzacji Lyndona oraz (przy jej pomocy) minimalnego suffixu oraz minimalnego przesunięcia cyklicznego. Ta faktoryzacja to unikalny podział słowa s na $w_1 w_2 \dots w_k$, że $w_1 \geq w_2 \geq \dots \geq w_k$ oraz w_i jest ściśle mniejsze od każdego jego suffixu. `duval` ("abacaba") == $\{0, 3\}$, $\{4, 5\}$, $\{6, 6\}$, `min_suffix` ("abacab") == "ab", `min_cyclic_shift` ("abacaba") == "aabacab".

```
V<pii> duval(vi s) {
    int n = ssize(s), i = 0;
    V<pii> ret;
    while(i < n) {
        int j = i + 1, k = i;
        while(j < n and s[k] <= s[j]) {
            k = (s[k] < s[j] ? i : k + 1);
            ++j;
        }
        while(i <= k) {
            ret.eb(i, i + j - k - 1);
            i += j - k;
        }
    }
    return ret;
}
vi min_suffix(vi s) {
    return {s.begin() + duval(s).back().fi, s.end()};
}
vi min_cyclic_shift(vi s) {
    int n = ssize(s);
    REP(i, n)
        s.eb(s[i]);
    for(auto [l, r] : duval(s))
        if(n <= r) {
            return {s.begin() + l, s.begin() + l + n};
        }
    assert(false);
}
```

manacher

#f87a5b

$\mathcal{O}(n)$, `radius[p][i]` = `rad` = największy promień palindromu parzystości p o środku i . $L = i - rad + 1$, $R = i + rad$ to palindrom. Dla [abaababab] daje [0030000020], [01001410000].

```
array<vi, 2> manacher(vi &n) {
    int n = ssize(n);
    array<vi, 2> radius = {{vi(n - 1), vi(n)}};
    REP(parity, 2) {
        int z = parity ^ 1, L = 0, R = 0;
        REP(i, n - z) {
            int &rad = radius[parity][i];
            if(i <= R - z)
```

```
                rad = min(R - i, radius[parity][L + (R - i - z)
                    ]);
            int l = i - rad + z, r = i + rad;
            while(0 <= l - 1 && r + 1 < n && in[l - 1] == in[
                r + 1])
                ++rad, ++r, --l;
            if(r > R)
                L = l, R = r;
        }
    }
    return radius;
}
```

pref

#103217

$\mathcal{O}(n)$, zwraca tablicę prefixo prefixową $[0, pref[i]] = [i, i + pref[i]]$.

```
vi pref(vi str) {
    int n = ssize(str);
    vi ret(n);
    ret[0] = n;
    int i = 1, m = 0;
    while(i < n) {
        while(m + i < n and str[m + i] == str[m])
            m++;
        ret[i++] = m;
        m = max(0, m - 1);
        for(int j = 1; ret[j] < m; m--)
            ret[i++] = ret[j++];
    }
    return ret;
}
```

squares

#e78cf9, includes: pref

$\mathcal{O}(n \log n)$, zwraca wszystkie skompresowane trójki $(start_l, start_r, len)$ oznaczające, że podslowa zaczynające się w $[start_l, start_r]$ o długości `len` są kwadratami, jest ich $\mathcal{O}(n \log n)$.

```
V<tuple<int, int, int>> squares(C vi &s) {
    V<tuple<int, int, int>> ans;
    V pos(ssize(s) + 2, -1);
    FOR(mid, 1, ssize(s) - 1) {
        int part = mid & ~(mid - 1), off = mid - part;
        int end = min(mid + part, ssize(s));
        V a(s.begin() + off, s.begin() + off + part),
            b(s.begin() + mid, s.begin() + end),
            ra(rall(a));
        REP(j, 2) {
            auto z1 = pref(ra), bha = b;
            bha.eb(-1);
            for(int x : a) bha.eb(x);
            auto z2 = pref(bha);
            for(auto *v : {&z1, &z2}) {
                v[0][0] = ssize(v[0]);
                v->eb(0);
            }
            REP(c, ssize(a)) {
                int l = ssize(a) - c, x = c - min(l - 1, z1[l])
                    ,
                    y = c - max(l - z2[ssize(b) + c + 1], j),
                    sb = (j ? end - y - l * 2 : off + x),
                    se = (j ? end - x - l * 2 + 1 : off + y + 1),
                    &p = pos[l];
                if (x > y) continue;
                if (p != -1 && get<1>(ans[p]) + 1 == sb)
                    get<1>(ans[p]) = se - 1;
            }
            else
                p = ssize(ans), ans.eb(sb, se - 1, l);
        }
        a = V(rall(b));
        b.swap(ra);
    }
}
```

```
    return ans;
}
```

suffix-array-interval

#a0655e, includes: suffix-array-short

$\mathcal{O}(t \log n)$, wyznaczenie przedziałów podstowa w tablicy suffixowej. Zwraca przedział $[l, r]$, gdzie dla każdego i w $[l, r]$, t jest podstwem $sa.sa[i]$ lub $[-1, -1]$ jeżeli nie ma takiego i .

```
pii get_substring_sa_range(C vi &s, C vi &sa, C vi &t)
{
    auto get_lcp = [&](int i) -> int {
        REP(k, ssize(t))
            if(i + k >= ssize(s) or s[i + k] != t[k])
                return k;
        return ssize(t);
    };
    auto get_side = [&](bool search_left) {
        int l = 0, r = ssize(sa) - 1;
        while(l < r) {
            int m = (l + r + not search_left) / 2, lcp =
                get_lcp(sa[m]);
            if(lcp == ssize(t))
                (search_left ? r : l) = m;
            else if(sa[m] + lcp >= ssize(s) or s[sa[m] + lcp]
                < t[lcp])
                l = m + 1;
            else
                r = m - 1;
        }
        return l;
    };
    int l = get_side(true);
    if(get_lcp(sa[l]) != ssize(t))
        return {-1, -1};
    return {l, get_side(false)};
}
```

suffix-array-long

#0c0515

$\mathcal{O}(n + \alpha)$, sa zawiera posortowane suffixy, zawiera pusty suffix, $lcp[i]$ to lcp suffixu $sa[i]$ i $sa[i + 1]$. Dla $s = aabaaab$, $sa = \{7, 3, 4, 0, 5, 1, 6, 2\}$, $lcp = \{0, 2, 3, 1, 2, 0, 1\}$

```
// BEGIN HASH 262977
void induced_sort(C vi &vec, int alpha, vi &sa,
    C V<bool> &sl, C vi &lms_idx) {
    vi l(alpha), r(alpha);
    for (int c : vec) {
        if (c + 1 < alpha)
            ++l[c + 1];
        ++r[c];
    }
    partial_sum(all(l), l.begin());
    partial_sum(all(r), r.begin());
    fill(all(sa), -1);
    RFOR(i, ssize(lms_idx) - 1, 0)
        sa[--r[vec[lms_idx[i]]]] = lms_idx[i];
    for (int i : sa)
        if (i >= 1 and sl[i - 1])
            sa[l[vec[i - 1]]++] = i - 1;
    fill(all(r), 0);
    for (int c : vec)
        ++r[c];
    partial_sum(all(r), r.begin());
    for (int k = ssize(sa) - 1, i = sa[k]; k >= 1; --k, i
        = sa[k])
        if (i >= 1 and not sl[i - 1])
            sa[--r[vec[i - 1]]] = i - 1;
}
vi sa_is(C vi &vec, int alpha) {
    C int n = ssize(vec);
    vi sa(n), lms_idx;
    V<bool> sl(n);
    RFOR(i, n - 2, 0) {
```

```
    sl[i] = vec[i] > vec[i + 1] or (vec[i] == vec[i +
        1] and sl[i + 1]);
    if (sl[i] and not sl[i + 1])
        lms_idx.eb(i + 1);
}
reverse(all(lms_idx));
induced_sort(vec, alpha, sa, sl, lms_idx);
vi new_lms_idx(ssize(lms_idx), lms_vec(ssize(lms_idx)
));
for (int i = 0, k = 0; i < n; ++i)
    if (not sl[sa[i]] and sa[i] >= 1 and sl[sa[i] - 1])
        new_lms_idx[k++] = sa[i];
int cur = sa[n - 1] = 0;
REP (k, ssize(new_lms_idx) - 1) {
    int i = new_lms_idx[k], j = new_lms_idx[k + 1];
    if (vec[i] != vec[j]) {
        sa[j] = ++cur;
        continue;
    }
    bool flag = false;
    for (int a = i + 1, b = j + 1; ++a, ++b) {
        if (vec[a] != vec[b]) {
            flag = true;
            break;
        }
        if ((not sl[a] and sl[a - 1]) or (not sl[b] and
            sl[b - 1])) {
            flag = not (not sl[a] and sl[a - 1] and not sl[
                b] and sl[b - 1]);
            break;
        }
    }
    sa[j] = (flag ? ++cur : cur);
}
REP (i, ssize(lms_idx))
    lms_vec[i] = sa[lms_idx[i]];
if (cur + 1 < ssize(lms_idx)) {
    vi lms_sa = sa_is(lms_vec, cur + 1);
    REP (i, ssize(lms_idx))
        new_lms_idx[i] = lms_idx[lms_sa[i]];
}
induced_sort(vec, alpha, sa, sl, new_lms_idx);
return sa;
}
vi suffix_array(C vi &s, int alpha) {
    vi vec(ssize(s) + 1);
    REP(i, ssize(s))
        vec[i] = s[i] + 1;
    vi ret = sa_is(vec, alpha + 2);
    return ret;
} // END HASH
vi get_lcp(C vi &s, C vi &sa) {
    int n = ssize(s), k = 0;
    vi lcp(n), rank(n);
    REP (i, n)
        rank[sa[i + 1]] = i;
    for (int i = 0; i < n; i++, k ? k-- : 0) {
        if (rank[i] == n - 1) {
            k = 0;
            continue;
        }
        int j = sa[rank[i] + 2];
        while (i + k < n and j + k < n and s[i + k] == s[j
            + k])
            k++;
        lcp[rank[i]] = k;
    }
    lcp.pop_back();
    lcp.insert(lcp.begin(), 0);
    return lcp;
}
```

suffix-array-short

#b8cca1

$\mathcal{O}(n \log n)$, zawiera posortowane suffixy, zawiera pusty suffix, $lcp[i]$ to lcp suffixu $sa[i - 1]$ i $sa[i]$. Dla $s = aabaaab$, $sa = \{7, 3, 4, 0, 5, 1, 6, 2\}$, $lcp = \{0, 0, 2, 3, 1, 2, 0, 1\}$

```
pair<vi, vi> suffix_array(vi s, int alpha = 26) {
    ++alpha;
    for(int &c : s) ++c;
    s.eb(0);
    int n = ssize(s), k = 0, a, b;
    vi x(all(s));
    vi y(n), ws(max(n, alpha)), rank(n);
    vi sa = y, lcp = y;
    iota(all(sa), 0);
    for(int j = 0, p = 0; p < n; j = max(1, j * 2), alpha
        = p) {
        p = j;
        iota(all(y), n - j);
        REP(i, n) if(sa[i] >= j)
            y[p++] = sa[i] - j;
        fill(all(ws), 0);
        REP(i, n) ws[x[i]]++;
        FOR(i, 1, alpha - 1) ws[i] += ws[i - 1];
        for(int i = n; i--;) sa[--ws[x[y[i]]]] = y[i];
        swap(x, y);
        p = 1, x[sa[0]] = 0;
        FOR(i, 1, n - 1) a = sa[i - 1], b = sa[i], x[b] =
            (y[a] == y[b] && y[a + j] == y[b + j]) ? p - 1 :
                p++;
    }
    FOR(i, 1, n - 1) rank[sa[i]] = i;
    for(int i = 0, j; i < n - 1; lcp[rank[i++]] = k)
        for(k && k-- , j = sa[rank[i] - 1];
            s[i + k] == s[j + k]; k++);
    lcp.erase(lcp.begin());
    return {sa, lcp};
}
```

suffix-automaton

#d7a7c7

$\mathcal{O}(n\alpha)$ (szybsze, ale więcej pamięci) albo $\mathcal{O}(n \log \alpha)$ (mapa), buduje suffix automaton. Wystąpienia wzorca, liczba różnych podstw, sumaryczna długość wszystkich podstw, leksykograficznie k -te podstowo, najmniejsze przesunięcie cykliczne, liczba wystąpień podstowa, pierwsze wystąpienie, najkrótsze niewystępujące podstowo, longest common substring wielu słów.

```
struct SuffixAutomaton {
    static constexpr int sigma = 26;
    using Node = array<int, sigma>; // map<int, int>
    Node new_node;
    V<Node> edges;
    vi link = {-1}, length = {0};
    int last = 0;
    SuffixAutomaton() {
        new_node.fill(-1); // -1 - stan nieistniejący
        edges = {new_node}; // dodajemy stan startowy,
            ktory reprezentuje puste slowo
    }
    void add_letter(int c) {
        edges.eb(new_node);
        length.eb(length[last] + 1);
        link.eb(0);
        int r = ssize(edges) - 1, p = last;
        while(p != -1 && edges[p][c] == -1) {
            edges[p][c] = r;
            p = link[p];
        }
        if(p != -1) {
            int q = edges[p][c];
            if(length[p] + 1 == length[q])
                link[r] = q;
            else {
                edges.eb(edges[q]);
                length.eb(length[p] + 1);
                link.eb(link[q]);
                int q_prim = ssize(edges) - 1;
```

```
                link[q] = link[r] = q_prim;
                while(p != -1 && edges[p][c] == q) {
                    edges[p][c] = q_prim;
                    p = link[p];
                }
            }
            last = r;
        }
        bool is_inside(vi &s) {
            int q = 0;
            for(int c : s) {
                if(edges[q][c] == -1)
                    return false;
                q = edges[q][c];
            }
            return true;
        }
    };
}
```

suffix-tree

#fdf937

$\mathcal{O}(n \log n)$ lub $\mathcal{O}(n\alpha)$. Dla słowa `abaab#` (hash jest aby to zawsze liście były stanami końzącymi) stworzy `sons[0]={{#,10},{a,4},{b,8}}`, `sons[4]={{a,5},{b,6}}`, `sons[6]={{#,7},{a,2}}`, `sons[8]={{#,9},{a,3}}`, reszta `sons`ów pusta, `slink[6]=8` i reszta `slink`ów 0 (gdzie `slink` jest zdefiniowany dla nie-liści jako wierzchołek zawierający ten suffix bez ostatniej literki), `up_edge_range[2]=up_edge_range[3]=(2,5)`, `up_edge_range[5]=(3,5)` i reszta jednoliterowa. Wierzchołek 1 oraz suffix wierzchołków jest roboczy. Zachodzi `up_edge_range[0]=(-1,-1)`, `parent[0]=0`, `slink[0]=1`.

```
struct SuffixTree {
    C int n;
    C vi &_in;
    V<map<int, int>> sons;
    V<pii> up_edge_range;
    vi parent, slink;
    int tv = 0, tp = 0, ts = 2, la = 0;
    void ukkadd(int c) {
        auto &lr = up_edge_range;
    suff:
        if (lr[tv].se < tp) {
            if (sons[tv].find(c) == sons[tv].end()) {
                sons[tv][c] = ts; lr[ts].fi = la; parent[ts++]
                    = tv;
                tv = slink[tv]; tp = lr[tv].se + 1; goto suff;
            }
            tv = sons[tv][c]; tp = lr[tv].fi;
        }
        if (tp == -1 || c == _in[tp])
            tp++;
        else {
            lr[ts + 1].fi = la; parent[ts + 1] = ts;
            lr[ts].fi = lr[tv].fi; lr[ts].se = tp - 1;
            parent[ts] = parent[tv]; sons[ts][c] = ts + 1;
            sons[ts][_in[tp]] = tv;
            lr[tv].fi = tp; parent[tv] = ts;
            sons[parent[ts]][_in[lr[ts].fi]] = ts; ts += 2;
            tv = slink[parent[ts - 2]]; tp = lr[ts - 2].fi;
            while (tp <= lr[ts - 2].se) {
                tv = sons[tv][_in[tp]]; tp += lr[tv].se - lr[tv
                    ].fi + 1;
            }
            if (tp == lr[ts - 2].se + 1)
                slink[ts - 2] = tv;
            else
                slink[ts - 2] = ts;
            tp = lr[tv].se - (tp - lr[ts-2].se) + 2; goto
                suff;
        }
    }
} // Remember to append string with a hash.
SuffixTree(C vi &in, int alpha)
    : n(ssize(in)), _in(in), sons(2 * n + 1),
```



```
up_edge_range(2 * n + 1, pair(0, n - 1)), parent(2
    * n + 1), slink(2 * n + 1) {
up_edge_range[0] = up_edge_range[1] = {-1, -1};
slink[0] = 1;
// When changing map to V, fill sons exactly here
// with -1 and replace if in ukkadd with sons[tv][c
] == -1.
REP(ch, alpha)
    sons[1][ch] = 0;
for(; la < n; ++la)
    ukkadd(in[la]);
}
};
```

wildcard-matching

#a35e01, includes: math/ntt
 $\mathcal{O}(n \log n)$, zwraca tablicę wystąpień wzorca. Alfabet od 0. Znaki zapytania to -1 . Mogą być zarówno w tekście jak i we wzrocu. Dla alfabetów większych niż 15 lepiej użyć bezpieczniejszej wersji.

```
// BEGIN HASH ee35a0
V<bool> wildcard_matching(vi text, vi pattern) {
for (int& e : text) ++e;
for (int& e : pattern) ++e;
reverse(all(pattern));
int n = ssize(text), m = ssize(pattern);
int sz = 1 << __lg(2 * n - 1);
vi a(sz), b(sz), c(sz);
auto h = [&](auto f, auto g) {
fill(all(a), 0);
fill(all(b), 0);
REP(i, n) a[i] = f(text[i]);
REP(i, m) b[i] = g(pattern[i]);
ntt(a, sz), ntt(b, sz);
reverse(all(pattern));
REP(i, sz) a[i] = mul(a[i], b[i]);
ntt(a, sz, true);
REP(i, sz) c[i] = add(c[i], a[i]);
};
h[[]](int x){return powi(x,3);},identity());
h[[]](int x){return sub[0, mul(2, mul(x, x))];}, [[]]
(int x){return mul(x, x);});
h(identity(),[])(int x){return powi(x,3);});
V<bool> ret(n - m + 1);
FOR(i, m, n) ret[i - m] = !c[i - 1];
return ret;
} // END HASH
V<bool> safer_wildcard_matching(vi text, vi pattern,
int alpha = 26) {
static mt19937 rng(0); // Can be changed.
int n = ssize(text), m = ssize(pattern);
V ret(n - m + 1, true);
vi v(alpha), a(n, -1), b(m, -1);
REP(iters, 2) { // The more the better.
REP(i, alpha) v[i] = int(rng()) % (mod - 1);
REP(i, n) if (text[i] != -1) a[i] = v[text[i]];
REP(i, m) if (pattern[i] != -1) b[i] = v[pattern[i]
]];
auto h = wildcard_matching(a, b);
REP(i, n - m + 1) ret[i] = ret[i] & h[i];
}
return ret;
}
```

Optymalizacje (9)

divide-and-conquer-dp

#e7e6da
 $\mathcal{O}(nm \log m)$, dla funkcji $cost(k, j)$ wylicza
 $dp(i, j) = \min_{0 \leq k < j} dp(i - 1, k - 1) + cost(k, j)$. Działa tylko wtedy, gdy $opt(i, j - 1) \leq opt(i, j)$, a jest to zawsze spetnione, gdy $cost(b, c) \leq cost(a, d)$ oraz $cost(a, c) + cost(b, d) \leq cost(a, d) + cost(b, c)$ dla $a \leq b \leq c \leq d$.

wildcard-matching divide-and-conquer-dp dp-1d1d fio knuth linear-knapsack

```
vll divide_and_conquer_optimization(int n, int m,
function<ll(int,int)> cost) {
vll dp_before(m);
auto dp_cur = dp_before;
REP(i, m)
    dp_before[i] = cost(0, i);
function<void(int,int,int,int)> compute = [&](int l,
int r, int optl, int optr) {
if (l > r)
return;
int mid = (l + r) / 2, opt;
pair<ll, int> best = {numeric_limits<ll>::max(),
-1};
FOR(k, optl, min(mid, optr))
    chmin(best, pair((k ? dp_before[k - 1] : 0) +
cost(k, mid), k));
tie(dp_cur[mid], opt) = best;
compute(l, mid - 1, optl, opt);
compute(mid + 1, r, opt, optr);
};
REP(i, n) {
compute(0, m - 1, 0, m - 1);
swap(dp_before, dp_cur);
}
return dp_before;
}
```

dp-1d1d

#ad7df5
 $\mathcal{O}(n \log n)$, $n > 0$ długość paska, $cost(i, j)$ koszt odcinka $[i, j]$ Dla $a \leq b \leq c \leq d$ $cost$ ma spełniać
 $cost(a, c) + cost(b, d) \leq cost(a, d) + cost(b, c)$. Dzieli pasek $[0, n]$ na odcinki $[0, cuts[0]], \dots, [cuts[i - 1], cuts[i]]$, gdzie $cuts.back() = n - 1$, aby sumaryczny koszt wszystkich odcinków był minimalny. $cuts$ to prawe końce tych odcinków. Zwraca $(opt_cost, cuts)$. Aby maksymalizować koszt zamienić nierówności tam, gdzie wskazane. Aby uzyskać $\mathcal{O}(n)$, należy przepisać $overtake$ w oparciu o dodatkowe założenia, aby chodził w $\mathcal{O}(1)$.

```
pair<ll, vi> dp_1d1d(int n, function<ll (int, int)>
cost) {
V<pair<ll, int>> dp(n);
vi lf(n + 2), rg(n + 2), dead(n);
V<vi> events(n + 1);
int beg = n, end = n + 1;
rg[beg] = end; lf[end] = beg;
auto score = [&](int i, int j) {
return dp[j].fi + cost(j + 1, i);
};
auto overtake = [&](int a, int b, int mn) {
int bp = mn - 1, bk = n;
while (bk - bp > 1) {
int bs = (bp + bk) / 2;
if (score(bs, a) <= score(bs, b)) // tu >=
    bk = bs;
else
    bp = bs;
}
return bk;
};
auto add = [&](int i, int mn) {
if (lf[i] == beg)
return;
events[overtake(i, lf[i], mn)].eb(i);
};
REP (i, n) {
dp[i] = {cost(0, i), -1};
REP (j, ssize(events[i])) {
int x = events[i][j];
if (dead[x])
continue;
dead[lf[x]] = 1; lf[x] = lf[lf[x]];
rg[lf[x]] = x; add(x, i);
}
if (rg[beg] != end)
```

```
    chmin(dp[i], pair(score(i, rg[beg]), rg[beg]));
// tu max
lf[i] = lf[end]; rg[i] = end;
rg[lf[i]] = i; lf[rg[i]] = i;
add(i, i + 1);
}
vi cuts;
for (int p = n - 1; p != -1; p = dp[p].se)
    cuts.eb(p);
reverse(all(cuts));
return pair(dp[n - 1].fi, cuts);
}
```

fio

#115ad1
FIO do wpychania kolanem. Nie należy wtedy używać cin/cout

```
#ifndef ONLINE_JUDGE
// write this when judge is on Windows
inline int getchar_unlocked() { return _getchar_nolock
(); }
inline void putchar_unlocked(char c) { _putchar_nolock(
c); }
#endif
// BEGIN HASH 1ed0dd
int fastin() {
int n = 0, c = getchar_unlocked();
while (isspace(c))
c = getchar_unlocked();
while (isdigit(c)) {
n = 10 * n + (c - '0');
c = getchar_unlocked();
}
return n;
} // END HASH
// BEGIN HASH 3abf5f
int fastin_negative() {
int n = 0, negative = false, c = getchar_unlocked();
while (isspace(c))
c = getchar_unlocked();
if (c == '-') {
negative = true;
c = getchar_unlocked();
}
while (isdigit(c)) {
n = 10 * n + (c - '0');
c = getchar_unlocked();
}
return negative ? -n : n;
} // END HASH
// BEGIN HASH 323fab
double fastin_double() {
double x = 0, t = 1;
int negative = false, c = getchar_unlocked();
while (isspace(c))
c = getchar_unlocked();
if (c == '-') {
negative = true;
c = getchar_unlocked();
}
while (isdigit(c)) {
x = x * 10 + (c - '0');
c = getchar_unlocked();
}
if (c == '.') {
c = getchar_unlocked();
while (isdigit(c)) {
t /= 10;
x = x + t * (c - '0');
c = getchar_unlocked();
}
}
return negative ? -x : x;
} // END HASH
// BEGIN HASH 0b2d96
void fastout(int x) {
```

```
if (x == 0) {
putchar_unlocked('0');
putchar_unlocked(' ');
return;
}
if (x < 0) {
putchar_unlocked('-');
x *= -1;
}
static char t[10];
int i = 0;
while (x) {
t[i++] = char('0' + (x % 10));
x /= 10;
}
while (--i >= 0)
    putchar_unlocked(t[i]);
putchar_unlocked(' ');
}
void nl() { putchar_unlocked('\n'); }
// END HASH
```

knuth

#221040
 $\mathcal{O}(n^2)$, dla tablicy $cost(i, j)$ wylicza
 $dp(i, j) = \min_{i \leq k < j} dp(i, k) + dp(k + 1, j) + cost(i, j)$. Działa tylko wtedy, gdy $opt(i, j - 1) \leq opt(i, j) \leq opt(i + 1, j)$, a jest to zawsze spełnione, gdy $cost(b, c) \leq cost(a, d)$ oraz $cost(a, c) + cost(b, d) \leq cost(a, d) + cost(b, c)$ dla $a \leq b \leq c \leq d$.

```
ll knuth_optimization(V<vll> cost) {
int n = ssize(cost);
V dp(n, vll(n, numeric_limits<ll>::max()));
V opt(n, vi(n));
REP(i, n) {
opt[i][i] = i;
dp[i][i] = cost[i][i];
}
for (int i = n - 2; i >= 0; --i)
    FOR(j, i + 1, n - 1)
        FOR(k, opt[i][j - 1], min(j - 1, opt[i + 1][j]))
            if (dp[i][j] >= dp[i][k] + dp[k + 1][j] + cost[i]
[j]) {
opt[i][j] = k;
dp[i][j] = dp[i][k] + dp[k + 1][j] + cost[i][
j];
}
}
return dp[0][n - 1];
}
```

linear-knapsack

#5afd26
 $\mathcal{O}(n \cdot \max(w_i))$ zamiast typowego $\mathcal{O}(n \cdot \sum(w_i))$, pamięć $\mathcal{O}(n + \max(w_i))$, plecak zwracający największą otrzymywalną sumę ciężarów $\leq$ bound.

```
ll knapsack(vi w, ll bound) {
erase_if(w, [=](int x){ return x > bound; });
{
ll sum = accumulate(all(w), 0LL);
if (sum <= bound)
return sum;
}
ll w_init = 0;
int b;
for (b = 0; w_init + w[b] <= bound; ++b)
    w_init += w[b];
int W = *max_element(all(w));
vi prev_s(2 * W, -1);
auto get = [&](vi &w, ll i) -> int& {
return v[i - (bound - W + 1)];
};
for (ll mu = bound + 1; mu <= bound + W; ++mu)
    get(prev_s, mu) = 0;
get(prev_s, w_init) = b;
```

```
FOR(t, b, ssize(w) - 1) {
    V curr_s = prev_s;
    for(ll mu = bound - W + 1; mu <= bound; ++mu)
        chmax(get(curr_s, mu + w[t]), get(prev_s, mu));
    for(ll mu = bound + w[t]; mu >= bound + 1; --mu)
        for(int j = get(curr_s, mu) - 1; j >= get(prev_s, mu); --j)
            chmax(get(curr_s, mu - w[j]), j);
    swap(prev_s, curr_s);
}
for(ll mu = bound; mu >= 0; --mu)
    if(get(prev_s, mu) != -1)
        return mu;
assert(false);
}
```

matroid-intersection

#080bd2
 $O(r^2 \cdot (init + n \cdot add))$, where r is max independent set. Find largest subset S of $[n]$ such that S is independent in both matroid A and B , given by their oracles, see example implementations below. Returns V V such that $V[i] = 1$ iff i -th element is included in found set; Zabrane z [https://github.com/KacperTopolski/kactl/tree/main/Zmienne w matroidach](https://github.com/KacperTopolski/kactl/tree/main/Zmienne%20matroidach) ustawiamy ręcznie aby "zainicjalizować" tylko jeśli mają komentarz co znaczą. W przeciwnym wypadku intersectMatroids zrobi robotę wołając `init`.

```
// BEGIN HASH c90feb
template<class T, class U>
V<bool> intersectMatroids(T& A, U& B, int n) {
    V<bool> ans(n);
    bool ok = 1;
    // NOTE: for weighted matroid intersection find
    // shortest augmenting paths first by weight change,
    // then by length using Bellman-Ford,
    // Speedup trick (only for unweighted):
    A.init(ans); B.init(ans);
    REP(i, n)
        if (A.canAdd(i) && B.canAdd(i))
            ans[i] = 1, A.init(ans), B.init(ans);
    //End of speedup
    while (ok) {
        V<vi> G(n);
        V<bool> good(n);
        queue<int> que;
        vi prev(n, -1);
        A.init(ans); B.init(ans); ok = 0;
        REP(i, n) if (!ans[i]) {
            if (A.canAdd(i)) que.emplace(i), prev[i] = -2;
            good[i] = B.canAdd(i);
        }
        REP(i, n) if (ans[i]) {
            ans[i] = 0;
            A.init(ans); B.init(ans);
            REP(j, n) if (i != j && !ans[j]) {
                if (A.canAdd(j)) G[i].eb(j); // -cost[j]
                if (B.canAdd(j)) G[j].eb(i); // cost[i]
            }
            ans[i] = 1;
        }
        while (!que.empty()) {
            int i = que.front();
            que.pop();
            if (good[i]) { // best found (unweighted =
                shortest path)
                ans[i] = 1;
                while (prev[i] >= 0) { // alternate matching
                    ans[i = prev[i]] = 0;
                    ans[i = prev[i]] = 1;
                }
                ok = 1; break;
            }
            for(auto j: G[i]) if (prev[j] == -1)
                que.emplace(j), prev[j] = i;
        }
    }
}
```

```
return ans;
} // END HASH
// Matroid where each element has color
// and set is independent iff for each color c
// #{elements of color c} <= maxAllowed[c].
struct LinOracle {
    vi color; // color[i] = color of i-th element
    vi maxAllowed; // Limits for colors
    vi tmp;
    // Init oracle for independent set S; O(n)
    void init(V<bool>& S) {
        tmp = maxAllowed;
        REP(i, ssize(S)) tmp[color[i]] -= S[i];
    }
    // Check if S+{k} is independent; time: O(1)
    bool canAdd(int k) { return tmp[color[k]] > 0; }
};
// Graphic matroid - each element is edge,
// set is independent iff subgraph is acyclic.
struct GraphOracle {
    V<pii> elems; // Ground set: graph edges
    int n; // Number of vertices, indexed [0;n-1]
    vi par;
    int find(int i) {
        return par[i] == -1 ? i : par[i] = find(par[i]);
    }
    // Init oracle for independent set S; ~O(n)
    void init(V<bool>& S) {
        par.assign(n, -1);
        REP(i, ssize(S)) if (S[i])
            par[find(elems[i].fi)] = find(elems[i].se);
    }
    // Check if S+{k} is independent; time: ~O(1)
    bool canAdd(int k) {
        return find(elems[k].fi) != find(elems[k].se);
    }
};
// Co-graphic matroid - each element is edge,
// set is independent iff after removing edges
// from graph number of connected components
// doesn't change.
struct CographOracle {
    V<pii> elems; // Ground set: graph edges
    int n; // Number of vertices, indexed [0;n-1]
    V<vi> G;
    vi pre, low;
    int cnt;
    int dfs(int v, int p) {
        pre[v] = low[v] = ++cnt;
        for(auto e: G[v]) if (e != p)
            chmin(low[v], pre[e] ? : dfs(e, v));
        return low[v];
    }
    // Init oracle for independent set S; O(n)
    void init(V<bool>& S) {
        G.assign(n, {});
        pre.assign(n, 0);
        low.resize(n);
        cnt = 0;
        REP(i, ssize(S)) if (!S[i]) {
            pii e = elems[i];
            G[e.fi].eb(e.se);
            G[e.se].eb(e.fi);
        }
        REP(v, n) if (!pre[v]) dfs(v, -1);
    }
    // Check if S+{k} is independent; time: O(1)
    bool canAdd(int k) {
        pii e = elems[k];
        return max(pre[e.fi], pre[e.se]) != max(low[e.fi], low[e.se]);
    }
};
// Matroid equivalent to linear space with XOR
struct XorOracle {
    vll elems; // Ground set: numbers
```

```
vll base;
// Init for independent set S; O(n+r^2)
void init(V<bool>& S) {
    base.assign(63, 0);
    REP(i, ssize(S)) if (S[i]) {
        ll e = elems[i];
        REP(j, ssize(base)) if ((e >> j) & 1) {
            if (!base[j]) {
                base[j] = e;
                break;
            }
            e ^= base[j];
        }
    }
}
// Check if S+{k} is independent; time: O(r)
bool canAdd(int k) {
    ll e = elems[k];
    REP(i, ssize(base)) if ((e >> i) & 1) {
        if (!base[i]) return 1;
        e ^= base[i];
    }
    return 0;
}
};
```

random

```
#77ae04
Szybsze rand.

uint32_t xorshf96() {
    static uint32_t x = 123456789, y = 362436069, z = 521288629;
    x ^= x << 16; x ^= x >> 5; x ^= x << 1;
    uint32_t t = x; x = y; y = z; z = t ^ x ^ y;
    return z;
}
```

sos-dp

#947fac
 $O(n2^n)$, dla tablicy $A[i]$ oblicza tablicę $F[mask] = \sum_{i \subseteq mask} A[i]$, czyli sumę po podmaskach. Może też liczyć sumę po nadmaskach. `sos_dp(2, {4, 3, 7, 2})` zwraca $\{4, 7, 11, 16\}$, `sos_dp(2, {4, 3, 7, 2}, true)` zwraca $\{16, 5, 9, 2\}$.

```
vll sos_dp(int n, vll A, bool nad = false) {
    int N = (1 << n);
    if (nad) REP(i, N / 2) swap(A[i], A[(N - 1) ^ i]);
    auto F = A;
    REP(i, n)
        REP(mask, N)
            if ((mask >> i) & 1)
                F[mask] += F[mask ^ (1 << i)];
    if (nad) REP(i, N / 2) swap(F[i], F[(N - 1) ^ i]);
    return F;
}
```

Utils (10)

dzien-probny

```
#b68ef5, includes: data-structures/ordered-set
Rzeczy do przetestowania w dzień próbny.

// alternatywne żmnoenie ll, gdyby na wypadek gdyby nie
// było __int128
ll llmul(ll a, ll b, ll m) {
    return (a * b - (ll)((long double) a * b / m) * m + m) % m;
}

void test_int128() {
    __int128 x = (1llu << 62);
    x *= x;
    string s;
    while(x) {
```

```
s += char(x % 10 + '0');
    x /= 10;
}
assert(s == "61231558446921906466935685523974676212")
;
}
void test_float128() {
    __float128 x = 4.2;
    assert(abs(double(x * x) - double(4.2 * 4.2)) < 1e-9)
;
}
void test_clock() {
    long seed = chrono::system_clock::now().time_since_epoch().count();
    (void) seed;
    auto start = chrono::system_clock::now();
    while(true) {
        auto end = chrono::system_clock::now();
        int ms = int(chrono::duration_cast<chrono::milliseconds>(end - start).count());
        if(ms > 420)
            break;
    }
}
void test_rd() {
    // czy jest sens to testowac?
    mt19937_64 my_rng(0);
    auto rd = [&](int l, int r) {
        return uniform_int_distribution<int>(l, r)(my_rng);
    };
    assert(rd(0, 0) == 0);
}
void test_policy() {
    ordered_set<int> s;
    s.insert(1);
    s.insert(2);
    assert(s.order_of_key(1) == 0);
    assert(*s.find_by_order(1) == 2);
}
void test_math() {
    constexpr long double pi = acosl(-1);
    assert(3.14 < pi && pi < 3.15);
}
```

python

```
#a75dc7
Przykładowy kod w Pythonie z różną funkcjonalnością.

fib_mem = [1] * 2
def fill_fib(n):
    global fib_mem
    while len(fib_mem) <= n:
        fib_mem.append(fib_mem[-2] + fib_mem[-1])
def main():
    assert list(range(3, 6)) == [3, 4, 5]
    s = set()
    s.add(5)
    for x in s:
        print(x)
    s = [2 * x for x in s]
    print(eval("s[0] + 10"))
    m = {}
    m[5] = 6
    assert 5 in m
    assert list(m) == [5]
    line_list = list(map(int, input().split()))
    print(line_list)
    print(' '.join(["a", "b", str(5)]))
    while True:
        try:
            line_int = int(input())
        except Exception as e:
            break
main()
```